



Chhatrapati Shivaji Institute of Technology

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ENERGY AUDIT, ENVIRONMENTAL AUDIT & GREEN AUDIT REPORT

OF

**CHHATRAPATI SHIVAJI
INSTITUTE OF TECHNOLOGY,
DURG**

For the Year of 2023

ENERGY AUDIT, ENVIRONMENTAL AUDIT & GREEN AUDIT REPORT OF CHHATRAPATI SHIVAJI INSTITUTE OF TECHNOLOGY, DURG



**Chhatrapati Shivaji Institute of Technology, Balod Rd, Shivaji Nagar, Durg,
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Prepared By



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We express our sincere gratitude to **Chhatrapati Shivaji Institute of Technology** for giving us the opportunity to be a part of their mission towards Energy Conservation. We are thankful to all officers and employees of Chhatrapati Shivaji Institute of Technology with whom we have interacted during the field study for their whole-hearted support in undertaking measurements and eagerness to assess the system/equipment efficiencies and saving potential. The willingness of these key personnel to participate in this program and acknowledge the call for energy efficiency is more than half the issues received.



Authorized Signatory

for, Audittech Industrial Services Private Limited

CERTIFICATE



This is to Certify that Chhatrapati Shivaji Group of Institution, Durg (Chhattisgarh) has conducted Energy Audit, Environmental Audit & Green Audit of their campus and has submitted necessary data in addition, credentials for scrutiny the activities and measures carried out by the university have been verified based on the report submitted and was found to be acceptable.

The efforts taken by the CSIT Management, Faculties and students towards Green Building, Environment and sustainability is highly appreciated and commendable.

Aashish Bafna



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EXECUTIVE SUMMARY

Energy Audit, Environmental & Green Audit is the process of identifying and determining whether institutions' practices are eco-friendly and sustainable. The main objective to carry out a Energy Audit, Environmental & Green Audit is to check green practices followed by the College and to conduct a well-formulated audit report to understand where we stand on a scale of environmental soundness. The initiative taken by Chhatrapati Shivaji Institute of Technology to conduct a Energy Audit, Environmental Audit & Green Audit of the College campus is a commendable sustainable goal. The strategies followed were the preparation of questionnaires and subsequent action plans to implement the project. Questionnaires prepared to conduct the Energy Audit, Environmental & Green Audit were based on the guidelines, rules, acts, and formats set by the Government of India, Ministry of Environment and Forest, New Delhi, and Central Pollution Control Board, New Delhi. Questionnaires were prepared for solid waste, energy, water, hazardous waste, and e-waste. For audit purposes, analysis of suitable data is required, for the same study area is grouped into various Blocks and Departments. The audit was carried out for solid waste, electricity and energy, water and wastewater, hazardous waste, air quality, and green inventory including carbon footprints. It also indicates the green initiatives taken by universities to save environmental resources. The “Energy Audit, Environmental Audit & Green Audit” also presents the “Environmental Management Plan”.

1. INTRODUCTION

1.1 ENERGY AUDIT, ENVIRONMENTAL & GREEN AUDIT AN EFFECTIVE EFFORTS TOWARDS ENVIRONMENT SUSTAINABILITY & ENERGY CONSERVATION

Modernization and industrialization are the two important outputs of the twentieth century that have made human life more luxurious and comfortable. Simultaneously, they are responsible for voracious use of natural resources, exploitation of forests and wildlife, producing massive solid waste, polluting the scarce and sacred water resources, and finally making our mother Earth ugly and inhospitable. Today, people are getting more familiar with global issues like global warming, greenhouse effect, ozone depletion, climate change, etc. Now, it is considered as a final call by mother Earth to walk on the path of sustainable development. The time has come to wake up, unite and combat together for a sustainable environment.

Considering the present environmental problems of pollution and excessive use of natural resources, Honorable Prime Minister, Shri. Narendra Modi has declared the Mission of Swachh Bharat Abhiyan. Also, College Grants Commission has mentioned the “Green Campus, Clean Campus” mission mandatory for all higher educational institutes. As environmental sustainability is becoming an increasingly important issue for the nation, the role of higher educational institutions in relation to environmental sustainability is more prevalent.

Energy Audit, Environmental & Green Audit is the most efficient ecological tool to solve such environmental problems. It is a process of regular identification, quantification, documenting, reporting and monitoring of environmentally important components in a specified area. Through this process, the regular environmental activities are monitored within and outside of the concerned sites which have direct and indirect impacts on the surroundings. A Energy Audit, Environmental & Green Audit can be one of the initiatives for such institutes to account for their energy, water resource use as well as wastewater, solid waste, hazardous waste generation. The Energy Audit, Environmental & Green Audit process can play an important role in the promotion of environmental awareness and sensitization about resource use. It can create consciousness towards ecological values and ethics. Through the Energy Audit, Environmental & Green Audit, one can get direction about how to improve the condition of the environment.

1.2 WHY ENERGY AUDIT, ENVIRONMENTAL & GREEN AUDIT

Energy Audit, Environmental & Green Auditing is the process of identifying and determining whether an institution's practices are eco-friendly and sustainable. Traditionally, we are good and efficient users of natural resources. However, over the period of time excess use of resources like energy, water, chemicals are become habitual for everyone especially, in common areas. Now, it is necessary to check whether our processes are consuming more than the required resources? Whether we are handling waste carefully? Energy Audit, Environmental & Green Audit regulates all such practices and gives an efficient way of natural resource utilization. In the era of climate change and resource depletion, it is necessary to verify the processes and convert them into green and clean ones. The Energy Audit, Environmental & Green Audit provides an approach for it. It also increases overall consciousness among the people working in institutions towards an environment.

1.3 GOALS OF ENERGY AUDIT, ENVIRONMENTAL & GREEN AUDIT

College has conducted a Energy Audit, Environmental & Green Audit with specific goals as:

- Assess facility of different types of waste management.
- Increase environmental awareness throughout campus.
- Identification and documentation of green practices followed by college.
- Identify strengths and weaknesses in green practices.
- Conduct a survey to know the ground reality about green practices.
- Analyze and suggest solutions for problems identified from the survey.
- Identify and assess environmental risk.
- The long-term goal of the environmental audit program is to collect baseline data of environmental parameters and resolve environmental issues.
- To motivate staff for optimized sustainable use of available resources.

1.4 OBJECTIVES OF ENERGY AUDIT, ENVIRONMENTAL & GREEN AUDIT

- To examine the current practices which can impact the environment such as resource utilization, waste management, etc.
- To prepare an Environmental Statement Report on green practices followed by different departments, support services, and administration building.
- To set goals, vision, and mission for green practices on the campus.
- To identify and analyze significant environmental issues.
- To establish and implement Environmental Management Plan in various departments.
- To assess for better performance in green practices and its valuation.

1.5 ABOUT CRITERIA 7 OF NAAC

Universities are playing a key role in the development of human resources worldwide. Higher education institutes campus run various activities with the aim to percolate the knowledge along with practical dimension among the society. Likewise, different technological solutions related to the environment are also provided by the higher education institutes. Different types of evolutionary methods are used to assess the problem concerning the environment. It includes Environmental Impact Assessment (EIA), Social Impact Assessment (SIA), Carbon Footprint Mapping, Energy Audit, Environmental & Green Audit, etc.

National Assessment and Accreditation Council (NAAC) is a self-governing organization that rated the institutions according to the scores assigned at the time of accreditation of the institution. Energy Audit, Environmental & Green Audit has become a mandatory procedure for educational institutes under Criterion VII of NAAC. The intention of the Energy Audit, Environmental & Green Audits is to upgrade the environmental condition inside and around the institution. It is performed by considering environmental parameters like water and wastewater accounting, energy conservation, waste management, air, noise monitoring, etc. for making the institution eco-friendlier.

Students are the major strength of any academic institution. Practicing green action in any educational institution will inculcate the good habit of caring for natural resources in students. Many environmental activities like plantation and nurturing saplings and trees, Cleanliness drives, Bird watching camps, no vehicle day, Rainwater harvesting, etc. will make the students good citizens of the country. Through Energy Audit, Environmental & Green Audit, higher educational institutions can ensure that they contribute towards the reduction of global warming through Carbon Footprint reduction measures.

1.6 BENEFITS OF ENERGY AUDIT, ENVIRONMENTAL & GREEN AUDIT TO AN EDUCATIONAL INSTITUTE

There are many advantages of Energy Audit, Environmental & Green Audit to an Educational Institute.

- It would help to protect the environment in and around the campus.
- Recognize the cost-saving methods through waste minimization and energy conservation.
- Empower the organization to frame a better environmental performance.
- It portrays a good image of the institution through its clean and green campus.
- More efficient resource management
- To create a green campus
- To enable waste management through reduction of waste generation, solid and waste
- To create plastic-free campus and evolve health consciousness among the stakeholder
- Recognize the cost-saving methods through waste minimizing and managing
- Authenticate conformity with the implemented laws
- Empower the organizations to frame a better environmental performance
- Enhance the alertness for environmental guidelines and duties
- Impart environmental education through systematic environmental management approach and improving environmental standards
- Benchmarking for environmental protection initiatives
- Financial savings through a reduction in resource use
- Development of ownership, personal and social responsibility for the College and its environment
- Developing an environmental ethic and value systems in youngsters.
- Energy Audit, Environmental & Green Auditing should become a valuable tool in the management and monitoring of environmental and sustainable development programs of the College.
- Finally, it will help to build a positive impression through green initiatives for the upcoming NAAC visit.

1.7 INTRODUCTION OF AUDITING FIRM

M/s. Audittech is an empaneled Accredited Energy Audit Firm from the Bureau of Energy Efficiency, Ministry of Power, Government of India. It is one of the fast-growing Energy Audit & Energy services providing company executed several projects covering all the energy Intensive Sectors & states of India. The directors and associate team members are very well experienced in the field of Energy Audit and executed more than 100 no's Detailed Energy Audit & More than 10 No's of Green Audit so far. The associate team and experts are highly qualified and experienced in the field of Energy Audit and Services. Individual credential of each member in the field of Energy Audit is very rich due to their past association with the very reputed organization of Energy Audit Services. The company has Head office at Balod (C.G.) & Branch offices at Raipur, Durg, Bhopal, Mumbai, and Delhi.

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Directors Details

Sr. No.	Name	Designation / Technical Experience	Technical Experience /Qualification
1	Mr. Aashish Bafna	Managing Director - 13yrs	B.E (E&I)., MBA(Energy Management), Certified Energy Auditor, Surveyor & Loss Assessor
2	Mr. Rakesh Khichariya	Director- 25Yrs	B.E (Elect.), Accredited Energy Auditor
3	Mr. Ramesh Patel	Director- 25Yrs	B.E.(Mech), Government approved Valuer, Competent Person for Factory Act
4	Mr. Isshant Chainani	Director- 14 Yrs	B.E. (Elect & telecom)
5	Mrs. Shikha Golchha	Director- 13 Yrs	B.E., MBA (Finance)

M/s. Audittech Recognized as “Startup” from Department of Promotion of Industry and Internal Trade, Ministry of Commerce & Industry, Government of India and also Registered in Ministry of Micro, Small & Medium Enterprises (MSME) as Micro unit.

ENERGY AUDIT TEAM

The energy audit team involved in the energy audit of CSIT, Durg details are in the following.

SN	Name	Designation/ Qualification	Experience	Contact Details
1	Mr. Rakesh Khichariya	Accredited Energy Auditor (AEA-0295)	25 yrs.	9827411444
2	Mr. Aashish Bafna	Certified Energy Auditor (EA-28916)	13 yrs.	9827143100
3	Mr. Isshant Chainani	Certified Energy Manager	14 yrs.	9407702444
4	Mr. Mahaveer Bafna	Certified Energy Manager	6 Yrs.	8962369293
5.	Mr. Retesh Verma	Energy Engineer	2 Yrs.	8770912822
6.	Mr. Kishor Sarwa	Energy Engineer	2 Yrs.	8435528535
7	Mr. Yeeshu kumar	Energy Engineer	1 Yrs.	9753954109

LIST OF INSTRUMENTS

Following are the instrument used at the time of the Energy Audit.

Sr. No.	Instrument	Make/Sr. No.
1	Power & Harmonics Analyzer, 1 Set (With CT, PT) HT	Krykard ALM 31/ 123673RCH
2	Power & Harmonics Analyzer, 1 Set (With CT, PT) HT	Krykard ALM 20/ 28107280
3	Temperature gun 1 Set (infrared Thermometer)	MECO 550 T/ IRT550T_17120136
4	Lux Meter 1Set (Digital Lux Meter)	MECO G 930P/201704004601
5	Ultrasonic Flow Meter	Chinese/ 28107280

1.8 ABOUT CHHATRAPATI SHIVAJI INSTITUTE OF TECHNOLOGY

Chhatrapati Shivaji Institute of Technology is a distinguished engineering and technology institute, potentially located in Durg, Chhattisgarh. Established in the year 1999, affiliated to Chhattisgarh Swami Vivekanand Technical University. The institute offers a range of undergraduate programs in engineering and technology. With facilities Computer Lab, Library, Seminar Hall, Auditorium / more, and a dynamic learning environment, the institute focuses on practical learning and research. Chhatrapati Shivaji Institute of Technology emphasizes industry collaboration and innovation, preparing students for successful careers in the rapidly evolving technological landscape. The institute's dedicated faculty and emphasis on overall development contribute to its reputation as a premier institution in engineering education.

The present location of College is Located at Durg in Chhattisgarh. The Address of The Institute Is Durg-Balod, Dallirajhara Road at Shivaji Nagar near Kolihapuri, Dist. Durg (C.G.), 4 Kms away from the Collectorate and 6 Kms from Railway Station, Durg. The grandeur of the location provides a unique environment for education in close proximity with nature which is farfetched from the hum-drum of the city.

Over the years, it has steadily evolved itself as one of the leading centers of comprehensive learning with its exposure to and linkages with its peers at national and international levels. With its state-of-the-art infrastructure, well-equipped laboratories, and highly qualified and dedicated faculty, the College is committed to the task of harnessing and cultivating the capabilities of young students with a view to enabling them to carve suitable space for themselves in the modern economic world. The College Building Campus is categorized mainly into 4 Building Blocks. The details of all departments are mentioned in the below table of Buildings:

- Academic Buildings
- Facilities Buildings
- Residential Buildings

1.8.1 CAMPUS BUILDING & DEPARTMENTS

Name of the Building Blocks and their coding	
Building / Blocks	Name of the Departments
Departments	Department of Computer Science and Engineering
	Department of Electrical and Electronics Engineering
	Department of Mechanical Engineering
	Department of Information Technology
	Department of Mechatronics
	Department of Civil Engineering
	Department of Artificial Intelligence and Data Science
Labs	Computer Organization & Architecture Lab
	Software Laboratory (Sci Lab/ MATLAB)
	Machine Learning Lab
	Automobile Engineering Lab
	Industrial Engineering & Management Lab
	Mechanical Measurement and Metrology Lab
	Robotics Lab
	Electrical and Electronics Engineering Laboratory
	Power Electronics Devices and Circuits Laboratory
	High Voltage Engineering Laboratory
	Electrical Measurements & Measuring Instruments Laboratory
	Surveying Field Work Lab
	Geology Lab
	Environment Engineering Lab
	Simulation Lab

Facilities	
Facilities	Central Workshop
	Computer Centre
	Canteen and Cafeteria
	Student Activity Centre
	Seminar Hall
	Administrative Building
	Health Care Center
	Bank and ATM
Residential Buildings	
Residential Building	Boys Hostel
	Girls Hostel
	Staff Quarters

The College has also adopted the 'Green Campus' system for environmental conservation and sustainability. The 'Green Campus' has been active for the last several years as an assembly group of sub committees along with the Horticulture section that actively promotes the various projects. The College administration works on several activities for 'Green Campus' including Renewable Energy, Water Conservation, Tree Plantation, Waste Management, Paperless Work, etc.

1.8.2 CAMPUS INFRASTRUCTURE

Chhatrapati Shivaji Institute of Technology has a very good and systematic building infrastructure. All classrooms are fully ventilated and comfortable. The College from the outside looks great and is quite unique keeping the architecture of other universities in mind.

CSIT, DURG – CAMPUS HIGHLIGHTS

1.8.3.1 INTERNAL QUALITY ASSURANCE CELL

In pursuance of its Action Plan for performance evaluation, assessment & accreditation & quality up-gradation of institutions of higher education, NAAC proposes that every accredited institution should establish an Internal Quality Assurance Cell (IQAC) as a post-accreditation quality sustenance measure. Since quality enhancement is a continuous process, the IQAC will become a part of the institution's system & work towards realization of the goals of quality enhancement & sustenance.

OBJECTIVES:

- To develop a system for conscious, consistent & catalytic improvement in the overall academic and administrative performance of institution.
- Making efforts towards promoting student – centric atmosphere for holistic academic excellence.
- To enhance the quality of staff by promoting and encouraging overall faculty development.

FUNCTIONS OF THE IQAC ARE:

- Development and application of quality benchmarks/parameters for various academic and administrative activities of the institution.
- Facilitating the creation of a learner-centric environment conducive to quality education and faculty maturation to adopt the required knowledge and technology for participatory teaching and learning process.
- Arrangement for feedback response from students, parents and other stakeholders on quality-related institutional processes.
- Dissemination of information on various quality parameters of higher education.
- Organization of inter and intra institutional workshops, seminars on quality related themes and promotion of quality circles.
- Documentation of the various programmes/activities leading to quality improvement.
- Acting as a nodal agency of the Institution for coordinating quality-related activities, including adoption and dissemination of best practices.
- Development and maintenance of institutional database through MIS for the purpose of maintaining/enhancing the institutional quality.
- Development of Quality Culture in the institution.
- Preparation of the Annual Quality Assurance Report (AQAR) as per guidelines and parameters of NACC, to be submitted to NACC.

1.8.3.2 SPORTS AND CULTURE PARK

CSIT inculcates a high standard of ethical and moral values amongst the student's community to attain this and train students to be morally, culturally and ethically to be on the right side of every aspect. Industrial visits, tours, excursions and study are part of college life. Students are given equal opportunity to show their talent in other art forms like Dance, Art, Singing, Music, Drama, Debate, Quiz, Writing and

Elocution etc. by organizing competitions during the year. Quite a few stars emerge from these competitions. The college felicitates outstanding students for their performance. Cultural events are organized every year by student associations which give students a platform to showcase their talent in cultural fields. Apart from these events the students participate in various state and college level program where students have brought laurels to the college.

SPORTS

Sports facilities are an integral part of any educational activity to keep students healthy, both physically and mentally. In addition to organizing the Annual Sports routinely, intra-college league matches of basketball, cricket, football, chess, volleyball, badminton, table tennis etc. are framed every year. Each year, the much-awaited inter-collegiate volleyball tournament attracts participants from all over the state.

OUTDOOR FACILITIES

Cricket / Football Ground
Basket Ball Court
Volley Ball Court
Handball Court

INDOOR FACILITIES

Carrom Tables
Chess Setup
Table Tennis Court
Badminton Court



GREEN AREA AT CENTRAL BLOCK



BOYS HOSTEL



ENTRANCE GARDEN AND THE STATUE OF CHHATRAPATI SHIVAJI MAHARAJ



Campus Entrance Path Way with Green Surrounding



Girls Hostel and the Surrounding Green Patch



Entrance



Shaheed Memorial Hall



Library



Swami Vivekanand Auditorium



Play Ground

2. ENERGY AUDIT, ENVIRONMENTAL & GREEN AUDIT METHODOLOGY

2.1 Pre-Audit Stage

A pre-audit meeting provided an opportunity to reinforce the scope and objectives of the audit and pre-audit discussions were held on the basis of green initiatives taken and the current scenario of the College campus. This meeting is an important prerequisite for the Energy Audit, Environmental & Green Audit because it is the first opportunity to understand the concerns. The meeting was an opportunity to gather the information that the audit team can study before arriving on the site. The audit protocol and audit plan were handed over at this meeting and discussed in advance of the audit itself. The pre-audit meeting was conducted successfully and necessary documents were collected directly from the College before the initiation of the audit processes. The actual planning of audit processes was discussed in the pre-audit meeting. An Audit team was also selected in this meeting with the help of staff and the College management. The audit protocol and audit plan were handed over at this meeting and discussed in advance of the audit itself.

2.2 Management Commitment

The Management of the College has shown a commitment towards Energy Audit, Environmental & Green Auditing during the pre-audit meeting. They were ready to encourage all green activities. It was decided to promote all activities that are environmentally friendly such as awareness programs on the environment, campus farming, planting more trees on the campus, etc., after the Energy Audit, Environmental & Green Auditing. The management of the College was willing to formulate policies based on a Energy Audit, Environmental & Green Auditing report.

2.3 Objectives of the study

A clean and healthy environment aids effective learning and provides a conducive learning environment. There are various efforts around the world to address environmental education issues. Energy Audit, Environmental & Green Audit is the most efficient and ecological way to manage environmental problems. It is a kind of professional care that is the responsibility of each individual who is part of economic, financial, social, environmental factors. It is necessary to conduct a Energy Audit, Environmental & Green Audit on a College campus because students become aware of the Energy Audit, Environmental & Green Audit, its advantages to saving the planet and they become social and responsible citizens of our country. Thus, Energy Audit, Environmental & Green Audit becomes necessary at the College level. The broad objectives are as follows.

- Diagnosing the environmental problems to eliminate them.
- Environmental education through a systematic environmental management approach.
- Improving environmental standards.
- Benchmarking for environmental protection initiatives.
- Efficient utilization of resources.
- Financial savings through a reduction in resource use.
- Curriculum enrichment through practical experience.
- Development of ownership, personal and social responsibility for the College and its environment.
- Developing environmental ethics and value systems in young people.
- Providing certain recommendations based on environmental audit reports.
- Ensuring compliance, not only with laws, regulations, and standards but also with company policies and the requirements of an Environmental Management System (EMS) standard.
- Enabling environmental problems and risks to be anticipated.
- To demonstrate that College is aware of its impact upon the environment.

2.4 Audit Stage

Energy Audit, Environmental & Green Audit was done with the help of co-associates involving different student groups, teaching, and non-teaching staff. The Energy Audit, Environmental & Green Audit began with the teams walking through all the different facilities at the College, determining the different types of appliances and utilities as well as measuring the usage per item (Watts indicated on the appliance or measuring water from a tap) and identifying the relevant consumption patterns (such as how often an appliance is used) and their impacts. The staff and learners were interviewed to get details of usage, frequency, or general characteristics of certain appliances. Data collection was done in the sectors such as Energy, Waste, Green Area, Carbon footprint, and Water use. College records and documents were verified several times to clarify the data received through surveys and discussions.

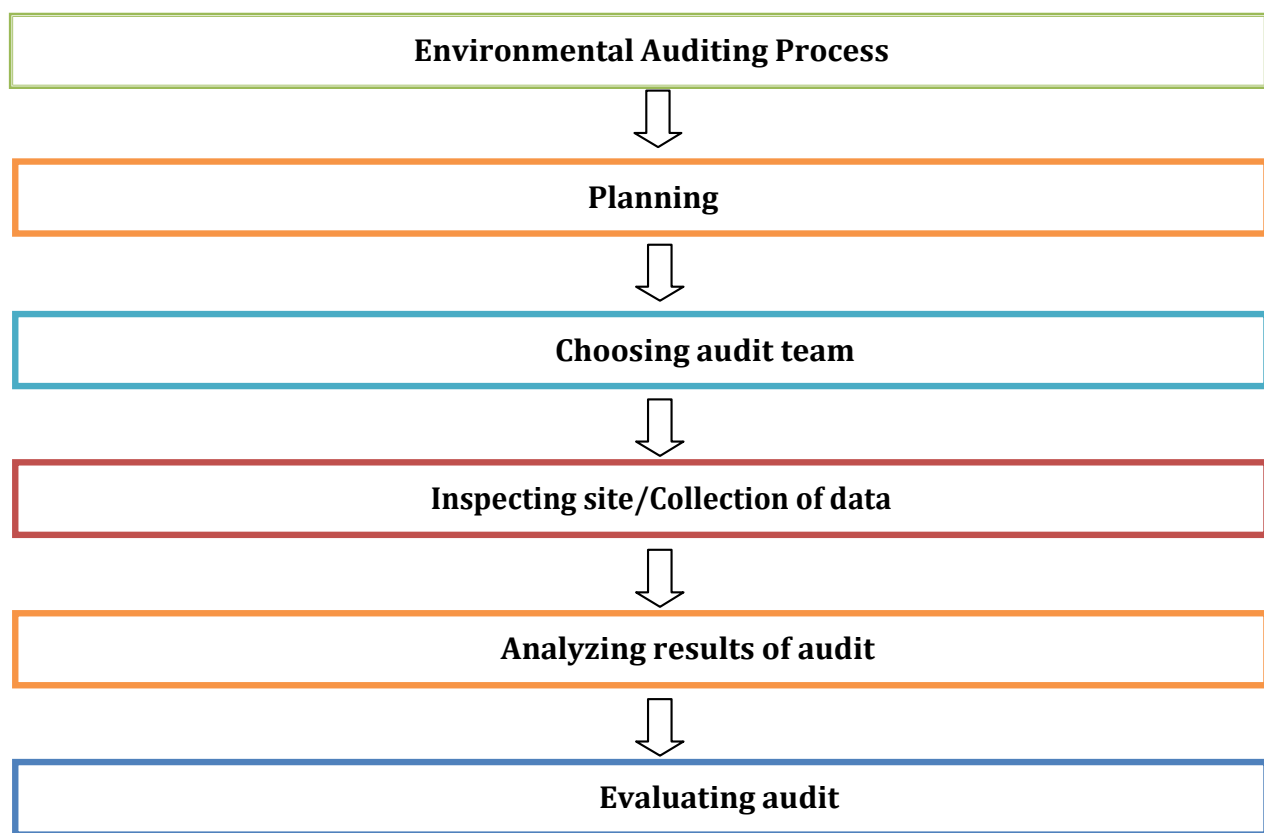
2.5 Methodology

The Management of the College has shown a commitment towards Energy Audit, Environmental & Green Auditing during the pre-audit meeting. They were ready to encourage all green activities. It was decided to promote all activities that are environmentally friendly such as awareness programs on the environment, campus farming, planting more trees on the campus, etc., after the Energy Audit, Environmental & Green Auditing. The management of the College was willing to formulate policies based on a Energy Audit, Environmental & Green Auditing report. In order to perform Energy Audit, Environmental & Green Audits, the methodology included different tools such as preparation of questionnaires, physical inspection of the campus, observation, and review of the documentation, interviewing key persons, and data analysis, measurements, and recommendations. The study covered the following areas to summarize the present status of environmental management on the campus:

- Energy Management
- Water Management
- Waste Management
- Environment Management

Methodology – Step by Step

The audit process was carried out in three phases. At first, all the secondary data required for the study was collected from various sources, like concerned departments such as engineering cell, Internal Quality Assurance Cell, Department of Physics, etc. A broad reference work was carried out to clear the idea of Energy Audit, Environmental & Green Auditing. Different case studies and methodologies were studied and the following methodology was adopted for the present audit. The methodology of the present study is based on onsite visits, personal observations, and questionnaires survey tools. Initially, based on data requirements, sets of questionnaires were prepared. The surveyors then visited all the departments of the College and the questionnaires were filled. The generated data is subsequently gathered and used for further analysis. From the outcome of the overall study, a final report is prepared.



2.5.1 Survey by Questionnaire

Baseline data for Energy Audit, Environmental & Green Audit report preparation was collected by questionnaire survey method. Questionnaires are prepared to conduct the Energy Audit, Environmental & Green Audit in the College campus based on the guidelines, rules, acts, and formats prepared by the Ministry of Environment, Forest and Climate Change, New Delhi, Central Pollution Control Board, and other statutory organizations. Most of the guidelines and formats are based on broad aspects and some of the issues or formats were not applicable for the College campus. Therefore, using these guidelines and formats, combinations, modifications, and restructuring were done and sets of questionnaires were prepared for solid waste, energy, water, hazardous waste, and e-waste data.

All the questionnaires are a group of modules. The first module is related to the general information of the concerned department, which broadly includes the name of the department, month and year, the total number of students and employees, visitors of the department, average working days and office timings, etc. The next module is related to the present consumption of resources like water, energy, or the handling of solid and hazardous waste. Maintaining records of the handling of solid and hazardous waste is much important in Energy Audit, Environmental & Green Audits. There are possibilities of loss of

resources like water, energy due to improper maintenance, and assessment of this kind of probability is necessary for the Energy Audit, Environmental & Green Audit. One separate module is based on the questions related to this aspect. Another module is related to maintaining records, like records of disposal of solid waste, records of solid waste recovery, etc. For better convenience of the surveyor, some statistics like basic energy consumption characteristics for electrical equipment, etc. were provided with the questionnaires.

2.5.2 Onsite visit and observations

Chhatrapati Shivaji Institute of Technology has a vast built-up area comprising various departments under various academic buildings, staff quarters, many facilities including Academic Buildings, Health Centre, Council Hall, Auditorium, other facilities and separate boy's and girl's hostels. All these amenities have different kinds of infrastructure as per their requirement. All these buildings were visited by the surveyors and the present condition is checked with the help of the questionnaires. Personal observations were made during the onsite visit. All the amenities were clubbed in, as per their similarities and differences, which makes the survey and further analysis easier. For the data compilation purpose, the College Departments and support services were clubbed into Three Types of Buildings and given names as academic buildings, Facilities Buildings, and Hostel Buildings. The details of the Buildings are as follows:

Sr. No.	Name of the Buildings
1.	Academic Buildings
2.	Facilities Buildings
3.	Hostel Buildings

After the collection of secondary data, the reviews related to each environmental factor were taken by the Energy Audit, Environmental & Green Audit team. The data were tabulated, analyzed and graphs were prepared. Depending upon the observations and data collected, interpretations were made. The lacunas and good practices were documented. The Environmental Management Plan (EMP) was prepared for the next academic year in order to have better environmental sensitization. Finally, all the information was compiled in the form of the Energy Audit, Environmental & Green Audit Report.

2.5.3 Data analysis and final report preparation

Proper analysis and presentation of data produced from work is a vital element. In the case of a Energy Audit, Environmental & Green Audit, the filled questionnaires of the survey from each group were tabulated as per their modules, in Excel spreadsheets. The tabulated data is then used for further analysis. For a better understanding of the results and to avoid complications, averages, and percentages of the tables were calculated. A graphical representation of these results was made to give a quick idea of the status. Interpretation of the overall outcomes was made which incorporates all the primary and secondary data, references, and interrelations within. Final report preparation was done using this interpretation.

3. WATER & WASTE WATER AUDIT

Water is a precious natural national resource available with a fixed quantum. The availability of water is decreasing due to the increasing population of the nation; as per capita availability of utilized water is going down. Due to the ever-rising standard of living of people, industrialization, urbanization, demand for freshwater is increasing day by day. The unabated discharge of industrial effluent in the available water bodies is reducing the quality of these ample sources of water continuously. Hence, the national mission on water conservation was declared by the Honorable Prime Minister Narendra Modi as 'Jal Shakti Abhiyan' and appealed to all citizens to collectively address the problem of water shortage, by conserving every drop of water and suggesting conducting water audits for all sectors of water use. Water audit can be defined as a qualitative and quantitative analysis of water consumption to identify means of reducing, reusing, and recycling water. Water Audit is nothing but an effective measure for minimizing losses, optimizing various uses, and thus enabling considerable conservation of water in the irrigation sector, domestic, power, and industrial sectors. A water audit is a technique or method which makes it possible to identify ways of conserving water by determining any inefficiency in the system of water distribution. The measurement of water losses due to different uses in the system or any utility is essential to implement water conservation measures in such an establishment.

3.1 Importance of Water Audit

- Systematic process
- May yield some surprising results
- Easier to work on solutions when the problems are identified.
- Attracting mechanism can be put into place.
- It is observed that a number of factors like climate, culture, food habits, work and working conditions, level and type of development, and physiology determine the requirement of water. The community which has a population between 20,000 to 1, 00,000 requires 100 to 150 liters per person (capita) per day. The communities with a population over 1, 00,000 require 150 to 200 liters per person (capita) per day. As per the standards provided by WHO Regional Office for Southeast Asia Schools require 2 liters of water per student for drinking purposes; 10-15 liters per student for Water-flush toilets. Administration requires (Staff Accommodation not included) 50 liters per person per day.

3.2 Water Audit

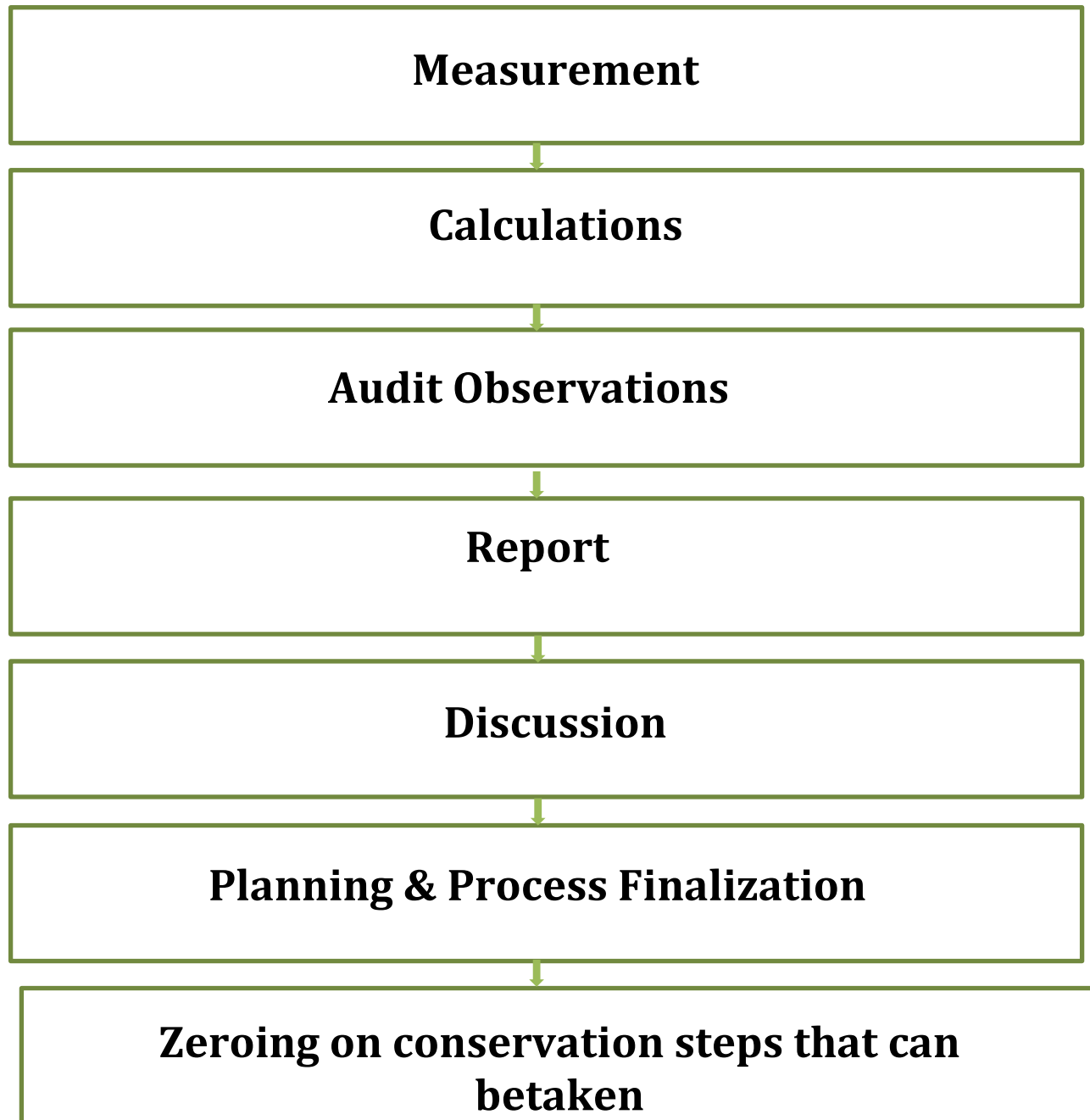
Water usage can be defined as water used for all activities which are carried out on campus from different water sources. This includes usage in all residential facilities, academic buildings, on-campus, and on-grounds. Wastewater is referred to as the water which is transported off the campus. The wastewater includes sewerage, residence water used in cooking, showering, clothes washing as well as wastewater from chemical and biological laboratories which ultimately go down in the sink or drainage system.

College water resources: -

The major resource for the water in the College is a self-reliant water boring system installed on the campus. There are Ground Water Tube wells installed on the campus with 7x1.5 HP which operate to fill the overhead tanks on the individual buildings. To fulfill the need for the supply of the campus, overhead storage tanks are available with capacity.

Sewage Water Treatment Plant: -

College has Installed Sewage water Treatment plant for recycling of water. Water from Toilets, kitchen & Other sources go to Sewage water treatment plant via drainage and after treating of water at STP the water then used for Gardening Purpose.



Water Audit Methodology & Steps

3.3 Sustainable Water Practices Watershed Management Practices

Underground water is the only one source of drinking water at CSIT. There are about seven borewells at the campus. The water is collected in the tanks located atop, and a regular water supply is maintained.

Observation:

1. Water in the College is not directly drained and is used again after treatment or recycling for watering the plants.
2. Sewage Treatment plant (STP) is installed in the back side of college for reducing the water wastage.
3. Water conservation/management practices have already been implemented in the college.
4. Rain water harvesting is also practiced by the college for recharging of the water table.



Rain Water Harvesting Spot



Rain Water Harvesting Points

Recommendations:

1. Filling of Water tank should be done on off peak time to save energy & Electricity Bill Cost.
2. Use of BEE star rated Energy Efficient Pumps in place of old & inefficient Pumps.

4. ENERGY AUDIT

Energy is one of the major inputs for the economic development of any country. The fundamental goal of energy management is to produce goods and provide services with the least cost and least environmental effect. Also, it can be said as “the strategy of adjusting and optimizing energy, using system and procedure so as to reduce energy requirements per unit of output while holding constant or reducing total costs producing the output from these systems”. The energy audit is key to a systematic approach for decision-making in the area of energy management. It attempts to balance the total energy inputs with its use and serves to identify all the energy streams in a facility.

4.1 Energy Audit

Energy resources utilized by all the departments, support services, and the administrative buildings of CSIT College, include Electricity, Solar Roof Top Systems, and Diesel Generators installed on the campus.

4.1.1 Energy Audit Objectives

Primary

- The first objective is to acquire and analyze data and find the necessary consumption pattern of these facilities.
- The second objective will be to calculate the wastage pattern based on the results of the first objective.
- The final objective is to find and implement solutions that are acceptable and feasible.

Secondary

- This would be our first exposure to this field hence experience gain would be vital.
- This project will precede many follow up projects and hence helps to gain technical and management exposure required for future energy projects.
- It is sure to help create a repertoire of vital contacts hence will develop interaction with alumni, faculty and students.

4.2 Sources of Energy

CSIT, Durg withdraws Energy from Followings:

- a. Electricity from CSPDCL
- b. High Speed Diesel (High Speed Diesel used as a fuel for Diesel Generator which is run whenever power supply is not available and for transportation)
- c. Solar Energy (Capacity 200 KW)

The Following are the Major consumers of electricity in the facility

- Lightning
- Air Conditioner
- Fans
- Computers
- Other Lab Equipment
- Bore well pumps

Indirect Benefits of Energy Audit

Every time the Energy Audit is carried out it rekindles the interest in Energy Conservation as an important function. Energy Auditors sharing their experience and knowledge with the Plant Personnel helps in fueling the innovative ideas for further action of reduction in Specific Power consumption (SPC). Any loose connections or heating of cables come to timely vision. For an external agency due to unbiased vision, a few points for Energy Conservation may be visible each time they perform the audit and this would help in achieving further saving. Inform any irregularities in Energy meter HT connections for rectification.

Executive Summary

S. No.	Proposed Energy Saving	Investment (In Rs. Lakhs)	Electricity Saving (kWh/ Annum)	Estimated Cost Saving (Rs. Lakhs)	Simple Payback Period (Yrs.)
A. Zero Investment					
1	Reduction of the existing Contract Demand	0.00	0.00	1.43	Immediate
	Sub- Total (A)	0.00	0.00	1.43	Immediate
B. Low Investment					
2	Replacement of Old CFL Bulbs(12W) with Led Bulbs(9W) in phase wise manner	0.15	1318	0.09	1.56
3	Replacement of Conventional Tubelights(40W) with Led Batten Lights(20W) in Phase wise manner	0.39	9360.0	0.66	0.6
	Sub- Total (B)	0.54	10677.60	0.76	0.71
C. High Investment					
4	Replacement of Normal Ceiling Fan of 60W with 30 W BLDC Fan	13.45	54983.60	3.90	3.45
5	Replacement of Non-Star AC with 1.5 Ton BEE 5 Star Rated AC	25.56	68160.00	4.84	5.28
	Sub- Total (C)	39.01	123143.60	8.74	4.46
	Grand- Total (A+B+C)	39.55	133821.20	10.93	3.62

Total Investment	39.55	In Rs. Lakhs
Total Monetary Saving	10.93	In Rs. Lakhs
Annual Energy Saving	1.34	Lakh kWh
Simple Payback Period	3.62	Yrs.
Total Thermal Saving	11.51	TOE
Reduction in CO2 Emission	105.72	Ton/ Annum

4.3 ELECTRICITY BILLING

Electricity Bill Summary for the year 2021-2023 are as follows.

Table 4.3. 1Energy Consumption details

For Consumer number: 1000312/374600-1000312

ELECTRICITY BILL									
Months	Consumption Unit (KWH)	Consumption Unit (KVAH)	Contract Demand (KVA)	Billing Demand (KVA)	Maximum Demand (KVA)	Power Factor	Energy Charge (Rs.)	Demand Charges (Rs.)	Total Payable Amount (Rs.)
Sep-21	23620	24978	270	220	111	0.95	167737	39330	254450
Oct-21	22512	24978	270	220	96.6	0.90	162437	39330	238432
Dec-21	14118	16120	270	220	50.52	0.88	105944	39330	174467
Mar-22	16305	17886	270	220	90.72	0.91	118385	39330	188700
Apr-22	27583	28611.00	270	220	123	0.96	194460	41800	285970
Jun-22	33096	33940.00	270	220	144.36	0.98	224481	82500	364820
Jul-22	24522	26067.00	270	220	89.4	0.94	176705	41800	264050
Aug-22	25278	26679.00	270	220	98.04	0.95	181099	41800	276538
Dec-22	27847	29693.00	270	220	52.44	0.94	133727	41800	242071
Jan-23	19444	21298.00	270	220	72.48	0.91	145138	41800	259090
Mar-23	19003	20695.00	270	220	72.6	0.92	140078	41800	244140
Apr-23	29886	30840.00	270	220	142.68	0.97	211156	82500	374310
Total/Avg.	283214	301785			95.32	0.93	1961347	573120	3167038
	23601.17	25148.75			144.36		163445.5833	47760	263919.8333

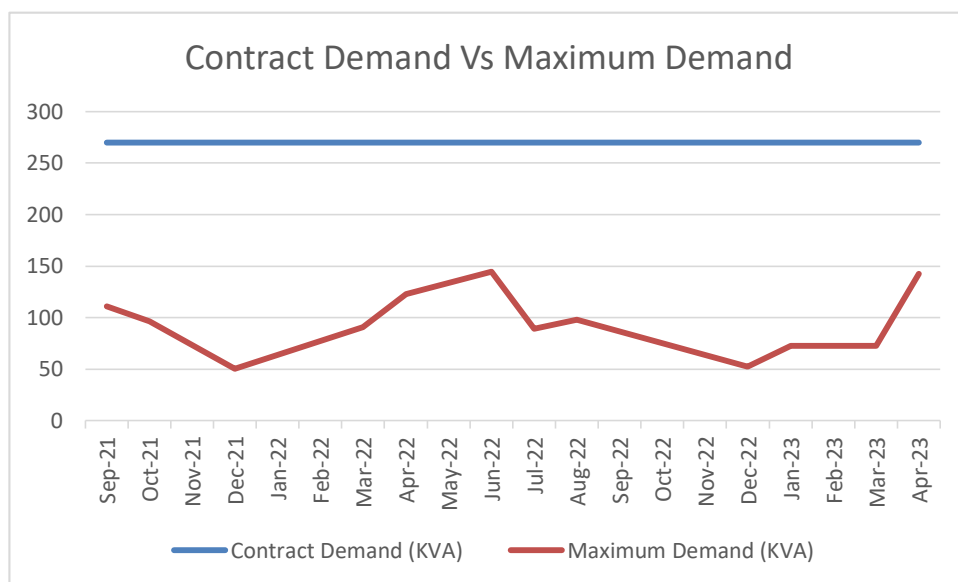
OBSERVATION:

The recorded average demand is of 95 KVA and Maximum demand is 144.36 KVA occurred at the month of June 2022. The Contract Demand is 270 KVA. It is recommended to reduce contract demand which will save cost on the total monthly bill value.

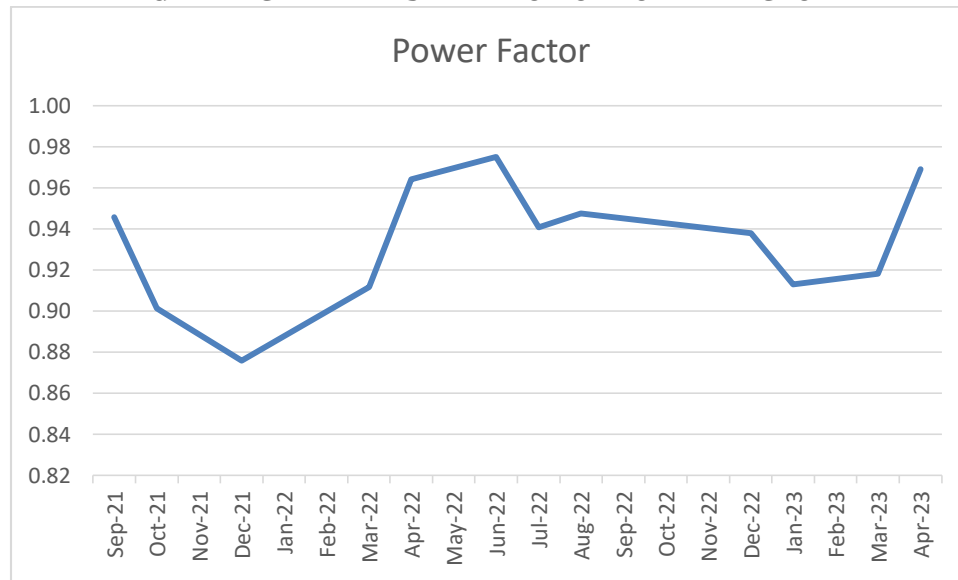
Table 4.3. 2 Maximum and Contract Demands Value

Particular	Value	UOM
Existing Contract demand	270	KVA
Existing minimum billing demand	220	KVA
Existing Normal Demand charges	190	Rs. /KVA
Fixed Normal demand charges	41800	Rs.
Maximum Demand Recorded	144	KVA
Proposed contract Demand	185	KVA
Proposed Minimum billing demand	157	KVA
Fixed Normal Demand Charges per month after contract Demand reduced	29877.5	Rs.
Annual Cost saving	1.4307	Rs. Lakhs
Investment	NIL	
Simple payback year	Immediate	

GRAPHICAL REPRESENTATION OF MAXIMUM DEMAND VS CONTRACT DEMAND



GRAPHICAL REPRESENTATION OF POWER FACTOR



OBSERVATION

The average Power factor recorded for the year of 2021-23 is 0.93 which is satisfactory.

4.4 LIGHTING SYSTEM

Lighting is an essential service in all industries, Universities, Hospitals, Malls, etc. Innovation and continuous improvement in the field of lighting, have given rise to tremendous energy-saving opportunities in this area. Lighting is an area, which provides a major scope to achieve energy efficiency at the design stage, by incorporation of modern energy-efficient lamps, luminaries, and gears, apart from good operational practices.

Basic Terms in Lighting System and Features

Lamps

Lamp is equipment, which produces light. The most commonly used lamps are described briefly as follows:

Incandescent lamps

Incandescent lamps produce light by means of a filament heated to incandescence by the flow of electric current through it. The principal parts of an incandescent lamp, also known as GLS (General Lighting Service) lamp include the filament, the bulb, the fill gas and the cap.

Reflector lamps

Reflector lamps are basically incandescent, provided with a high-quality internal mirror, which follows exactly the parabolic shape of the lamp. The reflector is resistant to corrosion, thus making the lamp maintenance free and output efficient.

Gas discharge lamps

The light from a gas discharge lamp is produced by the excitation of gas contained in either a tubular or elliptical outer bulb. The most commonly used discharge lamps are as follows:

Fluorescent Tube lamps (FTL)

Compact Fluorescent Lamps (CFL)

Mercury Vapor Lamps

Sodium Vapor Lamps

Metal Halide Lamps

4.4.1 ENERGY SAVING STRATEGY

Introduction of Light-emitting-diode or LED in place of Conventional Lighting Sources

A light transmitting diode (LED) is a semiconductor light source. LED's are utilized as marker lights as a part of numerous gadgets, and are progressively utilized for lighting. Early LED's emitted low power red light, however current adaptations are accessible over the unmistakable, ultraviolet and infrared wavelengths, with high splendor. The LED is focused around the semiconductor diode. At the point when a diode is forward predisposition (exchanged on), electrons can recombine with gaps inside the gadget, discharging vitality as photons. This impact is called electroluminescence and the color of the light (comparing to the vitality of the photon) is dictated by the vitality hole of the semiconductor. LED's present numerous focal points over radiant light sources including easier vitality utilization, longer lifetime, enhanced heartiness, more modest size, quicker exchanging, and more stupendous sturdiness and dependability. Notwithstanding, they are moderately unmanageable and require more exact current and hotness administration than customary light sources. Current LED items for general lighting are more unreasonable to purchase than fluorescent light wellsprings of equivalent yield.

Advantages of LED Lighting System -

- LEDs have an extremely long lifespan relative to every other lighting technology (including LPS and fluorescent lights but especially compared to incandescent lights). New LEDs can last 50,000 to 100,000 hours or more. The typical lifespan for an incandescent bulb, by comparison, is 1-5% as long at best (roughly 1,200 hours).
- LEDs are extremely energy efficient relative to every other commercially available lighting technology. There are several reasons for this including the fact that they waste very little energy in the form of infrared radiation (heat), and they emit light directionally (over 180 degrees versus 360 degrees which means there are far fewer losses from the need to redirect or reflect light).
- Very high light quality
- Very low maintenance costs and hassle

Automatic Control of Street Lights

Automatic control of street lights is designed to turn on and turn off street lights automatically. This project checks the amount of light. If light is 80 percent available, it automatically turns off street lights. But if amount of light is less than 80 percent, this project will automatically turn on street lights. one can also

adjust it according to its requirement. Light sensor is used to detect intensity of light. Microcontroller is used interfaced with light sensor to sense amount of light available. Control signal is generated with the help of microcontroller after analyzing amount of light. Control signal generated by microcontroller is used to turn on transistor which in turn energize the relay coil and relay turn on the street light.

Daylight utilization

Daylight is of extraordinary impact on energy savings. The measures made in this respect especially include investment costs and almost no future costs, compared with other measures (daylight is for free). Daylight can be controlled by use of venetian blinds, roller blinds, marquises, redirecting, and special glazing and so on, in existing buildings. The options how to increase daylight utilization are also installation of skylights into a ceiling structure, installation of light guides, regular window cleaning, painting the walls, change in a space geometry and a space utilization purpose.

Lighting system maintenance

The regular maintenance is very important for the efficiency of the lighting system. Therefore, the maintenance must be carried out in regular intervals. A correctly scheduled maintenance plan helps to maintain the illumination, reduce investment and operational costs and operate the system safely. The plan of maintenance is to include.

- Interval of maintenance execution.
- Description of activities performed within the framework of a regular maintenance.
- Description of activities performed within the framework of extraordinary maintenance (service action).
- Way of luminaries and surfaces cleaning.

Solar Lighting System

Solar street lighting system is ideal for street lighting in remote villages. The system is provided with battery storage backup sufficient to operate the light for 10-11 hours daily. The system is provided with automatic ON/OFF time switch for dusk to dawn operation and overcharge / deep discharge prevention cut-off with LED indicators.

The SPV modules are reported to have a service life of 15-20 years. Tubular Batteries provided with the solar street lighting system require lower maintenance; have longer life and give better performance.

4.5 INVENTORY DETAILS

The audit team has done the Inventory with Wattage analysis of the different types of lighting installed and the other electrical equipment across the campus. The below table shows the electrical equipment install at the College.

Table 4.5. 1 Inventory details of lights Building wise

Sl. No.	Electrical Equipment	Quantity	Wattage/HP	Total load (KW)
1	Tube light (Single)	195	40	7.8
2	Tube lights (LED)	294	20	5.88
3	LED Bulb	275	9	2.475
4	CFL Lights	183	12	2.196
5	LED Bulb Library	189	12	2.268

Observation:

1. Most of the lights in the College are LEDs.
2. Street lights were LED.
3. Overall lighting of College is good.

Recommendation:

1. General maintenance should be done regularly.
2. The street lights should be solar powered.
3. It is Recommend to Install **Astronomical Switch** with timer in all the street lights.

Note: It is stated that Astronomical Switch lightings can give up to 3-5 % savings on Lighting Power Consumption.

4.6 CEILING FANS

Ceiling Fan is the major part which consumes electricity and however, it is very useful in household, universities, offices, etc. Hence, Innovation and continuous improvement in the field of fans, have given rise to tremendous energy-saving opportunities in this area. The fan is an area, which provides a major scope to achieve energy efficiency at the design stage, by incorporation of modern energy-efficient Fans, BLDC Fans, smart Fans, apart from good operational practices.

Table 4.6. 1 Inventory details of Fans Building wise

S.No	Room NO.	Ceiling Fan of 65 W	Total Load (in KW)
1	179	2	0.13
2	177	6	0.39
3	182	6	0.39
4	183	6	0.39
5	184	6	0.39
6	223	5	0.325
7	320	4	0.26
8	178	2	0.13
9	173	2	0.13
10	181	1	0.065
11	169	6	0.39
12	170	6	0.39
13	171	6	0.39
14	319	2	0.13
15	167	4	0.26
16	114	4	0.26
17	115	4	0.26
18	116	9	0.585
19	117	7	0.455
20	118	4	0.26
21	119	5	0.325
22	120	3	0.195
23	122	26	1.69
24	129	14	0.91
25	131	5	0.325
26	209	4	0.26
27	Pylons Office	1	0.065
28	154	2	0.13
29	152	6	0.39
30	153	6	0.39
31	156	6	0.39
32	159	3	0.195

33	204	2	0.13
34	113	3	0.195
35	205	4	0.26
36	207	8	0.52
37	225	6	0.39
38	228	7	0.455
39	214	1	0.065
40	224	4	0.26
41	203	2	0.13
42	229-230	8	0.52
43	155	8	0.52
44	Sambhaji HALL	8	0.52
45	SMH	14	0.91
46	S-2	8	0.52
47	226	2	0.13
48	227	1	0.065
49	222	8	0.52
50	VIVEKANAND HALL	21	1.365
51	301	1	0.065
52	302	6	0.39
53	305	2	0.13
54	308	2	0.13
55	311	4	0.26
56	312	2	0.13
57	313	4	0.26
58	314	4	0.26
59	316	4	0.26
60	317	4	0.26
61	161	2	0.13
62	Digital Valuation Cell	2	0.13
63	226	2	0.13
64	227	2	0.13
65	232	4	0.26
66	215	1	0.065
67	172	6	0.39
68	237	1	0.065
69	180	6	0.39
70	223	6	0.39
71	306	6	0.39
72	307	6	0.39
73	310	6	0.39
74	315	6	0.39
75	325	4	0.26

76	327	1	0.065
77	329	1	0.065
78	330	1	0.065
79	332	1	0.065
80	334	7	0.455
81	335	6	0.39
82	337	7	0.455
83	338	7	0.455
84	339	1	0.065
85	321	6	0.39
86	322	6	0.39
87	323	6	0.39
88	324	6	0.39
89	328	6	0.39
90	331	6	0.39
91	333	6	0.39
92	336	6	0.39
93	Guard Room	2	0.13
94	Mandir Room	1	0.065
95	130	5	0.325
96	133	2	0.13
97	134	1	0.065
98	135	1	0.065
99	136	1	0.065
100	137	1	0.065
101	141	1	0.065
102	142	1	0.065
103	Café	1	0.065
104	217	11	0.715
105	145	1	0.065
106	146	2	0.13
107	Lounge	4	0.26
108	148	4	0.26
109	149	1	0.065
110	150	1	0.065
111	Principal office	4	0.26
112	T&P Cell	2	0.13
113	T&P Cell	2	0.13
114	Admission Cell	4	0.26
115	168	1	0.065
116	188	1	0.065
117	190	12	0.78
118	216	1	0.065

119	Library	9	0.585
	Total	538	34.97

Observation:

1. There are 538 number of 65-Watt fan which consumes 34.97 kW of electricity.
2. It is observed that fans installed at College are old & conventional which needs to be replace by BLDC (Brushless Direct Current) Fans.

Recommendation:

1. It is recommended to replace all the 65 W ceiling fans with energy efficient BLDC fan, in phase wise manner.

Table 4.6. 2 Replacement of Normal Ceiling Fan of 65 W with 30 W BLDC Fan

Replacement of Normal Ceiling Fan of 65 W with 30 W BLDC Fan		
Particulars		Units
Total Number of 65-Watt Ceiling Fan	538	Number
Measured Watt	65	Watts
Total Watts	34970	Watts
Proposed watt after replacement	30	Watts
Total Watt After Replacement	16140	Watts
Operating Hours in a day	8	Hours
Estimated Energy Saving after Replacement Annual kWh	54983.6	kWh
Per Unit Cost as Per CSPDCL Bill	7.1	Rs
Estimated Cost Saving Per Year	3.90384	Rs. Lacs
Cost of Per Fan	2500	Rs
Total Investment Cost	13.45	Rs. Lacs
Payback	3.45	Year

BLDC Fan

A BLDC Fan is a DC motor a magnet synchronous electric motor that is driven by electrical energy (DC) electricity and it accomplishes an electronically controlled commutation system (commutation is the process of manufacturing motility force in the motor by dynamical part currents through it at appropriate times) rather than an automatically commutation system. BLDC motors also are referred to as tetragon magnet motors that are positioned around the mechanical device. The electronics contain a driving algorithm that drives the BLDC motor could be an electric motor that keeps running on the provision of electric power. In this motor, there's the straight relationship amongst current and torsion moreover within the between

voltage and rate of BLDC motor. This straight relationship is that the main reason that the BLDC motor offers wonderful leads to the routine ceiling fans. BLDC Fans consume low power.

BLDC fan takes in AC voltage and internally converts it into DC victimization SMPS. DC voltage as input that is reborn victimization the heartbeat breadth Modulation Techniques to regulate the excitation of the coils to come up with the motion in a prescribed fashion. we've projected AN electronic management for the PWM generation and motion detection and management through the "hall effect" sensors that square measure embedded within the BLDC Motor. there's additionally some way to use the "back emf" generated by the excitation on the "third leg" of the motor for conniving the relative position for the motion management. This sensor less or back electromotive force methodology, whereas it reduces the value of the motor, has bound challenges and disadvantages and isn't being thought of here for the actual application.

Table 4.6. 3 Comparison of BLDC Fans & Ceiling Fan

Speed of Fans (RPM)	BLDC Fans (watts)	Ordinary Fans (Watts)
140	3.8	13
210	7.7	24
270	13.8	30
310	22.7	40
370	35.8	55
410	40	80

4.7 Air Conditioner Details

Table 4.7. 1 Details Air Conditioner

S. No.	Room No.	Air Conditioner of 1.5 Ton (1800 W)	Air Conditioner of 4 Ton (6500 W)	Total Load (in KW)
1	179	1		1.8
2	319	2		3.6
3	167	3		5.4
4	158	1		1.8
5	159	1		1.8
6	214	1		1.8
7	155	4		7.2
8	SambhaJI HALL	2		3.6
9	S-2	7	2	25.6
10	226	4		7.2
11	227	4		7.2
12	222	3		5.4
13	VIVEKANAND HALL	9	2	29.2
14	301	1		1.8
15	314	6		10.8
16	160	4		7.2
17	232	2		3.6
18	327	1		1.8
19	329	1		1.8
20	145	1		1.8
21	146	1		1.8
22	147	1		1.8
23	148	1		1.8
24	150	1		1.8
25	151	1		1.8
26	157	1		1.8
27	Board room	2		3.6
28	163	1		1.8
29	164	1		1.8
30	166	2		3.6
31	176	1		1.8
32	Library		5	32.5
	Total	71	9	186.3

Table 4.7. 2 Ac's Energy Efficiency Ratio on Sample Basis

1	Type of Air-conditioner	UOM	Window	Split	Tower Ac
2	Capacity	TR	1.5	1.5	4.0
4	Set Temperature	°C	24.0	21.0	24.0
5	Inlet Air Dry Bulb Temp. (DBT)	°C	21.4	23	23.19
6	Inlet Air Wet Bulb Temp. (WBT)	°C	20.3	17.4	19.5
7	Inlet Air Relative Humidity (RH)	%	78.0	73.0	78.3
8	Outlet Air DBT	°C	19.0	18.4	11.4
9	Outlet Air WBT	°C	15.0	13.0	8.3
10	Outlet Air RH	%	82.0	78.4	76.3
11	Temp. differential of Inlet DBT & Outlet DBT	°C	2.4	4.6	11.8
12	Avg. Air Velocity	m/sec.	6.0	2.8	2.5
13	Air discharge area	m ²	0.040	0.040	0.080
14	Avg. Air flow	m ³ /h	864	403	720
15	Density of inlet Air	kg/m ³	1.199	1.193	1.192
16	Enthalpy at Inlet Air	kCal/kg of dry air	12.36	11.69	12.067
17	Enthalpy at Outlet Air	kCal/kg of dry air	9.47	9.696	9.565
18	Net Refrigeration Effect	TR	0.99	0.32	0.71
		kW	3.48	1.11	2.50
19	Percent Loading of Air-conditioner	%	66.00	21.14	17.75
20	Power Input to Air-conditioner	kW	2.00	1.80	2.53
21	Compressor Motor Efficiency	%	80.0	80.0	80.0
22	Drive Transmission Efficiency	%	100.0	100.0	100.0
23	Shaft Power of Compressor	kW	1.60	1.44	2.02
24	Specific Power Consumption (SPC)	kW/TR	2.02	5.68	3.56
25	Energy Efficiency Ratio (EER)	W/W	2.18	0.77	1.24

Observations:

1. ACs are old and conventional type.

Recommendations:

1. It is recommended to replace the conventional ACs with five-star inverter AC, in phase wise manner.

4.8 TRANSFORMER LOAD PROFILE

The Table below shows the Load Profile of Transformer.

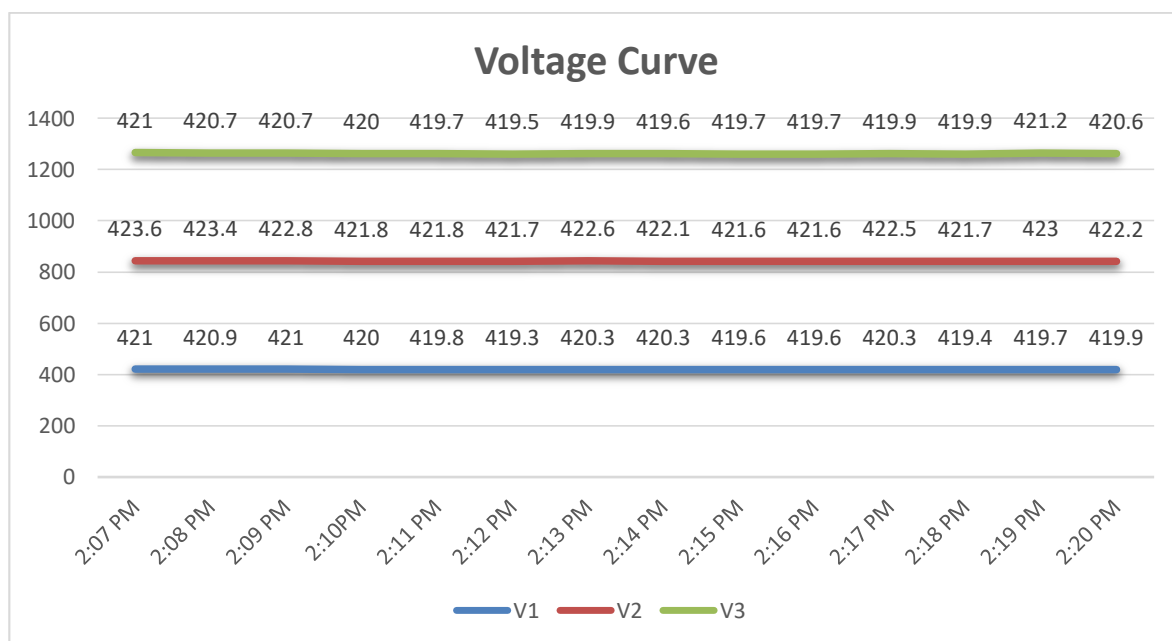
Table 4.8. 1 Load profile of Transformer

CSIT Transformer									
SN.	Name of the drive	Drive Status	Design	Measured Parameters			Calculated Parameters		
	Equipment Name		Transformer Rating KVA	Avg. Voltage (V)	Avg. Current (A)	PF	Calculated kVA	Calculated Values (KW)	Loading (%)
1	Transformer Near Boys Hostel	LT	250.00	419.84	95.33	0.91	69.32	63.08	27.73

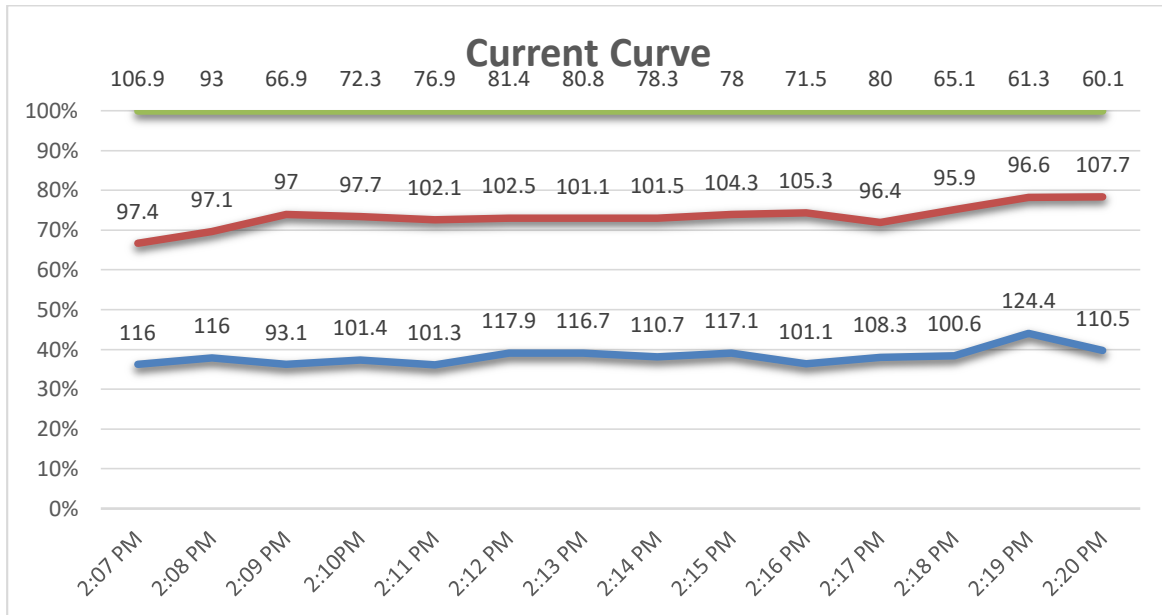
Observations:

1. Above table is based on recording of loading for Six hours.
2. Loading percentage of all transformers were very less because of non-availability of students because of semester break.

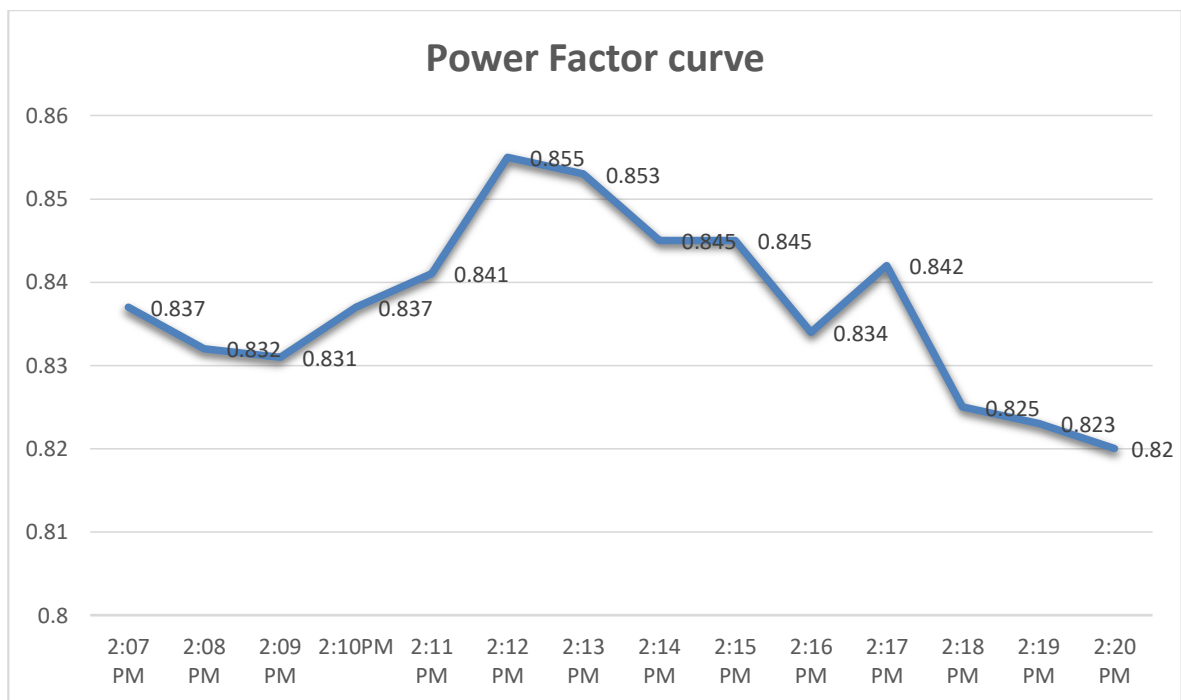
Graphical Representation of Voltage

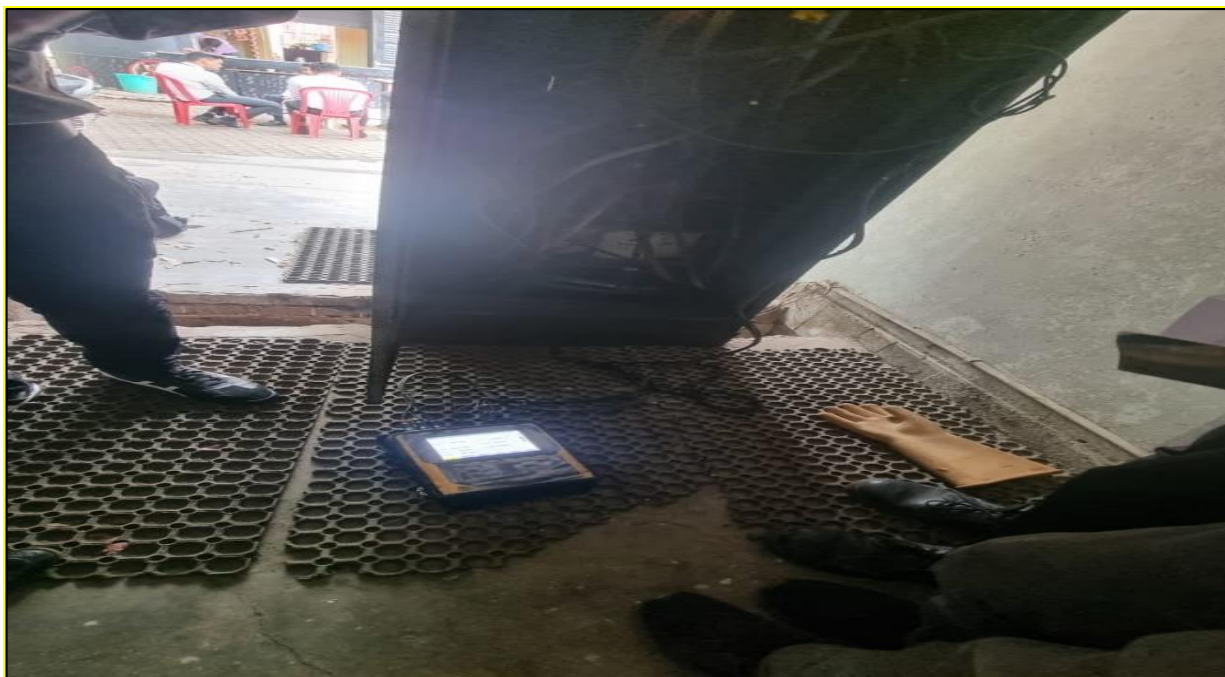


Graphical Representation of Current



Graphical Representation of Power factor





Recording of Electrical Parameters of Transformer



Cables Connection at the Transformer

4.9 POWER QUALITY

Power Quality & Harmonics

Equipment based on frequency conversion techniques generates harmonics. With the increased use of such equipment, harmonics- related problems have been enhanced. The harmonic currents generated by different types of loads travel back to the source. While traveling back to the source, they generate harmonic voltages, following simple Ohm's Law. Harmonic voltages, which appear on the system bus, are harmful to other equipment connected to the same bus. In general, sensitive electronic equipment connected to this bus will be affected. The Harmonics Level on the HT side of the Transformers was measured, details of which is as under: -

Particulars	TR
Voltage Harmonics (VTHD)	
"R"Phase	1.61
"Y"Phase	1.26
"B"Phase	1.17
Current Harmonics (ATHD)	
"R"Phase	13.38
"Y"Phase	12.62
"B"Phase	14

- Maximum Individual Frequency Voltage Harmonic: 3%
- Total Harmonic Distortion of the Voltage: 5%

harmonic current limitations

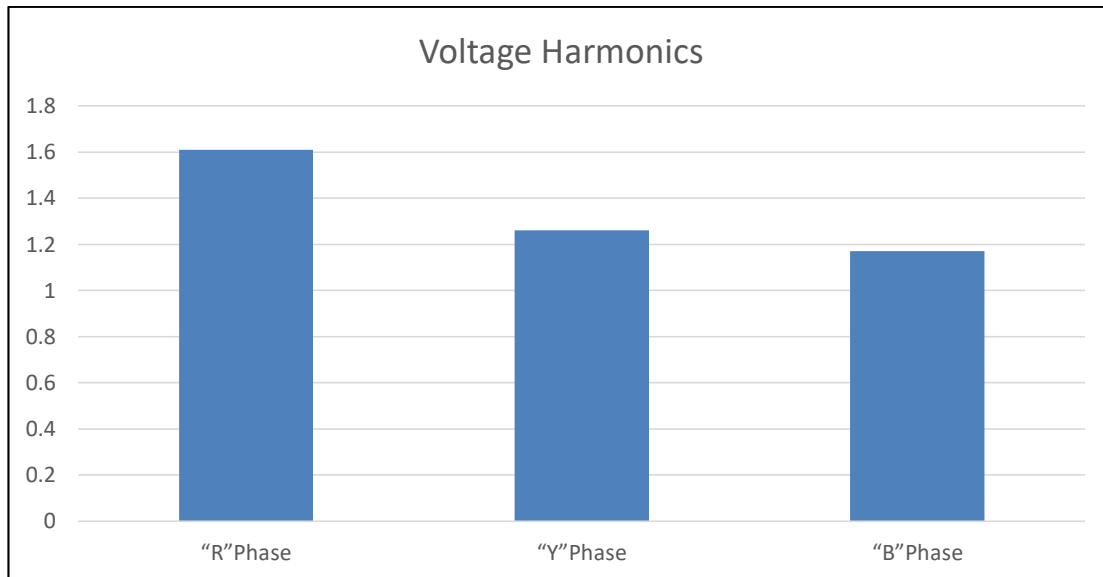
Maximum Harmonic Current Distortion in Percent of IL 120 Volt through 69 KV						
Individual Harmonic Order (Odd Harmonics)						
ISC/IL	h<11	11<h<17	17<h<23	23<h<35	35<h	TDD
<20*	4.0	2.0	1.5	0.6	0.3	5.0
20<50	7.0	3.5	2.5	1.0	0.5	8.0
50<100	10.0	4.5	4.0	1.5	0.7	12.0
100<1000	12.0	5.5	5.0	2.0	1.0	15.0
>1000	15.0	7.0	6.0	2.5	1.4	20.0

Even harmonics are limited to 25% of the odd harmonic limits
TDD refers to Total Demand Distortion based on the average demand current at the fundamental frequency and measured at the PCC (Point of Common Coupling).

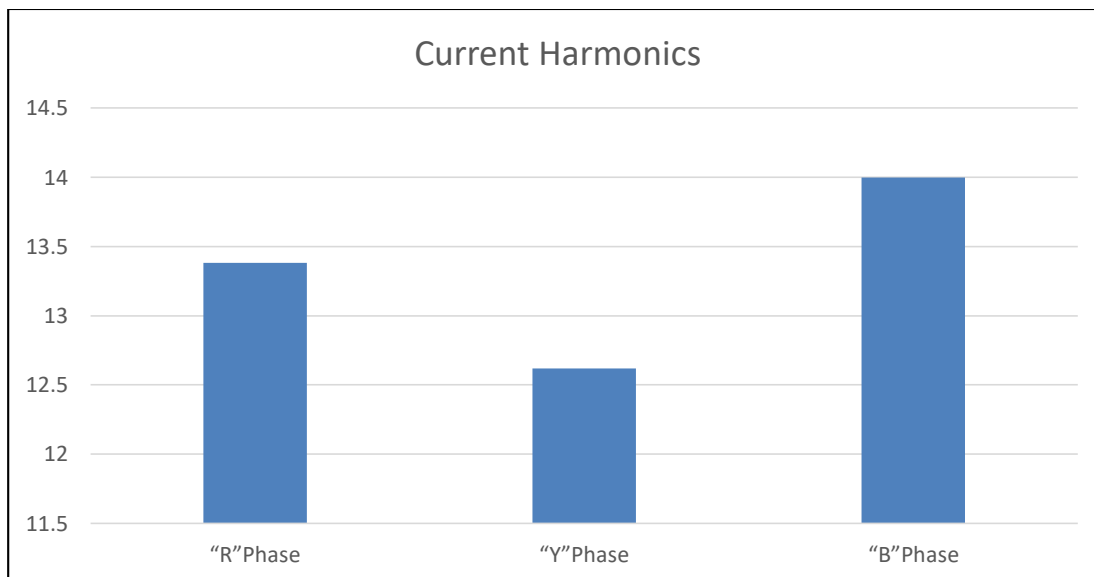
*All power generation equipment is limited to these values of current distortion regardless of ISC/ IL value.

ISC = Maximum short-circuit current at PCC.
IL = Maximum demand load current (fundamental) at the PCC.
h = Harmonic number.

Graphical Representation of Voltage Harmonics (V THD) of Main Incomer



Graphical Representation of Voltage Harmonics (V THD) of Main Incomer



Observations & Suggestions

As detailed above, the voltage harmonics levels were around 1.11-1.61% and the levels of the current harmonics were 12-14%. The Overall harmonics are within limits. If the Harmonics level is on the higher side, then appropriate harmonic filters may have to be installed in the system. Different technologies are available mitigating the harmonics from the system. These include Detuned or broadband harmonic filters: these filter banks are tuned to a frequency just

below predominant harmonic frequency is say, 5th, it is normal practice to tune the filters to 189 Hz, or 3.78th harmonic, in 50 Hz systems.

Active Harmonic Filters: these units are designed in such a manner that, they will inject harmonic frequencies in the system, which will be in anti-phase of the load harmonic frequencies. This will effectively free the source being loaded due to harmonics.

Major Causes of Harmonics

Devices that draw non-sinusoidal currents when a sinusoidal voltage is applied create harmonics. Frequently these are devices that convert AC to DC. Some of these devices are listed below:

- Electronic Switching Power Converters
- Computers, Uninterruptible power supplies (UPS), Solid-state rectifiers
- Electronic process control equipment, PLC's, etc.
- Electronic lighting ballasts, including light dimmer
- Reduced voltage motor controllers
- Arcing Devices
- Discharge lighting, e.g. Fluorescent, Sodium and Mercury vapor
- Transformers operating near saturation level
- Magnetic ballasts (Saturated Iron core)
- Induction heating equipment, Chokes, Motors, Appliances
- TV sets, air conditioners, washing machines, microwave ovens
- Fax machines, photocopiers, printers
- These devices use power electronics like SCRs, diodes, and thyristors, which are a growing percentage of the load in industrial power systems.

Many problems can arise from harmonic currents in a power system. Some problems are easy to detect; others exist and persist because harmonics are not suspected. Higher RMS current and voltage in the system are caused by harmonic currents, which can result in any of the problems listed below:

Blinking of Incandescent Lights	Transformer Saturation
Capacitor Failure	Harmonic Resonance
Circuit Breakers Tripping	Inductive Heating and Overload
Conductor Failure	Inductive Heating
Electronic Equipment Shutting down	Voltage Distortion
Flickering of Fluorescent Lights	Transformer Saturation
Fuses Blowing for No Apparent Reason	Inductive Heating and Overload
Motor Failures (overheating)	Voltage Drop
Electromagnetic Load Failures	Inductive Heating
Overheating of Metal Enclosures	Inductive Heating
Power Interference on Voice Communication	Harmonic Noise
Transformer Failures	Inductive Heating

4.10 SOLAR POWER IN COLLEGE

There were two Solar System Installed of Capacity 100 KW each in CSIT at –

- 1 Engineering Block
- 2 Administrative Building

It is worth noting that during the duration of audit, one of the solar stations was not in use as there were issues reported, i.e., of battery failure and some damaged panels.



Solar System Installed at Institute Building



Solar System and Installed Devices at the Institute

Observation: -

1. It is Observed that there are total 2 units having capacity 100 KWP of Solar power installed at the college out of which only 1 unit is in operating condition and other unit is in non-operation damage condition.

Recommendations: -

1. It is recommended to repair or replace the old unit and start the operation of 100 KWp solar panel for generation of Renewable Energy.

4.11 GENERAL TIPS FOR ENERGY CONSERVATION IN DIFFERENT UTILITIES SYSTEMS

4.11.1 ELECTRICITY

- Schedule your operations to maintain a high load factor
- Minimize maximum demand by tripping loads through a demand controller
- Use standby electric generation equipment for on-peak high load periods.
- Correct power factor to at least 0.99 under rated load conditions.
- Set transformer taps to optimum settings.
- Shut off unnecessary computers, printers, and copiers at night.

4.11.2 MOTORS

- Properly size to the load for optimum efficiency.
- (High efficiency motors offer of 4 - 5% higher efficiency than standard motors)
- Check alignment.
- Provide proper ventilation, (For every 10°C increase in motor operating temperature over recommended peak, the motor life is estimated to be halved)
- Check for under-voltage and over-voltage conditions.
- Balance the three-phase power supply.
- (An Imbalanced voltage can reduce 3 - 5% in motor input power)
- Demand efficiency restoration after motor rewinding.

4.11.3 FANS

- Use smooth, well-rounded air inlet cones for fan air intakes.
- Avoid poor flow distribution at the fan inlet.
- Minimize fan inlet and outlet obstructions.
- Clean screens, filters, and fan blades regularly
- Use aero foil-shaped fan blades.
- Minimize fan speed.

- Use low-slip or flat belts.
- Check belt tension regularly.
- Eliminate variable pitch pulleys.
- Use variable speed drives for large variable fan loads.
- Use energy-efficient motors for continuous or near-continuous operation
- Eliminate leaks in ductwork.
- Minimize bends in ductwork
- Turn fans off when not needed

4.11.4 PUMPS

- Operate pumping near best efficiency point.
- Modify pumping to minimize throttling.
- Adapt to wide load variation with variable speed drives or sequenced control of smaller units.
- Stop running both pumps -- add an auto-start for an on-line spare or add a booster pump in the problem area.
- Use booster pumps for small loads requiring higher pressures.
- Increase fluid temperature differentials to reduce pumping rates.
- Repair seals and packing to minimize water waste.
- Balance the system to minimize flows and reduce pump power requirements.
- Use siphon effect to advantage: don't waste pumping head with a free-fall (gravity) return.

4.11.5 LIGHTING

- Reduce excessive illumination levels to standard levels using switching, delamping, etc. (Know the electrical effects before doing delamping.)
- Aggressively control lighting with clock timers, delay timers, photocells, and/or occupancy sensors.
- Install efficient alternatives to incandescent lighting, mercury vapor lighting, etc.
- Efficiency (lumens/watt) of various technologies range from best to worst approximately

as follows: low pressure sodium, high pressure sodium, metal halide, fluorescent, mercury vapor, incandescent.

- Select ballasts and lamps carefully with high power factor and long-term efficiency in mind.
- Upgrade obsolete fluorescent systems to Compact fluorescents and electronic ballasts
- Consider lowering the fixtures to enable using less of them.
- Consider day lighting, skylights, etc.
- Consider painting the walls a lighter color and using less lighting fixtures or lower wattages.
- Use task lighting and reduce background illumination.
- Re-evaluate exterior lighting strategy, type, and control. Control it aggressively.
- Change exit signs from incandescent to LED.

4.11.6 DG SETS

- Optimize loading
- Use waste heat to generate steam/hot water/power an absorption chiller or preheat process or utility feeds.
- Use jacket and head cooling water for process needs
- Clean air filters regularly
- Insulate exhaust pipes to reduce DG set room temperatures.

4.11.7 WATER & WASTE WATER

- Recycle water, particularly for uses with less-critical quality requirements.
- Recycle water, especially if sewer costs are based on water consumption.
- Balance closed systems to minimize flow and reduce pump power requirements.
- Eliminate once-through cooling with water.
- Use the least expensive type of water that will satisfy the requirement.
- Test for underground water leaks. (It's easy to do over a holiday shutdown.)
- Check water overflow pipes for proper operating level.

- Automate blow down to minimize it.
- Provide proper tools for wash down- especially self-closing nozzles.
- Reduce flows at water sampling stations.
- Eliminate continuous overflow at water tanks.
- Promptly repair leaking toilets and faucets.
- Use water restrictors on faucets, showers, etc.
- Use self-closing type faucets in restrooms.

4.11.8 ENERGY MANAGEMENT STRATEGY

Energy Management should be seen as a continuous process. Strategies should be reviewed annually and revised as necessary. The key activities suggested have been outlined below:

Set Up an Energy Monitoring and Reporting System

Successful energy management requires the establishment of a system to collect/ analyze and report the energy costs and consumption pattern. This will enable an overview of energy use and its related costs, as well as facilitating the identification of savings that might otherwise not be detected. The system needs to record both historical and ongoing energy use, as well as cost information from billing data, and capable of producing summary reports on a regular basis. This information will provide the means by which trends can be analyzed and reviewed for corrective measures.

Implement A Staff Awareness and Training Program

A key ingredient to the success of an energy management program is maintaining a high level of awareness among staff. This can be achieved in a number of ways, including formal training, newsletters, posters and publications. It is important to communicate program plans and case studies that demonstrate savings, and to report results at least at 12-month intervals. Staff may need training from specialists on energy saving practices and equipment.

5. SOLID WASTE AUDIT

Solid waste is the unwanted or useless solid material generated from human activities in a residential, industrial, or commercial area. Solid waste management reduces or eliminates the adverse impact on the environment and human health. A number of processes are involved inefficiently managing waste for an organization. It is necessary to manage the solid waste properly to reduce the load on the waste management system. Solid waste generation and its management is a burning issue in current days. The rate of generation of solid waste is very high and yet we do not have adequate technology to manage the generated waste. Unscientific handling of solid waste can create threats to public health and environmental safety issues. Thus, it is necessary to manage solid waste properly to reduce the load on the waste management system. The purpose of this audit is to find out the quantity, volume, type, and current management practice of solid waste generation in the CSIT College campus. This report will help for further solid waste management and to go for green campus development.

5.1 GENERATION OF SOLID WASTE IN CSIT COLLEGE

The Institute has a clear policy for waste management. It encourages reuse and recycling of the waste, unless impossible. There are different types of waste are recorded such as paper waste, plastic waste, biodegradable waste etc. However biodegradable waste is recycled through the vermicomposting process previously, but now this process is not functioning. The wastes generated in the campus include

- (i) kitchen wastes,
- (ii) liquid waste (residential and eateries),
- (iii) sewage and sludge,
- (iv) plastic wastes,
- (v) cans and bottles,
- (vi) papers including packaging materials
- (vii) garden leaves and

The College is committed to ensuring that all forms of wastes generated are handled based on the RRRR (Reduce, Reuse, Recycle, Recover) principles following appropriate source segregation protocols including safe disposal of chemical and hazardous wastes. A preliminary survey reveals the domination of biodegradable components (volume basis) over the non-biodegradable counterparts on the campus. The students' hostels share the highest amount of solid waste mostly dominated by food/kitchen wastes (a substantial amount of papers, plastics, metals are also seen

with waste also generated in hostels) followed by residential areas, academic buildings, construction sites (occasionally), open areas including gardens and roads.

5.2 WASTE MANAGEMENT

Biodegradable Waste Management

Handling of the solid waste:

College has taken initiative for Biodegradable Waste Management to compost using processes like Dry & Wet Waste Management.

The mess wardens and the servers are strict towards the waste of food. The students are encouraged to take food as per their appetite. Though in a small quantity the vegetable peels and leftover food from the mess is collected in drums and is taken to nearby domestic cattle shelter for feeding. This practice is done twice i.e. after the lunch and dinner at the hostel.

Vermicomposting Unit

Vermicomposting technology relies upon the conjoint action of earthworms and microorganisms to rapidly transform varied types of solid wastes. Considering the simplicity and flexibility of the technology, a vermicomposting unit was established in the College. The prime objectives are to recycle biodegradable waste fractions in a sustainable manner and curtail the cost of purchasing organic manure from the market for landscaping ventures. The compost is usually taken by nearby farmers for their fields.

Liquid waste:

The major amount of liquid waste is sewerage water from academic block and hostels. The institute has installed an STP for the treatment of waste water. The treated waste water is used for maintain the greenery at the campus using a separate supply line. The water treated water has contributed in growing a variety of plants. The lush green area that occupies approximately 40 % of the total area remains green throughout the year due to the proper supply of water from the STP. The dewatered sludge and compost is given to farmers for farming.

Initiatives taken by the College for Waste Management

- The stationary waste generated at CSIT is sent for recycling and reuse.
- Hazardous waste generated in a solid and liquid state during experiments in the laboratory is disposed of properly.
- Biodegradable waste is a major solid waste generated on campus which is further treated by vermicompost technology.

Recommendations

- Provision of installation of garbage unit should be introduced where the multilevel segregation of various wastes such as paper, construction, glass, metal scrap, and food waste should be done. Further various waste recycling plans for different types of waste should be introduced.
 - Recycling facilities should be introduced and should be supported by City Municipality and private suppliers, including glass, cans, white, colored, and brown paper, plastic bottles, batteries, print cartridges, cardboard, and furniture.



Dust bins

6. ENVIRONMENT QUALITY AUDIT

This includes the plants, greenery, and sustainability of the campus to ensure that the buildings conform to green standards. This also helps in ensuring that the Environmental Policy is enacted, enforced, and reviewed using various environmental awareness programs.

6.1 ENVIRONMENT QUALITY AUDIT

The CSIT campus is well known for its lush green vicinity and beauty. The campus has a green cover of about 40%. It comprises of herbs, shrubs and various trees. The fauna includes herbs like Tulsi, Haldi, croton, aloe vera, lemon grass etc.; shrubs like- amla, bottle-brush, eliche, lemon and trees like guava, mango, neem, bel etc. Activities organized to create greenery and its conservation at the College campus is as follows-

- Plantation of diversified species, Uses of medicinal plants, Identification of plants species
- Waste management plan and disposal facility

Various departmental activities are being carried out every year such as: -

- Plantations and other Landscaping Activities
- Maintenance of Gardens and Landscape
- Maintenance of Plantations

There are three types of vegetation in the College campus, which are-

- i) Big Trees.
- ii) Herbs
- iii) Shrubs

6.1.1 PLANTATIONS

Towards the sustainable land use practice, approximately 5446 plant saplings of different species have been planted in various sites through routine and special plantation drives. This program helps in encouraging an eco-friendly environment that provides pure oxygen within the institute and awareness among villagers. The plantation program includes various types of indigenous species of ornamental and medicinal wild plant species.

Herbs

S. No	Common Name	Botanical Name	Number
1	Bay Leaves	Laurus nobilis	1
2	Brilliantly hot	Capsicum annuum	1
3	Dalchini	Cinnamomum zeylanicum	1
4	Cloves	Syzygium aromaticum	1
5	Cardamom	Elettaria cardamomum	4
6	Aloe vera	Aloe vera	10
7	Kali Mirch	Piper nigrum	20
8	Kalanchoe pinnata	Bryophyllum pinnatum	50
9	Chirayita	Swertia	4
10	Caraway	Trachyspermum ammi	3
11	Wandering Jew	Commelina benghalensis	40
12	Tulsi	Ocimum tenuiflorum	20
13	Lemon grass	Cymbopogon citratus	1

Shrubs

S. No	Common Name	Botanical Name	Number
1	Century plant	Agave americana	100
2	Taupata	Coprosma	50
3	Copper leaf	Acalypha wilkesiana	500
4	Ming aralia	Polyscias fruticosa	10
5	Sissoo spinach	Alternanthera sessilis	20
6	Garden privet	Ligustrum ovalifolium	30
7	Banana tree	Musa paradisiaca Linn	2
8	Spider plant	Chlorophytum comosum	400
9	Mountain Blechnum	Blechnum Orientale	5
10	Mosaic Plant	Peppy flora	20
11	Bodhi tree	Ficus religiosa	15
12	Dudhmogre	Delo nix regia	3
13	Jasmine	Jasminum sambac	100
14	Ixora	Ixora coccinea	40
15	Pedilanthus	Euphorbia	200
16	Corn Palm	Dracaena	250
17	Golden Durante	Durante repens	1000
18	Bougainvillea	Nyctaginaceae	80
19	Bird-of-paradise	Strelitziaceae	5
20	Adenium	Adenium obesum	10
21	Lily	Lilium	5
22	Spurge	Euphorbia	100
23	Creeping zinnia	Zinnia	400
24	Sunflowers	Helianthus	100

25	Blanket flowers	Gaillardia	80
26	Love Leena	Beaucarnea recurvata	17
27	Mini tagar	Tabernaemontana coronaria nana	80
28	Cord moss	Funaria	2
29	Fig trees	Ficus	50
30	Chinese spinach	Amaranthus tricolor	200
31	Fenishpama	Mimosa pudica	7
32	Vridhadaru	Argyreia speciosa	8
33	Garlic Vine	Mansoa alliacea	4
34	Lobster-claws	Heliconias	5
35	Button flower	Centaurea cyanus	50
36	Verbena	Verbena officinalis	50
37	Rhoeo	Tradescantia spathacea	150
38	Ferns	Polypodiopsida	50
39	King sago	Cycas	5
40	Song of India	Dracaena reflexa	100
41	Mardhravita	Mandragora officinarum	2
42	Bamboo	Bambusa vulgaris	100
43	Puding	Croton	20
44	Sensitive plant	Mimosa pudica	30
45	Angelica	Angelica archangelica	5
46	Norwegian angelica	Angelica archangelica	10
47	Black lily	Dracunculus vulgaris	50
48	Areca palm	Dypsis lutescens	50
49	Ribbon grass	Phalaris arundinacea	50
50	February daphne	Daphne mezereum	10
51	Spider lily	Hymenocallis littoralis	20
52	Rose	Rosa	40
53	Miss lily	Lilium hybrid	30
54	Harsingar	Nyctanthes arbor-tristis	2
55	Rhapis palm	Rhapis excelsa	20

Trees

S. No	Common Name	Botanical Name	Number
1	Stone apple	Aegle marmelos	2
2	Bur flower	Neolamarckia cadamba	1
3	Banyan	Ficus benghalensis	2
4	Ficus religiosa	Moraceae	2
5	Rudraksha	Elaeocarpus angustifolius	1
6	Nerium Oleander	Cascabela thevetia	4
7	Gooseberry	Phyllanthus emblica	2
8	Mango	Mangifera indica	4

9	Golden shower	Cassia fistula	10
10	Water palm	Araliaceae	40
11	Palmyra Palm	Borassus flabellifer	50
12	Spanish cherry	Mimusops elengi	12
13	Almond	Prunus dulcis	70
14	Foxtail millet	Setaria italic	2
15	Indian cork tree	Millingtonia hortensis	40
16	Crepe myrtle	Lagerstroemia indica	12
17	Mehandi	Lawsonia inermis	1
18	Royal Poinciana	Delo nix regia	12
19	Christmas Tree	Araucaria columnaris	3
20	Chinaberry	Melia azedarach	2
21	Coconut	Cocos nucifera	3
22	Blackboard tree	Alstonia scholaris	25
23	Champak	Magnolia Champaca	15
24	Ashoka	Saraca asoca	3
25	Yellow Flame	Peltophorum	100
26	Nimba tree	Azadirachta indica	2
27	Sagwan	Tectona grandis	20
28	Nilgiri	Myrtaceous	40
29	Arjun	Terminalia arjuna	2
30	Jequirity bean	Abrus precatorius	15
31	Gum Arabic tree	Acacia senegal	2
32	Black plum	Syzygium cumini	1
33	Mulberries	Morus	2
34	Sandalwood	Santalum album	1
35	Leamon	Rutaceae	2
36	Black Currant	Carissa carandas	2
37	Jackfruit	Artocarpus heterophyllus	25
38	Madras thorn	Pithecellobium dulce	5
39	Rose mallow	Hibiscus	10
40	Bird Lime Tree	Cordia dichotoma	1



Green Area View near Labs



Green Area View from Workshop



Greenery at the backyard and play ground



Greenery along the hostel and staff quarters way



Greenery at E- Block

Maintenance of Gardens and Landscape

In addition to new plantation drives and landscaping/beautification activities, all essential maintenance work (like lawn, hedge, existing plants /shrubs, growing of seasonal flowers) for previously developed flower and other gardens, as well as other locations of the College campus, is done regularly under the supervision of Horticulture Section.

Maintenance of Plantations

Apart from the maintenance of gardens, all previously planted trees (like roadside and other plantations) in different locations of the College campus are regularly nurtured by cleaning, fertilization, watering, etc.

House Plants

House plants do not just look good – they can make us feel good, too. Studies have shown that house plants-

- Boost our mood, productivity, concentration, and creativity
- Reduce our stress, fatigue, sore throats, and colds
- Help clean indoor air by absorbing toxins, increasing humidity & producing oxygen
- Add life to a sterile space, give privacy and reduce noise levels

Campus Involvement

For sustainable use of resources and for the mission of “GO-GREEN” it is necessary that the students, faculty, and administration welcome it. CSIT has an environment that invites opportunities to better its community through campus organizations. The green initiative started on the campus many years ago. The College students are actively participating and solely concerned with the environment. These students under the guidance of faculties strive to create an environmentally friendly campus. Their purpose is to create awareness and eventually act on that awareness. College is also actively conducting environmental awareness programs on campus regularly.

6.2 AMBIENT AIR QUALITY AT CSIT

Sr. No	Parameters	Unit	Result	Requirement permissible limits as per NAAQS/CPCB
1	Particulate Matter, PM10	µg/m ³	85.40	100
2	Particulate Matter, PM 2.5	µg/m ³	40.5	60
3	Sulphur Dioxide (as O ₂)	µg/m ³	7.9	80
4	Carbon Monoxide (as O)	mg/m ³	0.883	04
5	Oxide of Nitrogen (as O ₂)	µg/m ³	23.40	80

Central Pollution Control Board, New Delhi has set guidelines to monitor and analyze the air pollution quality parameters. The trees cover on the campus are the leading sources to absorb CO₂ and release enough fresh O₂ across the College Campus. The result shows that CSIT Campus's air quality status is good as compared to other locations. It is identified that CSIT campus is a green campus. CSIT campus observed minimum air pollution as compared to other Ambient Air Pollution Centers located in different parts of the city.

7. CARBON FOOTPRINTS

Carbon is the basis of life on mother Earth. It is incorporated into the plants through photosynthesis, consumed by animal species through the food, present in the form of carbon dioxide (CO₂) in the atmosphere, locked into the rocks like limestone, and compressed into the different fossil fuels such as coal and oil. As CO₂ levels in the atmosphere continue to increase, most climate designs or projects that the oceans of the world and trees will keep soaking up more than half of CO₂. The plants on land and in the sea, taken up carbon over many years increased the percentage discharged during decay, and this increased carbon became locked away as fossil fuels beneath the surface of the planet. At the start of the 21st century, we brought growing concern about global warming, climate change, food security, poverty, and population growth. In the 21st century, more carbon has been released into the atmosphere than that has been absorbed. CO₂ is a principal component causing global warming. Atmospheric carbon dioxide levels have increased to 40 % from preindustrial levels to more than 390 parts per million CO₂. On this background, it is a need of time to cover the research areas interrelated with climate change.

Carbon foot prints

In today's world, one of the biggest issues faced by all of us is global warming. Global warming refers to an increase in the average global temperature of mother Earth. The main cause of global warming is an increase in the concentration of greenhouse gases (GHGs) in the atmosphere due to anthropogenic activities and their level is determined with the help of global warming potential (GWP) and expressed as Carbon Footprint (CF). Carbon Footprint is another phenomenon used for GHGs or carbon dioxide emission in terms of CO₂ equivalents. There are various definitions of carbon footprint are in literature. But the most recognized definition given by Wiedmann is “the Carbon footprint is the measure of carbon dioxide emissions directly or indirectly caused by an activity or accumulated over the life stages of a product.” In other words, “A carbon footprint is the total greenhouse gas (GHG) emissions caused directly and indirectly by an individual, organization, event or product.”

As CSIT is considered an institutional organization, various energy resources like electricity, solar rooftop systems are used. It is necessary to calculate the carbon footprint of the College to upgrade the Clean Developmental Mechanism (CDM) in various processes. All the data from the various sources were collected from all the sectors where energy resources are used. The collected data is calculated by using standard emission factors.

Efforts for Carbon Neutrality

Air pollution is a matter of serious concern on the campus owing to its urban location. CSIT College as a responsible institution understands the importance of its carbon footprint and developed a plan to reduce greenhouse gas emissions in all its activities. Strictly ban on burning of dried leaves and waste paper in College.

Electricity carbon footprint

In the College, electricity is used for various purposes like residential, office use, and laboratories. The total electricity used in the College liberates mass kg of CO₂ per year. The laboratory equipment consumes the highest electricity which emits a large amount of carbon CO₂ per year. The solar panels are installed on the roof of various buildings produce electricity from solar panels which further saves ample mass of CO₂ per year.

Paper footprint

The papers are used in the institution for various purposes like exam answer sheets, circulars, notices, office work, etc. The papers are responsible for the emission of CO₂. The College used a. On the CSIT Campus, various departments follow paperless methods of communication to reduce the footprint by the use of papers.

Conclusion

India's CO₂ emission is increased by an estimated 4.6 % in 2017, despite a turbulent year for its economy. The carbon footprint of the nation is measured per person; India's emissions are still very low – at only 1.8 tons of CO₂ per capita which is much lower than the world average of 4.2 tons. But those emissions have been increasing steadily, with an average growth rate over the past decade of 6 %. The universities are the organizations which are having large areas which consume high quantities of electricity and LPG for many purposes. The CSIT College Campus emits approx. 24,000 tons of CO₂ per year approximately. The present Clean Development Mechanism (CDM) practices to reduce the 18.10 tons CO₂ per year approximately.

Recommendations

- The food waste generated from College hostel mess, guest house, canteens, and staff quarters should be converted into biogas which can be further utilized for hostel kitchens.
- The solar battery-operated vehicles should be used on the campus to overcome the vehicle

footprint.

- Green computing or E- work is helping the organization to reduce footprint very effectively.
- The solar energy-based street lamps on campus will reduce carbon footprint.
- The awareness should be made among the faculty, students, and other employees regarding Clean Development Mechanism (CDM) to reduce the consumption of electricity and natural resources.
- “Carbon Sequestration” survey should be conducted on the campus. Carbon sequestration is a process of converting atmospheric carbon i.e. CO₂ into other sinks of carbon such as vegetation, soil, ocean, etc. in various forms to mitigate global warming audit is one of the important clauses of the Kyoto Protocol.

Carbon Sequestration

While transforming ourselves from regional universities to global universities, the need of such universities to face the global future challenges and try to find out possible solutions for them. It is a social and environmental responsibility of Government Institutes, Universities, National and International Organizations to respond positively to various global issues at the local level and should collate the generated knowledge into the society. Global warming and climate change are current environmental issues that need to be addressed scientifically and efficiently. As Universities are provided with skillful human resources supported by analytical infrastructure, it is our duty to bring such ideas into practice.

8. GREEN INITIATIVES

CSIT is located in the area which is one of the important wilder areas of Durg city with its precious biodiversity. The major portion is covered with vegetation. The College aims to protect and conserve its biodiversity, fresh and clean ambiance through many initiatives. The College has taken the following green initiatives to protect and conserve nature.

Plantation and Nurturing Programme

Many plantation drives are taken by the College on its campus. Every year on 5th June i.e. World Environment Day, the CSIT takes Plantation activity. College has taken many plantation drives. The Horticulture Section looks after tree plantation activities. The trees are watered by students of various Departments. They nurture these trees throughout the year. Students of various departments and students make the plantation and nurturing program successful. A total of 23 plant saplings of different species (like ornamental, fruit and medicinal plant, etc.) were planted in various sites of the College campus during this year's environment day program.

Green computing practice

Being an academic institution, papers are used for various purposes like exam answer sheets, circulars, notices, office work, document printing, and Xeroxing. Since the trees are cut for paper manufacturing, the sequestration of carbon is reduced increasing carbon footprint. To cut down the carbon footprint, the College administration and various departments follow paperless methods of communication by using emails, online forms submission, etc. The paperless work was helpful in reducing tons of CO₂. The tons of biomass are saved by this green computing practice.

Solar Electricity Generation

The College has installed a 200KW capacity Solar Power Plant for electricity generation which produces electricity and sends it to the local grid which is helpful for an electricity bill reduction. Most of the buildings are constructed considering the need for Light and ventilation which reduces the use of electricity. The air conditioners are used only in essential conditions in the laboratories and offices to reduce electricity consumption.

Conferences and workshops on Environmental Sustainability

CSIT College organizes Conferences and Workshops based on the theme of environmental sustainability.

Energy Conservation Day: -

On 14th December of 2022, which is National Energy Conservation Day was celebrated at CSIT-Durg premises. The CSVTU, Bhilai, and CREDA, Raipur jointly organized and conducted the activities, which included speech, essay, and quiz contests. The prizes that were awarded to the winners of the various competitions each were worth of Rs. 8,000 and consisted of the articles that are listed below, according to the category in which they had placed.

9. CONCLUSIONS

Energy Audit, Environmental & Green Audit is one of the important tools to check the balance of natural resources and their judicious use. Energy Audit, Environmental & Green Auditing is the process of identifying and determining whether institutional practices are eco-friendly and sustainable. It is a process of regular identification, quantification, documenting, reporting and monitoring of environmentally important components in a specified area. The main objective to carry out a Energy Audit, Environmental & Green Audit is to check the green practices followed by the College and to conduct a well-defined audit report to understand whether the College is on the track of sustainable development. After completing the audit procedure of the College for green practices, there are the following conclusions, recommendations, and Environmental Management Plan (EMP) which can be followed by the College in the future for keeping campus environment friendly.

- College takes efforts to dispose of majority of waste by proper methods. Green computing i.e. Online payment systems, online circulars, and examination procedures are helpful for reducing the use of papers and ultimately reducing carbon footprint.
- Reducing the use of one-time use plastic bottles, cups, folders, pens, bouquets, decorative items will be useful to solve the problem of plastic pollution to some extent.
- Biodegradable waste is used efficiently for composting and vermicomposting.
- Use of LED lamps and Tube Lights is to be encouraged.
- Toilets and bathrooms are consuming more water in the departments. The replacement of old taps can be beneficial for solving this issue.
- The overall ambient air quality on the campus is good while some air quality issues that may arise due to developmental activities on the campus should be addressed. The sound levels on the campus are good.

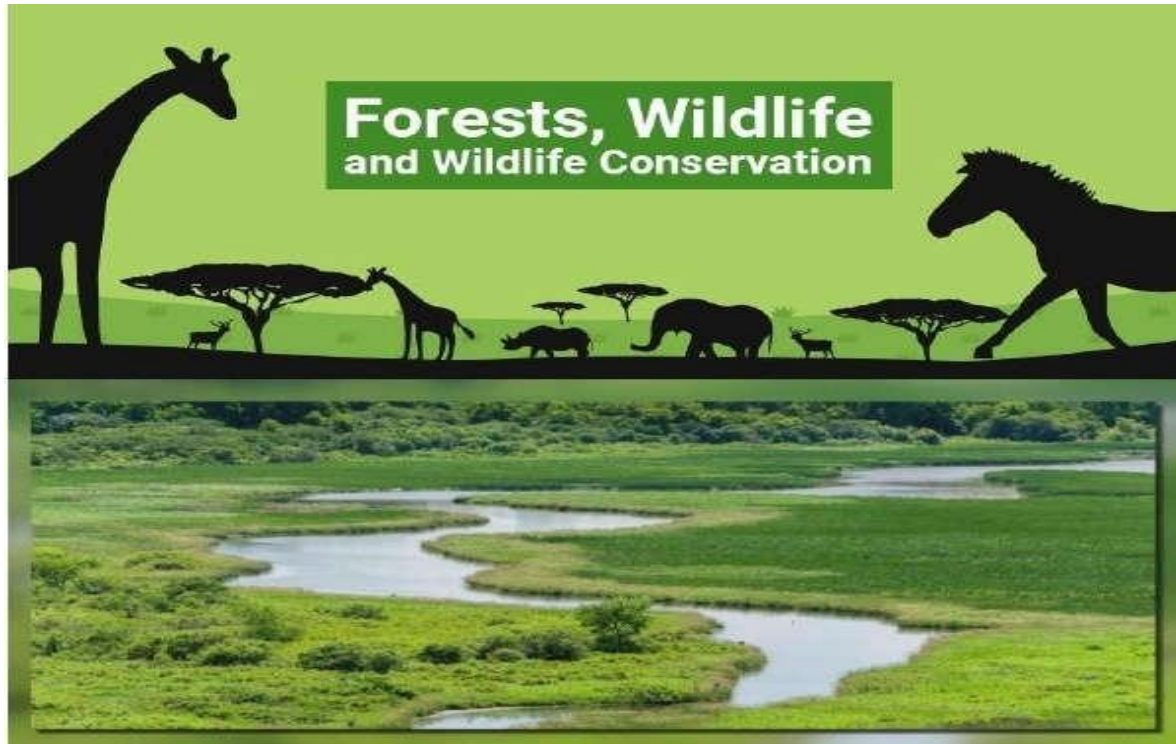
Key Recommendations & Environment Management Plan (EMP)

Following are some of the key recommendations for improving the campus environment and to be considered as Environment Management Plan (EMP).

- An environmental policy document has to be prepared with all the recommendations and current practices carried by the College.
- A frequent visit should be conducted to ensure that the generated waste is measured, monitored, and recorded regularly and information should be made available to the administration.
- The College should develop internal procedures to ensure its compliance with environmental legislation and responsibility should be fixed to carry out it in practice.
- The solid waste should be reused or recycled at maximum possible places.
- Installation of sensor-based electrification items like fans, lights, etc. can save electricity
- Regular checkups and maintenance of pipes, overhead tanks, and plumbing systems should be done by the engineering section to reduce overflow, leakages, and corruptions.
- Science laboratories large amount of water goes waste during the process of making distilled water; the system should develop to reuse this water for other purposes.
- No such processes or activities were observed at CSIT which can deteriorate the environmental quality.
- The said college is in continuous efforts to spread the environmental awareness programs among staff and students.
- It was also observed that the said college is keeping the environmental quality at priority in every developmental stage.

***DEVELOPMENT IS AN IMPORTANT ASPECT OF ANY
ORGANIZATION, COLLEGE, OR Institute. THIS
DEVELOPMENT BY CSIT IS ALWAYS ACHIEVED AT THE
EXPENSE OF ENVIRONMENTAL REHABILITATION.***

***WE ARE GLAD TO DECLARE THAT CSIT IS AN
ENVIRONMENT-FRIENDLY COLLEGE ALONG WITH MANY
GREEN DEVELOPMENT PROCESSES THAT ARE FAIRLY
PRACTICED BY THE COLLEGE.***





Chhatrapati Shivaji Institute of Technology

Approved by: AICTE, New Delhi | Affiliated to CSVTU, Bhilai

ENERGY AUDIT, ENVIRONMENTAL AUDIT & GREEN AUDIT REPORT

OF

**CHHATRAPATI SHIVAJI
INSTITUTE OF TECHNOLOGY,
DURG**

For the Year of 2022

ENERGY AUDIT, ENVIRONMENTAL AUDIT & GREEN AUDIT REPORT OF CHHATRAPATI SHIVAJI INSTITUTE OF TECHNOLOGY, DURG



**Chhatrapati Shivaji Institute of Technology, Balod Rd, Shivaji Nagar, Durg,
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April 2022

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We express our sincere gratitude to **Chhatrapati Shivaji Institute of Technology** for giving us the opportunity to be a part of their mission towards Energy Conservation. We are thankful to all officers and employees of Chhatrapati Shivaji Institute of Technology with whom we have interacted during the field study for their whole-hearted support in undertaking measurements and eagerness to assess the system/equipment efficiencies and saving potential. The willingness of these key personnel to participate in this program and acknowledge the call for energy efficiency is more than half the issues received.



Authorized Signatory

for, Audittech Industrial Services Private Limited

CERTIFICATE



This is to Certify that Chhatrapati Shivaji Group of Institution, Durg (Chhattisgarh) has conducted Energy Audit, Environmental Audit & Green Audit of their campus and has submitted necessary data in addition, credentials for scrutiny the activities and measures carried out by the university have been verified based on the report submitted and was found to be acceptable.

The efforts taken by the CSIT Management, Faculties and students towards Green Building, Environment and sustainability is highly appreciated and commendable.

Aashish Bafna



(Director)

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EXECUTIVE SUMMARY

Energy Audit, Environmental & Green Audit is the process of identifying and determining whether institutions' practices are eco-friendly and sustainable. The main objective to carry out a Energy Audit, Environmental & Green Audit is to check green practices followed by the College and to conduct a well-formulated audit report to understand where we stand on a scale of environmental soundness. The initiative taken by Chhatrapati Shivaji Institute of Technology to conduct a Energy Audit, Environmental & Green Audit of the College campus is a commendable sustainable goal. The strategies followed were the preparation of questionnaires and subsequent action plans to implement the project. Questionnaires prepared to conduct the Energy Audit, Environmental & Green Audit were based on the guidelines, rules, acts, and formats set by the Government of India, Ministry of Environment and Forest, New Delhi, and Central Pollution Control Board, New Delhi. Questionnaires were prepared for solid waste, energy, water, hazardous waste, and e-waste. For audit purposes, analysis of suitable data is required, for the same study area is grouped into various Blocks and Departments. The audit was carried out for solid waste, electricity and energy, water and wastewater, hazardous waste, air quality, and green inventory including carbon footprints. It also indicates the green initiatives taken by universities to save environmental resources. The “Energy Audit, Environmental & Green Audit “ also presents the “Environmental Management Plan”.

1. INTRODUCTION

1.1 ENERGY AUDIT, ENVIRONMENTAL & GREEN AUDIT AN EFFECTIVE EFFORTS TOWARDS ENVIRONMENT SUSTAINABILITY & ENERGY CONSERVATION

Modernization and industrialization are the two important outputs of the twentieth century that have made human life more luxurious and comfortable. Simultaneously, they are responsible for voracious use of natural resources, exploitation of forests and wildlife, producing massive solid waste, polluting the scarce and sacred water resources, and finally making our mother Earth ugly and inhospitable. Today, people are getting more familiar with global issues like global warming, greenhouse effect, ozone depletion, climate change, etc. Now, it is considered as a final call by mother Earth to walk on the path of sustainable development. The time has come to wake up, unite and combat together for a sustainable environment.

Considering the present environmental problems of pollution and excessive use of natural resources, Honorable Prime Minister, Shri. Narendra Modi has declared the Mission of Swachh Bharat Abhiyan. Also, College Grants Commission has mentioned the “Green Campus, Clean Campus” mission mandatory for all higher educational institutes. As environmental sustainability is becoming an increasingly important issue for the nation, the role of higher educational institutions in relation to environmental sustainability is more prevalent.

Energy Audit, Environmental & Green Audit is the most efficient ecological tool to solve such environmental problems. It is a process of regular identification, quantification, documenting, reporting and monitoring of environmentally important components in a specified area. Through this process, the regular environmental activities are monitored within and outside of the concerned sites which have direct and indirect impacts on the surroundings. An Energy Audit, Environmental & Green Audit can be one of the initiatives for such institutes to account for their energy, water resource use as well as wastewater, solid waste, hazardous waste generation. The Energy Audit, Environmental & Green Audit process can play an important role in the promotion of environmental awareness and sensitization about resource use. It can create consciousness towards ecological values and ethics. Through the Energy Audit, Environmental & Green Audit, one can get direction about how to improve the condition of the environment.

1.2 WHY ENERGY AUDIT, ENVIRONMENTAL & GREEN AUDIT

Energy Audit, Environmental & Green Auditing is the process of identifying and determining whether an institution's practices are eco-friendly and sustainable. Traditionally, we are good and efficient users of natural resources. However, over the period of time excess use of resources like energy, water, chemicals are become habitual for everyone especially, in common areas. Now, it is necessary to check whether our processes are consuming more than the required resources? Whether we are handling waste carefully? Energy Audit, Environmental & Green Audit regulates all such practices and gives an efficient way of natural resource utilization. In the era of climate change and resource depletion, it is necessary to verify the processes and convert them into green and clean ones. The Energy Audit, Environmental & Green Audit provides an approach for it. It also increases overall consciousness among the people working in institutions towards an environment.

1.3 GOALS OF ENERGY AUDIT, ENVIRONMENTAL & GREEN AUDIT

College has conducted a Energy Audit, Environmental & Green Audit with specific goals as:

- Assess facility of different types of waste management.
- Increase environmental awareness throughout campus.
- Identification and documentation of green practices followed by college.
- Identify strengths and weaknesses in green practices.
- Conduct a survey to know the ground reality about green practices.
- Analyze and suggest solutions for problems identified from the survey.
- Identify and assess environmental risk.
- The long-term goal of the environmental audit program is to collect baseline data of environmental parameters and resolve environmental issues.
- To motivate staff for optimized sustainable use of available resources.

1.4 OBJECTIVES OF ENERGY AUDIT, ENVIRONMENTAL & GREEN AUDIT

- To examine the current practices which can impact the environment such as resource utilization, waste management, etc.
- To prepare an Environmental Statement Report on green practices followed by different departments, support services, and administration building.
- To set goals, vision, and mission for green practices on the campus.
- To identify and analyze significant environmental issues.
- To establish and implement Environmental Management Plan in various departments.
- To assess for better performance in green practices and its valuation.

1.5 ABOUT CRITERIA 7 OF NAAC

Universities are playing a key role in the development of human resources worldwide. Higher education institutes campus run various activities with the aim to percolate the knowledge along with practical dimension among the society. Likewise, different technological solutions related to the environment are also provided by the higher education institutes. Different types of evolutionary methods are used to assess the problem concerning the environment. It includes Environmental Impact Assessment (EIA), Social Impact Assessment (SIA), Carbon Footprint Mapping, Green audit, etc.

National Assessment and Accreditation Council (NAAC) is a self-governing organization that rated the institutions according to the scores assigned at the time of accreditation of the institution. Energy Audit, Environmental & Green Audit has become a mandatory procedure for educational institutes under Criterion VII of NAAC. The intention of the Energy Audit, Environmental & Green Audits is to upgrade the environmental condition inside and around the institution. It is performed by considering environmental parameters like water and wastewater accounting, energy conservation, waste management, air, noise monitoring, etc. for making the institution eco-friendlier.

Students are the major strength of any academic institution. Practicing green action in any educational institution will inculcate the good habit of caring for natural resources in students. Many environmental activities like plantation and nurturing saplings and trees, Cleanliness drives, Bird watching camps, no vehicle day, Rainwater harvesting, etc. will make the students good citizens of the country. Through Energy Audit, Environmental & Green Audit, higher educational institutions can ensure that they contribute towards the reduction of global warming through Carbon Footprint reduction measures.

1.6 BENEFITS OF GREEN AUDIT TO AN EDUCATIONAL INSTITUTE

There are many advantages of Energy Audit, Environmental & Green Audit to an Educational Institute.

- It would help to protect the environment in and around the campus.
- Recognize the cost-saving methods through waste minimization and energy conservation.
- Empower the organization to frame a better environmental performance.
- It portrays a good image of the institution through its clean and green campus.
- More efficient resource management
- To create a green campus
- To enable waste management through reduction of waste generation, solid and waste
- To create plastic-free campus and evolve health consciousness among the stakeholder
- Recognize the cost-saving methods through waste minimizing and managing
- Authenticate conformity with the implemented laws
- Empower the organizations to frame a better environmental performance
- Enhance the alertness for environmental guidelines and duties
- Impart environmental education through systematic environmental management approach and improving environmental standards
- Benchmarking for environmental protection initiatives
- Financial savings through a reduction in resource use
- Development of ownership, personal and social responsibility for the College and its environment
- Developing an environmental ethic and value systems in youngsters.
- Green auditing should become a valuable tool in the management and monitoring of environmental and sustainable development programs of the College.
- Finally, it will help to build a positive impression through green initiatives for the upcoming NAAC visit.

1.7 INTRODUCTION OF AUDITING FIRM

M/s. Audittech is an empaneled Accredited Energy Audit Firm from the Bureau of Energy Efficiency, Ministry of Power, Government of India. It is one of the fast-growing Energy Audit & Energy services providing company executed several projects covering all the energy Intensive Sectors & states of India. The directors and associate team members are very well experienced in the field of Energy Audit and executed more than 100 no's Detailed Energy Audit & More than 10 No's of Green Audit so far. The associate team and experts are highly qualified and experienced in the field of Energy Audit and Services. Individual credential of each member in the field of Energy Audit is very rich due to their past association with the very reputed organization of Energy Audit Services. The company has Head office at Balod (C.G.) & Branch offices at Raipur, Durg, Bhopal, Mumbai, and Delhi.

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Directors Details

Sr. No.	Name	Designation / Technical Experience	Technical Experience /Qualification
1	Mr. Aashish Bafna	Managing Director - 13yrs	B.E (E&I)., MBA(Energy Management), Certified Energy Auditor, Surveyor & Loss Assessor
2	Mr. Rakesh Khichariya	Director- 25Yrs	B.E (Elect.), Accredited Energy Auditor
3	Mr. Ramesh Patel	Director- 25Yrs	B.E.(Mech), Government approved Valuer, Competent Person for Factory Act
4	Mr. Isshant Chainani	Director- 14 Yrs	B.E. (Elect & telecom)
5	Mrs. Shikha Golchha	Director- 13 Yrs	B.E., MBA (Finance)

M/s. Audittech Recognized as “Startup” from Department of Promotion of Industry and Internal Trade, Ministry of Commerce & Industry, Government of India and also Registered in Ministry of Micro, Small & Medium Enterprises (MSME) as Micro unit.

ENERGY AUDIT TEAM

The energy audit team involved in the energy audit of CSIT, Durg details are in the following.

SN	Name	Designation/ Qualification	Experience	Contact Details
1	Mr. Rakesh Khichariya	Accredited Energy Auditor (AEA-0295)	25 yrs.	9827411444
2	Mr. Aashish Bafna	Certified Energy Auditor (EA-28916)	13 yrs.	9827143100
3	Mr. Isshant Chainani	Certified Energy Manager	14 yrs.	9407702444
4	Mr. Mahaveer Bafna	Certified Energy Manager	6 Yrs.	8962369293
5.	Mr. Retesh Verma	Energy Engineer	2 Yrs.	8770912822
6.	Mr. Kishor Sarwa	Energy Engineer	2 Yrs.	8435528535
7	Mr. Yeeshu kumar	Energy Engineer	1 Yrs.	9753954109

LIST OF INSTRUMENTS

Following are the instrument used at the time of the Energy Audit.

Sr. No.	Instrument	Make/Sr. No.
1	Power & Harmonics Analyzer, 1 Set (With CT, PT) HT	Krykard ALM 31/ 123673RCH
2	Power & Harmonics Analyzer, 1 Set (With CT, PT) HT	Krykard ALM 20/ 28107280
3	Temperature gun 1 Set (infrared Thermometer)	MECO 550 T/ IRT550T_17120136
4	Lux Meter 1Set (Digital Lux Meter)	MECO G 930P/201704004601
5	Ultrasonic Flow Meter	Chinese/ 28107280

1.8 ABOUT CHHATRAPATI SHIVAJI INSTITUTE OF TECHNOLOGY

Chhatrapati Shivaji Institute of Technology is a distinguished engineering and technology institute, potentially located in Durg, Chhattisgarh. Established in the year 1999, affiliated to Chhattisgarh Swami Vivekanand Technical University. The institute offers a range of undergraduate programs in engineering and technology. With facilities Computer Lab, Library, Seminar Hall, Auditorium / more, and a dynamic learning environment, the institute focuses on practical learning and research. Chhatrapati Shivaji Institute of Technology emphasizes industry collaboration and innovation, preparing students for successful careers in the rapidly evolving technological landscape. The institute's dedicated faculty and emphasis on overall development contribute to its reputation as a premier institution in engineering education.

The present location of College is Located at Durg in Chhattisgarh. The Address of The Institute Is Durg-Balod, Dallirajhara Road at Shivaji Nagar near Kolihapuri, Dist. Durg (C.G.), 4 Kms away from the Collectorate and 6 Kms from Railway Station, Durg. The grandeur of the location provides a unique environment for education in close proximity with nature which is farfetched from the hum-drum of the city.

Over the years, it has steadily evolved itself as one of the leading centers of comprehensive learning with its exposure to and linkages with its peers at national and international levels. With its state-of-the-art infrastructure, well-equipped laboratories, and highly qualified and dedicated faculty, the College is committed to the task of harnessing and cultivating the capabilities of young students with a view to enabling them to carve suitable space for themselves in the modern economic world. The College Building Campus is categorized mainly into 4 Building Blocks. The details of all departments are mentioned in the below table of Buildings:

- Academic Buildings
- Facilities Buildings
- Residential Buildings

1.8.1 CAMPUS BUILDING & DEPARTMENTS

Name of the Building Blocks and their coding	
Building / Blocks	Name of the Departments
Departments	Department of Computer Science and Engineering
	Department of Electrical and Electronics Engineering
	Department of Mechanical Engineering
	Department of Information Technology
	Department of Mechatronics
	Department of Civil Engineering
	Department of Artificial Intelligence and Data Science
Labs	Computer Organization & Architecture Lab
	Software Laboratory (Sci Lab/ MATLAB)
	Machine Learning Lab
	Automobile Engineering Lab
	Industrial Engineering & Management Lab
	Mechanical Measurement and Metrology Lab
	Robotics Lab
	Electrical and Electronics Engineering Laboratory
	Power Electronics Devices and Circuits Laboratory
	High Voltage Engineering Laboratory
	Electrical Measurements & Measuring Instruments Laboratory
	Surveying Field Work Lab
	Geology Lab
	Environment Engineering Lab
	Simulation Lab

Facilities	
Facilities	Central Workshop
	Computer Centre
	Canteen and Cafeteria
	Student Activity Centre
	Seminar Hall
	Administrative Building
	Health Care Center
	Bank and ATM
Residential Buildings	
Residential Building	Boys Hostel
	Girls Hostel
	Staff Quarters

The College has also adopted the 'Green Campus' system for environmental conservation and sustainability. The 'Green Campus' has been active for the last several years as an assembly group of sub committees along with the Horticulture section that actively promotes the various projects. The College administration works on several activities for 'Green Campus' including Renewable Energy, Water Conservation, Tree Plantation, Waste Management, Paperless Work, etc.

1.8.2 CAMPUS INFRASTRUCTURE

Chhatrapati Shivaji Institute of Technology has a very good and systematic building infrastructure. All classrooms are fully ventilated and comfortable. The College from the outside looks great and is quite unique keeping the architecture of other universities in mind.

CSIT, DURG – CAMPUS HIGHLIGHTS

1.8.3.1 INTERNAL QUALITY ASSURANCE CELL

In pursuance of its Action Plan for performance evaluation, assessment & accreditation & quality up-gradation of institutions of higher education, NAAC proposes that every accredited institution should establish an Internal Quality Assurance Cell (IQAC) as a post-accreditation quality sustenance measure. Since quality enhancement is a continuous process, the IQAC will become a part of the institution's system & work towards realization of the goals of quality enhancement & sustenance.

OBJECTIVES:

- To develop a system for conscious, consistent & catalytic improvement in the overall academic and administrative performance of institution.
- Making efforts towards promoting student – centric atmosphere for holistic academic excellence.
- To enhance the quality of staff by promoting and encouraging overall faculty development.

FUNCTIONS OF THE IQAC ARE:

- Development and application of quality benchmarks/parameters for various academic and administrative activities of the institution.
- Facilitating the creation of a learner-centric environment conducive to quality education and faculty maturation to adopt the required knowledge and technology for participatory teaching and learning process.
- Arrangement for feedback response from students, parents and other stakeholders on quality-related institutional processes.
- Dissemination of information on various quality parameters of higher education.
- Organization of inter and intra institutional workshops, seminars on quality related themes and promotion of quality circles.
- Documentation of the various programmes/activities leading to quality improvement.
- Acting as a nodal agency of the Institution for coordinating quality-related activities, including adoption and dissemination of best practices.
- Development and maintenance of institutional database through MIS for the purpose of maintaining/enhancing the institutional quality.
- Development of Quality Culture in the institution.
- Preparation of the Annual Quality Assurance Report (AQAR) as per guidelines and parameters of NACC, to be submitted to NACC.

1.8.3.2 SPORTS AND CULTURE PARK

CSIT inculcates a high standard of ethical and moral values amongst the student's community to attain this and train students to be morally, culturally and ethically to be on the right side of every aspect. Industrial visits, tours, excursions and study are part of college life. Students are given equal opportunity to show their talent in other art forms like Dance, Art, Singing, Music, Drama, Debate, Quiz, Writing and

Elocution etc. by organizing competitions during the year. Quite a few stars emerge from these competitions. The college felicitates outstanding students for their performance. Cultural events are organized every year by student associations which give students a platform to showcase their talent in cultural fields. Apart from these events the students participate in various state and college level program where students have brought laurels to the college.

SPORTS

Sports facilities are an integral part of any educational activity to keep students healthy, both physically and mentally. In addition to organizing the Annual Sports routinely, intra-college league matches of basketball, cricket, football, chess, volleyball, badminton, table tennis etc. are framed every year. Each year, the much-awaited inter-collegiate volleyball tournament attracts participants from all over the state.

OUTDOOR FACILITIES

Cricket / Football Ground
Basket Ball Court
Volley Ball Court
Handball Court

INDOOR FACILITIES

Carrom Tables
Chess Setup
Table Tennis Court
Badminton Court



GREEN AREA AT CENTRAL BLOCK



BOYS HOSTEL



ENTRANCE GARDEN AND THE STATUE OF CHHATRAPATI SHIVAJI MAHARAJ



Campus Entrance Path Way with Green Surrounding



Girls Hostel and the Surrounding Green Patch



Entrance



Shaheed Memorial Hall



Library



Swami Vivekanand Auditorium



Play Ground

2. ENERGY AUDIT, ENVIRONMENTAL & GREEN AUDIT METHODOLOGY

2.1 Pre-Audit Stage

A pre-audit meeting provided an opportunity to reinforce the scope and objectives of the audit and pre-audit discussions were held on the basis of green initiatives taken and the current scenario of the College campus. This meeting is an important prerequisite for the Energy Audit, Environmental & Green Audit because it is the first opportunity to understand the concerns. The meeting was an opportunity to gather the information that the audit team can study before arriving on the site. The audit protocol and audit plan were handed over at this meeting and discussed in advance of the audit itself. The pre-audit meeting was conducted successfully and necessary documents were collected directly from the College before the initiation of the audit processes. The actual planning of audit processes was discussed in the pre-audit meeting. An Audit team was also selected in this meeting with the help of staff and the College management. The audit protocol and audit plan were handed over at this meeting and discussed in advance of the audit itself.

2.2 Management Commitment

The Management of the College has shown a commitment towards Energy Audit, Environmental & Green Auditing during the pre-audit meeting. They were ready to encourage all green activities. It was decided to promote all activities that are environmentally friendly such as awareness programs on the environment, campus farming, planting more trees on the campus, etc., after the green auditing. The management of the College was willing to formulate policies based on a Energy Audit, Environmental & Green Auditing report.

2.3 Objectives of the study

A clean and healthy environment aids effective learning and provides a conducive learning environment. There are various efforts around the world to address environmental education issues. Energy Audit, Environmental & Green Audit is the most efficient and ecological way to manage environmental problems. It is a kind of professional care that is the responsibility of each individual who is part of economic, financial, social, environmental factors. It is necessary to conduct a Energy Audit, Environmental & Green Audit on a College campus because students become aware of the Energy Audit, Environmental & Green Audit, its advantages to saving the planet and they become social and responsible citizens of our country. Thus, Energy Audit, Environmental & Green Audit becomes necessary at the College level. The broad objectives are as follows.

- Diagnosing the environmental problems to eliminate them.
- Environmental education through a systematic environmental management approach.
- Improving environmental standards.
- Benchmarking for environmental protection initiatives.
- Efficient utilization of resources.
- Financial savings through a reduction in resource use.
- Curriculum enrichment through practical experience.
- Development of ownership, personal and social responsibility for the College and its environment.
- Developing environmental ethics and value systems in young people.
- Providing certain recommendations based on environmental audit reports.
- Ensuring compliance, not only with laws, regulations, and standards but also with company policies and the requirements of an Environmental Management System (EMS) standard.
- Enabling environmental problems and risks to be anticipated.
- To demonstrate that College is aware of its impact upon the environment.

2.4 Audit Stage

Energy Audit, Environmental & Green Audit was done with the help of co-associates involving different student groups, teaching, and non-teaching staff. The Energy Audit, Environmental & Green Audit began with the teams walking through all the different facilities at the College, determining the different types of appliances and utilities as well as measuring the usage per item (Watts indicated on the appliance or measuring water from a tap) and identifying the relevant consumption patterns (such as how often an appliance is used) and their impacts. The staff and learners were interviewed to get details of usage, frequency, or general characteristics of certain appliances. Data collection was done in the sectors such as Energy, Waste, Green Area, Carbon footprint, and Water use. College records and documents were verified several times to clarify the data received through surveys and discussions.

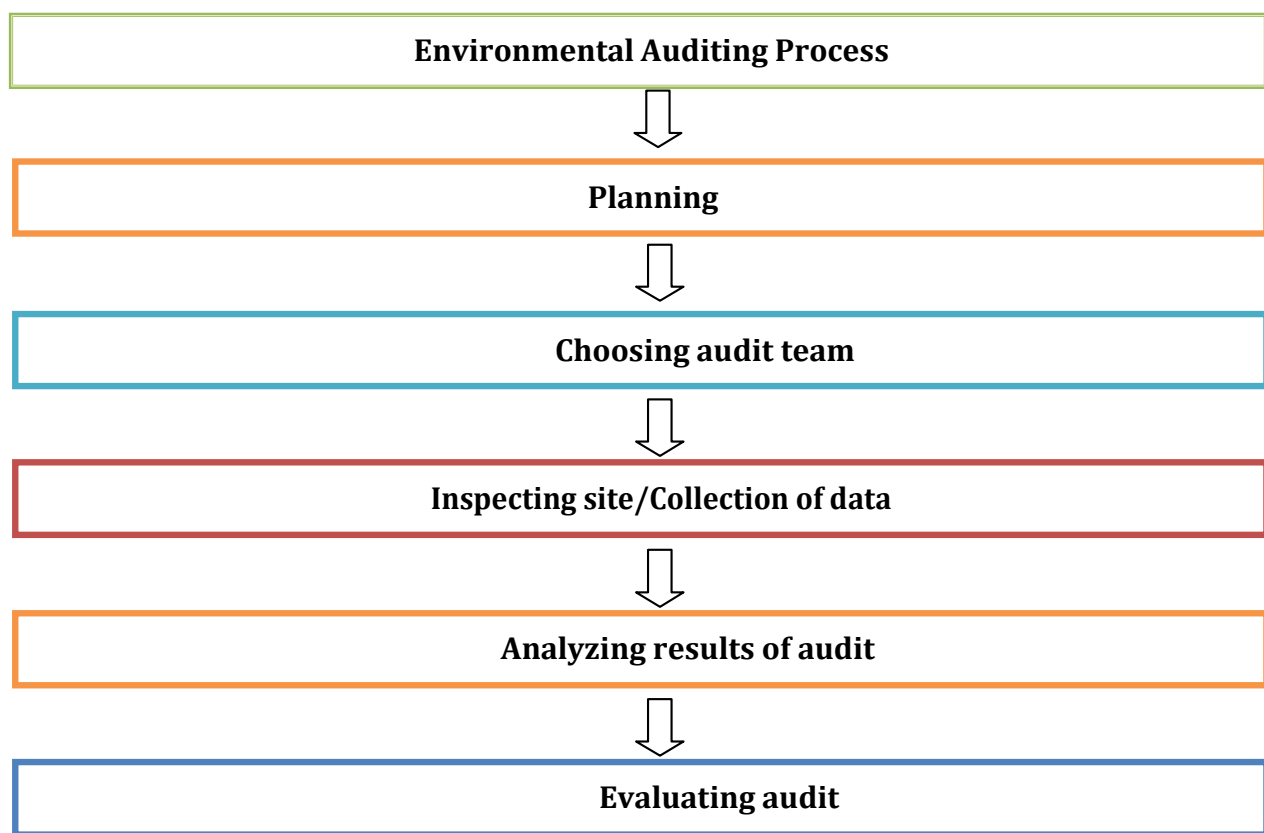
2.5 Methodology

The Management of the College has shown a commitment towards Energy Audit, Environmental & Green Auditing during the pre-audit meeting. They were ready to encourage all green activities. It was decided to promote all activities that are environmentally friendly such as awareness programs on the environment, campus farming, planting more trees on the campus, etc., after the Energy Audit, Environmental & Green Auditing. The management of the College was willing to formulate policies based on a Energy Audit, Environmental & Green Auditing report. In order to perform Energy Audit, Environmental & Green Audits, the methodology included different tools such as preparation of questionnaires, physical inspection of the campus, observation, and review of the documentation, interviewing key persons, and data analysis, measurements, and recommendations. The study covered the following areas to summarize the present status of environmental management on the campus:

- Energy Management
- Water Management
- Waste Management
- Environment Management

Methodology – Step by Step

The audit process was carried out in three phases. At first, all the secondary data required for the study was collected from various sources, like concerned departments such as engineering cell, Internal Quality Assurance Cell, Department of Physics, etc. A broad reference work was carried out to clear the idea of Energy Audit, Environmental & Green Auditing. Different case studies and methodologies were studied and the following methodology was adopted for the present audit. The methodology of the present study is based on onsite visits, personal observations, and questionnaires survey tools. Initially, based on data requirements, sets of questionnaires were prepared. The surveyors then visited all the departments of the College and the questionnaires were filled. The generated data is subsequently gathered and used for further analysis. From the outcome of the overall study, a final report is prepared.



2.5.1 Survey by Questionnaire

Baseline data for Energy Audit, Environmental & Green Audit report preparation was collected by questionnaire survey method. Questionnaires are prepared to conduct the Energy Audit, Environmental & Green Audit in the College campus based on the guidelines, rules, acts, and formats prepared by the Ministry of Environment, Forest and Climate Change, New Delhi, Central Pollution Control Board, and other statutory organizations. Most of the guidelines and formats are based on broad aspects and some of the issues or formats were not applicable for the College campus. Therefore, using these guidelines and formats, combinations, modifications, and restructuring were done and sets of questionnaires were prepared for solid waste, energy, water, hazardous waste, and e-waste data.

All the questionnaires are a group of modules. The first module is related to the general information of the concerned department, which broadly includes the name of the department, month and year, the total number of students and employees, visitors of the department, average working days and office timings, etc. The next module is related to the present consumption of resources like water, energy, or the handling of solid and hazardous waste. Maintaining records of the handling of solid and hazardous waste is much important in Energy Audit, Environmental & Green Audits. There are possibilities of loss of

resources like water, energy due to improper maintenance, and assessment of this kind of probability is necessary for the Energy Audit, Environmental & Green Audit. One separate module is based on the questions related to this aspect. Another module is related to maintaining records, like records of disposal of solid waste, records of solid waste recovery, etc. For better convenience of the surveyor, some statistics like basic energy consumption characteristics for electrical equipment, etc. were provided with the questionnaires.

2.5.2 Onsite visit and observations

Chhatrapati Shivaji Institute of Technology has a vast built-up area comprising various departments under various academic buildings, staff quarters, many facilities including Academic Buildings, Health Centre, Council Hall, Auditorium, other facilities and separate boy's and girl's hostels. All these amenities have different kinds of infrastructure as per their requirement. All these buildings were visited by the surveyors and the present condition is checked with the help of the questionnaires. Personal observations were made during the onsite visit. All the amenities were clubbed in, as per their similarities and differences, which makes the survey and further analysis easier. For the data compilation purpose, the College Departments and support services were clubbed into Three Types of Buildings and given names as academic buildings, Facilities Buildings, and Hostel Buildings. The details of the Buildings are as follows:

Sr. No.	Name of the Buildings
1.	Academic Buildings
2.	Facilities Buildings
3.	Hostel Buildings

After the collection of secondary data, the reviews related to each environmental factor were taken by the Energy Audit, Environmental & Green Audit team. The data were tabulated, analyzed and graphs were prepared. Depending upon the observations and data collected, interpretations were made. The lacunas and good practices were documented. The Environmental Management Plan (EMP) was prepared for the next academic year in order to have better environmental sensitization. Finally, all the information was compiled in the form of the Energy Audit, Environmental & Green Audit Report.

2.5.3 Data analysis and final report preparation

Proper analysis and presentation of data produced from work is a vital element. In the case of a Energy Audit, Environmental & Green Audit, the filled questionnaires of the survey from each group were tabulated as per their modules, in Excel spreadsheets. The tabulated data is then used for further analysis. For a better understanding of the results and to avoid complications, averages, and percentages of the tables were calculated. A graphical representation of these results was made to give a quick idea of the status. Interpretation of the overall outcomes was made which incorporates all the primary and secondary data, references, and interrelations within. Final report preparation was done using this interpretation.

3. WATER & WASTE WATER AUDIT

Water is a precious natural resource available with a fixed quantum. The availability of water is decreasing due to the increasing population of the nation; as per capita availability of utilized water is going down. Due to the ever-rising standard of living of people, industrialization, urbanization, demand for freshwater is increasing day by day. The unabated discharge of industrial effluent in the available water bodies is reducing the quality of these ample sources of water continuously. Hence, the national mission on water conservation was declared by the Honorable Prime Minister Narendra Modi as 'Jal Shakti Abhiyan' and appealed to all citizens to collectively address the problem of water shortage, by conserving every drop of water and suggesting conducting water audits for all sectors of water use. Water audit can be defined as a qualitative and quantitative analysis of water consumption to identify means of reducing, reusing, and recycling water. Water Audit is nothing but an effective measure for minimizing losses, optimizing various uses, and thus enabling considerable conservation of water in the irrigation sector, domestic, power, and industrial sectors. A water audit is a technique or method which makes it possible to identify ways of conserving water by determining any inefficiency in the system of water distribution. The measurement of water losses due to different uses in the system or any utility is essential to implement water conservation measures in such an establishment.

3.1 Importance of Water Audit

- Systematic process
- May yield some surprising results
- Easier to work on solutions when the problems are identified.
- Attracting mechanism can be put into place.
- It is observed that a number of factors like climate, culture, food habits, work and working conditions, level and type of development, and physiology determine the requirement of water. The community which has a population between 20,000 to 1, 00,000 requires 100 to 150 liters per person (capita) per day. The communities with a population over 1, 00,000 require 150 to 200 liters per person (capita) per day. As per the standards provided by WHO Regional Office for Southeast Asia Schools require 2 liters of water per student for drinking purposes; 10-15 liters per student for Water-flush toilets. Administration requires (Staff Accommodation not included) 50 liters per person per day.

3.2 Water Audit

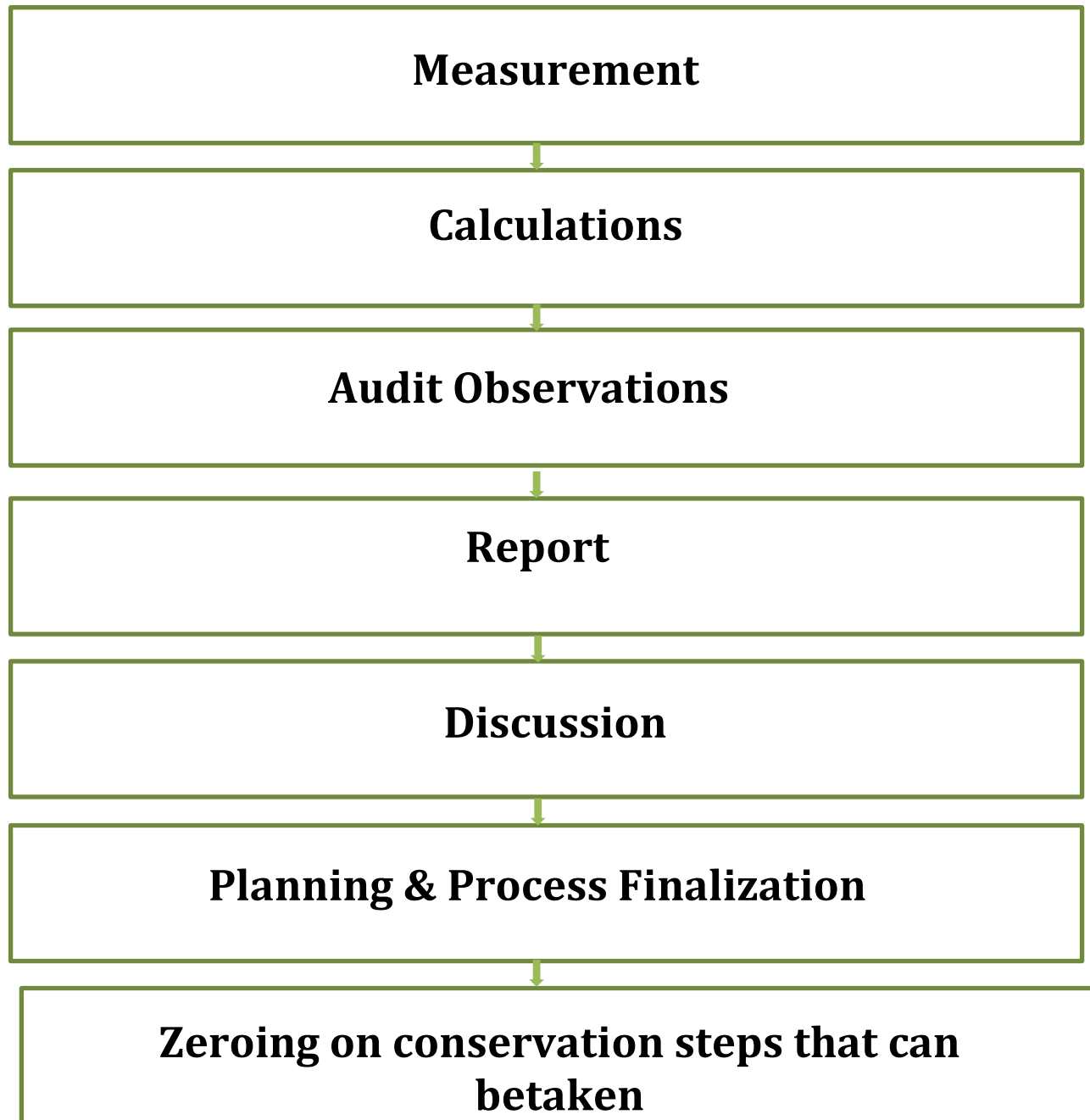
Water usage can be defined as water used for all activities which are carried out on campus from different water sources. This includes usage in all residential facilities, academic buildings, on-campus, and on-grounds. Wastewater is referred to as the water which is transported off the campus. The wastewater includes sewerage, residence water used in cooking, showering, clothes washing as well as wastewater from chemical and biological laboratories which ultimately go down in the sink or drainage system.

College water resources: -

The major resource for the water in the College is a self-reliant water boring system installed on the campus. There are Ground Water Tube wells installed on the campus with 7x1.5 HP which operate to fill the overhead tanks on the individual buildings. To fulfill the need for the supply of the campus, overhead storage tanks are available with capacity.

Sewage Water Treatment Plant: -

College has Installed Sewage water Treatment plant for recycling of water. Water from Toilets, kitchen & Other sources go to Sewage water treatment plant via drainage and after treating of water at STP the water then used for Gardening Purpose.



Water Audit Methodology & Steps

3.3 Sustainable Water Practices Watershed Management Practices

Underground water is the only one source of drinking water at CSIT. There are about seven borewells at the campus. The water is collected in the tanks located atop, and a regular water supply is maintained.

Observation:

1. Water in the College is not directly drained and is used again after treatment or recycling for watering the plants.
2. Sewage Treatment plant (STP) is installed in the back side of college for reducing the water wastage.
3. Water conservation/management practices have already been implemented in the college.
4. Rain water harvesting is also practiced by the college for recharging of the water table.



Rain Water Harvesting Spot



Rain Water Harvesting Points

Recommendations:

1. Filling of Water tank should be done on off peak time to save energy & Electricity Bill Cost.
2. Use of BEE star rated Energy Efficient Pumps in place of old & inefficient Pumps.

4. ENERGY AUDIT

Energy is one of the major inputs for the economic development of any country. The fundamental goal of energy management is to produce goods and provide services with the least cost and least environmental effect. Also, it can be said as “the strategy of adjusting and optimizing energy, using system and procedure so as to reduce energy requirements per unit of output while holding constant or reducing total costs producing the output from these systems”. The energy audit is key to a systematic approach for decision-making in the area of energy management. It attempts to balance the total energy inputs with its use and serves to identify all the energy streams in a facility.

4.1 Energy Audit

Energy resources utilized by all the departments, support services, and the administrative buildings of CSIT College, include Electricity, Solar Roof Top Systems, and Diesel Generators installed on the campus.

4.1.1 Energy Audit Objectives

Primary

- The first objective is to acquire and analyze data and find the necessary consumption pattern of these facilities.
- The second objective will be to calculate the wastage pattern based on the results of the first objective.
- The final objective is to find and implement solutions that are acceptable and feasible.

Secondary

- This would be our first exposure to this field hence experience gain would be vital.
- This project will precede many follow up projects and hence helps to gain technical and management exposure required for future energy projects.
- It is sure to help create a repertoire of vital contacts hence will develop interaction with alumni, faculty and students.

4.2 Sources of Energy

CSIT, Durg withdraws Energy from Followings:

- a. Electricity from CSPDCL
- b. High Speed Diesel (High Speed Diesel used as a fuel for Diesel Generator which is run whenever power supply is not available and for transportation)
- c. Solar Energy (Capacity 200 KW)

The Following are the Major consumers of electricity in the facility

- Lightning
- Air Conditioner
- Fans
- Computers
- Other Lab Equipment
- Bore well pumps

Indirect Benefits of Energy Audit

Every time the Energy Audit is carried out it rekindles the interest in Energy Conservation as an important function. Energy Auditors sharing their experience and knowledge with the Plant Personnel helps in fueling the innovative ideas for further action of reduction in Specific Power consumption (SPC). Any loose connections or heating of cables come to timely vision. For an external agency due to unbiased vision, a few points for Energy Conservation may be visible each time they perform the audit and this would help in achieving further saving. Inform any irregularities in Energy meter HT connections for rectification.

Executive Summary

S. No.	Proposed Energy Saving	Investment (In Rs. Lakhs)	Electricity Saving (kWh/ Annum)	Estimated Cost Saving (Rs. Lakhs)	Simple Payback Period (Yrs.)
A. Zero Investment					
1	Reduction of the existing Contract Demand	0.00	0.00	2.47	Immediate
	Sub- Total (A)	0.00	0.00	1.43	Immediate
B. Low Investment					
2	Replacement of Old CFL Bulbs(12W) with Led Bulbs(9W) in phase wise manner	0.15	1318	0.09	1.56
3	Replacement of Conventional Tubelights(40W) with Led Batten Lights(20W) in Phase wise manner	0.39	9360.0	0.66	0.6
	Sub- Total (B)	0.54	10677.60	0.76	0.71
C. High Investment					
4	Replacement of Normal Ceiling Fan of 60W with 30 W BLDC Fan	13.45	54983.60	3.90	3.45
5	Replacement of Non-Star AC with 1.5 Ton BEE 5 Star Rated AC	25.56	68160.00	4.84	5.28
	Sub- Total (C)	39.01	123143.60	8.74	4.46
	Grand- Total (A+B+C)	39.55	133821.20	11.97	3.02

Total Investment	39.55	In Rs. Lakhs
Total Monetary Saving	11.97	In Rs. Lakhs
Annual Energy Saving	1.34	Lakh kWh
Simple Payback Period	3.02	Yrs.
Total Thermal Saving	11.51	TOE
Reduction in CO2 Emission	105.72	Ton/ Annum

4.3 ELECTRICITY BILLING

Electricity Bill Summary for the year 2021-2022 are as follows.

Table 4.3. 1Energy Consumption details

For Consumer number: 1000312/374600-1000312

ELECTRICITY BILL									
Months	Consumption Unit (KWH)	Consumption Unit (KVAH)	Contract Demand (KVA)	Billing Demand (KVA)	Maximum Demand (KVA)	Power Factor	Energy Charge (Rs.)	Demand Charges (Rs.)	Total Payable Amount (Rs.)
Sep-21	23620	24978	270	220	111	0.95	167737	39330	254450
Oct-21	22512	24978	270	220	96.6	0.90	162437	39330	238432
Dec-21	14118	16120	270	220	50.52	0.88	105944	39330	174467
Mar-22	16305	17886	270	220	90.72	0.91	118385	39330	188700
Apr-22	27583	28611.00	270	220	123	0.96	194460	41800	285970
Total/Avg.	104138	112573			94.368	0.92	748963	199120	1142019
	20827.6	22514.6			123		149792.6	39824	228403.8

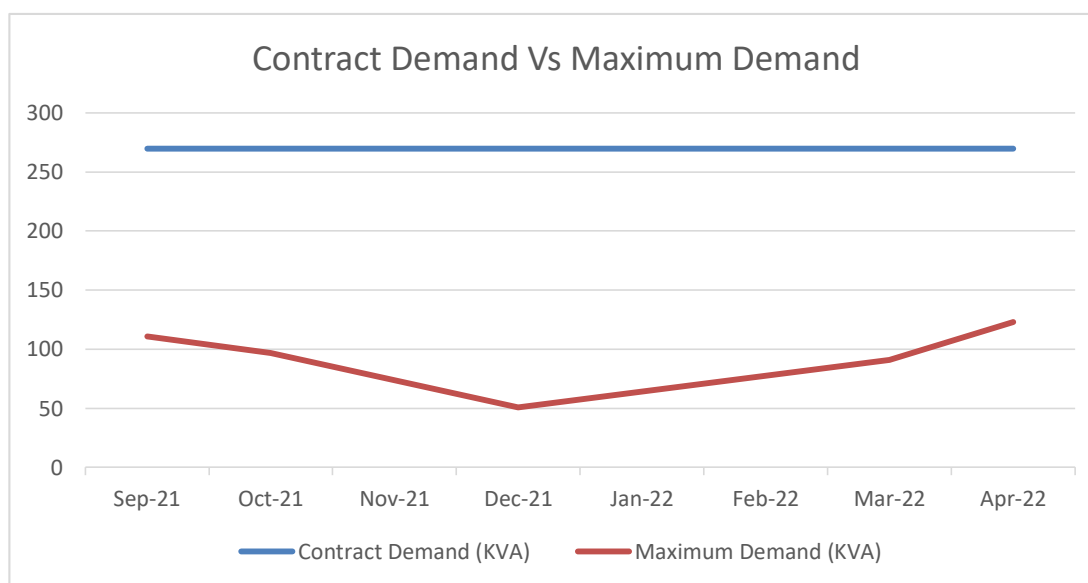
OBSERVATION:

The recorded average demand is of 95 KVA and Maximum demand is 123 KVA occurred at the month of April 2022. The Contract Demand is 270 KVA. It is recommended to reduce contract demand which will save cost on the total monthly bill value.

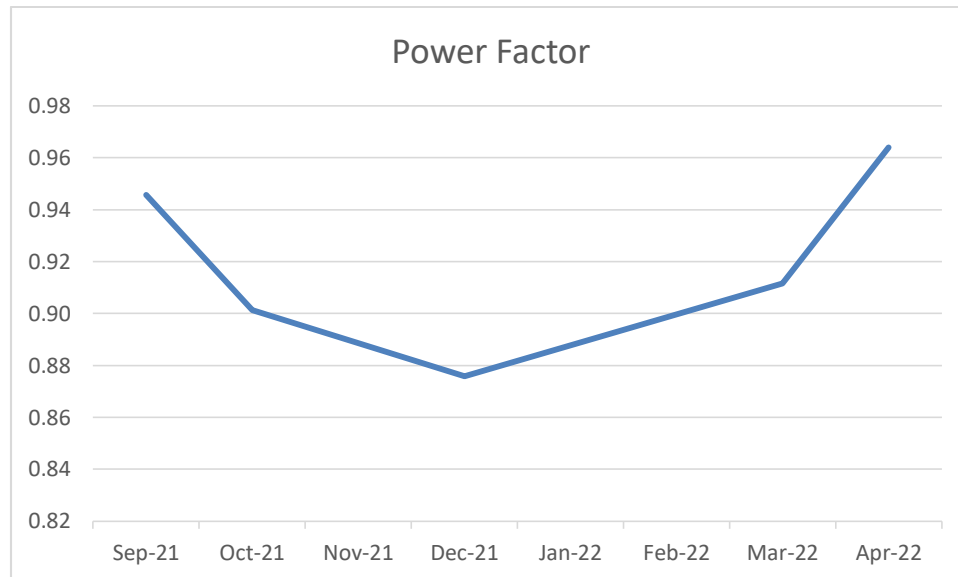
Table 4.3. 2 Maximum and Contract Demands Value

Particular	Value	UOM
Existing Contract demand	270	KVA
Existing minimum billing demand	160	KVA
Existing Normal Demand charges	190	Rs. /KVA
Fixed Normal demand charges	30400	Rs.
Maximum Demand Recorded	123	KVA
Proposed contract Demand	160	KVA
Proposed Minimum billing demand	136	KVA
Fixed Normal Demand Charges per month after contract Demand reduced	51000	Rs.
Annual Cost saving	2.472	Rs. Lakhs
Investment	NIL	
Simple payback year	Immediate	

GRAPHICAL REPRESENTATION OF MAXIMUM DEMAND VS CONTRACT DEMAND



GRAPHICAL REPRESENTATION OF POWER FACTOR



OBSERVATION

The average Power factor recorded for the year of 2021-24 is 0.95 which is satisfactory.

4.4 LIGHTING SYSTEM

Lighting is an essential service in all industries, Universities, Hospitals, Malls, etc. Innovation and continuous improvement in the field of lighting, have given rise to tremendous energy-saving opportunities in this area. Lighting is an area, which provides a major scope to achieve energy efficiency at the design stage, by incorporation of modern energy-efficient lamps, luminaries, and gears, apart from good operational practices.

Basic Terms in Lighting System and Features

Lamps

Lamp is equipment, which produces light. The most commonly used lamps are described briefly as follows:

Incandescent lamps

Incandescent lamps produce light by means of a filament heated to incandescence by the flow of electric current through it. The principal parts of an incandescent lamp, also known as GLS (General Lighting Service) lamp include the filament, the bulb, the fill gas and the cap.

Reflector lamps

Reflector lamps are basically incandescent, provided with a high-quality internal mirror, which follows exactly the parabolic shape of the lamp. The reflector is resistant to corrosion, thus making the lamp maintenance free and output efficient.

Gas discharge lamps

The light from a gas discharge lamp is produced by the excitation of gas contained in either a tubular or elliptical outer bulb. The most commonly used discharge lamps are as follows:

Fluorescent Tube lamps (FTL)

Compact Fluorescent Lamps (CFL)

Mercury Vapor Lamps

Sodium Vapor Lamps

Metal Halide Lamps

4.4.1 ENERGY SAVING STRATEGY

Introduction of Light-emitting-diode or LED in place of Conventional Lighting Sources

A light transmitting diode (LED) is a semiconductor light source. LED's are utilized as marker lights as a part of numerous gadgets, and are progressively utilized for lighting. Early LED's emitted low power red light, however current adaptations are accessible over the unmistakable, ultraviolet and infrared wavelengths, with high splendor. The LED is focused around the semiconductor diode. At the point when a diode is forward predisposition (exchanged on), electrons can recombine with gaps inside the gadget, discharging vitality as photons. This impact is called electroluminescence and the color of the light (comparing to the vitality of the photon) is dictated by the vitality hole of the semiconductor. LED's present numerous focal points over radiant light sources including easier vitality utilization, longer lifetime, enhanced heartiness, more modest size, quicker exchanging, and more stupendous sturdiness and dependability. Notwithstanding, they are moderately unmanageable and require more exact current and hotness administration than customary light sources. Current LED items for general lighting are more unreasonable to purchase than fluorescent light wellsprings of equivalent yield.

Advantages of LED Lighting System -

- LEDs have an extremely long lifespan relative to every other lighting technology (including LPS and fluorescent lights but especially compared to incandescent lights). New LEDs can last 50,000 to 100,000 hours or more. The typical lifespan for an incandescent bulb, by comparison, is 1-5% as long at best (roughly 1,200 hours).
- LEDs are extremely energy efficient relative to every other commercially available lighting technology. There are several reasons for this including the fact that they waste very little energy in the form of infrared radiation (heat), and they emit light directionally (over 180 degrees versus 360 degrees which means there are far fewer losses from the need to redirect or reflect light).
- Very high light quality
- Very low maintenance costs and hassle

Automatic Control of Street Lights

Automatic control of street lights is designed to turn on and turn off street lights automatically. This project checks the amount of light. If light is 80 percent available, it automatically turns off street lights. But if amount of light is less than 80 percent, this project will automatically turn on street lights. one can also

adjust it according to its requirement. Light sensor is used to detect intensity of light. Microcontroller is used interfaced with light sensor to sense amount of light available. Control signal is generated with the help of microcontroller after analyzing amount of light. Control signal generated by microcontroller is used to turn on transistor which in turn energize the relay coil and relay turn on the street light.

Daylight utilization

Daylight is of extraordinary impact on energy savings. The measures made in this respect especially include investment costs and almost no future costs, compared with other measures (daylight is for free). Daylight can be controlled by use of venetian blinds, roller blinds, marquises, redirecting, and special glazing and so on, in existing buildings. The options how to increase daylight utilization are also installation of skylights into a ceiling structure, installation of light guides, regular window cleaning, painting the walls, change in a space geometry and a space utilization purpose.

Lighting system maintenance

The regular maintenance is very important for the efficiency of the lighting system. Therefore, the maintenance must be carried out in regular intervals. A correctly scheduled maintenance plan helps to maintain the illumination, reduce investment and operational costs and operate the system safely. The plan of maintenance is to include.

- Interval of maintenance execution.
- Description of activities performed within the framework of a regular maintenance.
- Description of activities performed within the framework of extraordinary maintenance (service action).
- Way of luminaries and surfaces cleaning.

Solar Lighting System

Solar street lighting system is ideal for street lighting in remote villages. The system is provided with battery storage backup sufficient to operate the light for 10-11 hours daily. The system is provided with automatic ON/OFF time switch for dusk to dawn operation and overcharge / deep discharge prevention cut-off with LED indicators.

The SPV modules are reported to have a service life of 15-20 years. Tubular Batteries provided with the solar street lighting system require lower maintenance; have longer life and give better performance.

4.5 INVENTORY DETAILS

The audit team has done the Inventory with Wattage analysis of the different types of lighting installed and the other electrical equipment across the campus. The below table shows the electrical equipment install at the College.

Table 4.5. 1 Inventory details of lights Building wise

Sl. No.	Electrical Equipment	Quantity	Wattage/HP	Total load (KW)
1	Tube light (Single)	195	40	7.8
2	Tube lights (LED)	294	20	5.88
3	LED Bulb	275	9	2.475
4	CFL Lights	183	12	2.196
5	LED Bulb Library	189	12	2.268

Observation:

1. Most of the lights in the College are LEDs.
2. Street lights were LED.
3. Overall lighting of College is good.

Recommendation:

1. General maintenance should be done regularly.
2. The street lights should be solar powered.
3. It is Recommend to Install **Astronomical Switch** with timer in all the street lights.

Note: It is stated that Astronomical Switch lightings can give up to 3-5 % savings on Lighting Power Consumption.

4.6 CEILING FANS

Ceiling Fan is the major part which consumes electricity and however, it is very useful in household, universities, offices, etc. Hence, Innovation and continuous improvement in the field of fans, have given rise to tremendous energy-saving opportunities in this area. The fan is an area, which provides a major scope to achieve energy efficiency at the design stage, by incorporation of modern energy-efficient Fans, BLDC Fans, smart Fans, apart from good operational practices.

Table 4.6. 1 Inventory details of Fans Building wise

S.No	Room NO.	Ceiling Fan of 65 W	Total Load (in KW)
1	179	2	0.13
2	177	6	0.39
3	182	6	0.39
4	183	6	0.39
5	184	6	0.39
6	223	5	0.325
7	320	4	0.26
8	178	2	0.13
9	173	2	0.13
10	181	1	0.065
11	169	6	0.39
12	170	6	0.39
13	171	6	0.39
14	319	2	0.13
15	167	4	0.26
16	114	4	0.26
17	115	4	0.26
18	116	9	0.585
19	117	7	0.455
20	118	4	0.26
21	119	5	0.325
22	120	3	0.195
23	122	26	1.69
24	129	14	0.91
25	131	5	0.325
26	209	4	0.26
27	Pylons Office	1	0.065
28	154	2	0.13
29	152	6	0.39
30	153	6	0.39
31	156	6	0.39
32	159	3	0.195

33	204	2	0.13
34	113	3	0.195
35	205	4	0.26
36	207	8	0.52
37	225	6	0.39
38	228	7	0.455
39	214	1	0.065
40	224	4	0.26
41	203	2	0.13
42	229-230	8	0.52
43	155	8	0.52
44	Sambhaji HALL	8	0.52
45	SMH	14	0.91
46	S-2	8	0.52
47	226	2	0.13
48	227	1	0.065
49	222	8	0.52
50	VIVEKANAND HALL	21	1.365
51	301	1	0.065
52	302	6	0.39
53	305	2	0.13
54	308	2	0.13
55	311	4	0.26
56	312	2	0.13
57	313	4	0.26
58	314	4	0.26
59	316	4	0.26
60	317	4	0.26
61	161	2	0.13
62	Digital Valuation Cell	2	0.13
63	226	2	0.13
64	227	2	0.13
65	232	4	0.26
66	215	1	0.065
67	172	6	0.39
68	237	1	0.065
69	180	6	0.39
70	223	6	0.39
71	306	6	0.39
72	307	6	0.39
73	310	6	0.39
74	315	6	0.39
75	325	4	0.26

76	327	1	0.065
77	329	1	0.065
78	330	1	0.065
79	332	1	0.065
80	334	7	0.455
81	335	6	0.39
82	337	7	0.455
83	338	7	0.455
84	339	1	0.065
85	321	6	0.39
86	322	6	0.39
87	323	6	0.39
88	324	6	0.39
89	328	6	0.39
90	331	6	0.39
91	333	6	0.39
92	336	6	0.39
93	Guard Room	2	0.13
94	Mandir Room	1	0.065
95	130	5	0.325
96	133	2	0.13
97	134	1	0.065
98	135	1	0.065
99	136	1	0.065
100	137	1	0.065
101	141	1	0.065
102	142	1	0.065
103	Café	1	0.065
104	217	11	0.715
105	145	1	0.065
106	146	2	0.13
107	Lounge	4	0.26
108	148	4	0.26
109	149	1	0.065
110	150	1	0.065
111	Principal office	4	0.26
112	T&P Cell	2	0.13
113	T&P Cell	2	0.13
114	Admission Cell	4	0.26
115	168	1	0.065
116	188	1	0.065
117	190	12	0.78
118	216	1	0.065

119	Library	9	0.585
	Total	538	34.97

Observation:

1. There are 538 number of 65-Watt fan which consumes 34.97 kW of electricity.
2. It is observed that fans installed at College are old & conventional which needs to be replace by BLDC (Brushless Direct Current) Fans.

Recommendation:

1. It is recommended to replace all the 65 W ceiling fans with energy efficient BLDC fan, in phase wise manner.

Table 4.6. 2 Replacement of Normal Ceiling Fan of 65 W with 30 W BLDC Fan

Replacement of Normal Ceiling Fan of 65 W with 30 W BLDC Fan		
Particulars		Units
Total Number of 65-Watt Ceiling Fan	538	Number
Measured Watt	65	Watts
Total Watts	34970	Watts
Proposed watt after replacement	30	Watts
Total Watt After Replacement	16140	Watts
Operating Hours in a day	8	Hours
Estimated Energy Saving after Replacement Annual kWh	54983.6	kWh
Per Unit Cost as Per CSPDCL Bill	7.1	Rs
Estimated Cost Saving Per Year	3.90384	Rs. Lacs
Cost of Per Fan	2500	Rs
Total Investment Cost	13.45	Rs. Lacs
Payback	3.45	Year

BLDC Fan

A BLDC Fan is a DC motor a magnet synchronous electric motor that is driven by electrical energy (DC) electricity and it accomplishes an electronically controlled commutation system (commutation is the process of manufacturing motility force in the motor by dynamical part currents through it at appropriate times) rather than an automatically commutation system. BLDC motors also are referred to as tetragon magnet motors that are positioned around the mechanical device. The electronics contain a driving algorithm that drives the BLDC motor could be an electric motor that keeps running on the provision of electric power. In this motor, there's the straight relationship amongst current and torsion moreover within the between

voltage and rate of BLDC motor. This straight relationship is that the main reason that the BLDC motor offers wonderful leads to the routine ceiling fans. BLDC Fans consume low power.

BLDC fan takes in AC voltage and internally converts it into DC victimization SMPS. DC voltage as input that is reborn victimization the heartbeat breadth Modulation Techniques to regulate the excitation of the coils to come up with the motion in a prescribed fashion. we've projected AN electronic management for the PWM generation and motion detection and management through the "hall effect" sensors that square measure embedded within the BLDC Motor. there's additionally some way to use the "back emf" generated by the excitation on the "third leg" of the motor for conniving the relative position for the motion management. This sensor less or back electromotive force methodology, whereas it reduces the value of the motor, has bound challenges and disadvantages and isn't being thought of here for the actual application.

Table 4.6. 3 Comparison of BLDC Fans & Ceiling Fan

Speed of Fans (RPM)	BLDC Fans (watts)	Ordinary Fans (Watts)
140	3.8	13
210	7.7	24
270	13.8	30
310	22.7	40
370	35.8	55
410	40	80

4.7 Air Conditioner Details

Table 4.7. 1 Details Air Conditioner

S. No.	Room No.	Air Conditioner of 1.5 Ton (1800 W)	Air Conditioner of 4 Ton (6500 W)	Total Load (in KW)
1	179	1		1.8
2	319	2		3.6
3	167	3		5.4
4	158	1		1.8
5	159	1		1.8
6	214	1		1.8
7	155	4		7.2
8	SambhaJI HALL	2		3.6
9	S-2	7	2	25.6
10	226	4		7.2
11	227	4		7.2
12	222	3		5.4
13	VIVEKANAND HALL	9	2	29.2
14	301	1		1.8
15	314	6		10.8
16	160	4		7.2
17	232	2		3.6
18	327	1		1.8
19	329	1		1.8
20	145	1		1.8
21	146	1		1.8
22	147	1		1.8
23	148	1		1.8
24	150	1		1.8
25	151	1		1.8
26	157	1		1.8
27	Board room	2		3.6
28	163	1		1.8
29	164	1		1.8
30	166	2		3.6
31	176	1		1.8
32	Library		5	32.5
	Total	71	9	186.3

Table 4.7. 2 Ac's Energy Efficiency Ratio on Sample Basis

1	Type of Air-conditioner	UOM	Window	Split	Tower Ac
2	Capacity	TR	1.5	1.5	4.0
4	Set Temperature	°C	24.0	21.0	24.0
5	Inlet Air Dry Bulb Temp. (DBT)	°C	21.4	23	23.19
6	Inlet Air Wet Bulb Temp. (WBT)	°C	20.3	17.4	19.5
7	Inlet Air Relative Humidity (RH)	%	78.0	73.0	78.3
8	Outlet Air DBT	°C	19.0	18.4	11.4
9	Outlet Air WBT	°C	15.0	13.0	8.3
10	Outlet Air RH	%	82.0	78.4	76.3
11	Temp. differential of Inlet DBT & Outlet DBT	°C	2.4	4.6	11.8
12	Avg. Air Velocity	m/sec.	6.0	2.8	2.5
13	Air discharge area	m ²	0.040	0.040	0.080
14	Avg. Air flow	m ³ /h	864	403	720
15	Density of inlet Air	kg/m ³	1.199	1.193	1.192
16	Enthalpy at Inlet Air	kCal/kg of dry air	12.36	11.69	12.067
17	Enthalpy at Outlet Air	kCal/kg of dry air	9.47	9.696	9.565
18	Net Refrigeration Effect	TR	0.99	0.32	0.71
		kW	3.48	1.11	2.50
19	Percent Loading of Air-conditioner	%	66.00	21.14	17.75
20	Power Input to Air-conditioner	kW	2.00	1.80	2.53
21	Compressor Motor Efficiency	%	80.0	80.0	80.0
22	Drive Transmission Efficiency	%	100.0	100.0	100.0
23	Shaft Power of Compressor	kW	1.60	1.44	2.02
24	Specific Power Consumption (SPC)	kW/TR	2.02	5.68	3.56
25	Energy Efficiency Ratio (EER)	W/W	2.18	0.77	1.24

Observations:

1. ACs are old and conventional type.

Recommendations:

1. It is recommended to replace the conventional ACs with five-star inverter AC, in phase wise manner.

4.8 TRANSFORMER LOAD PROFILE

The Table below shows the Load Profile of Transformer.

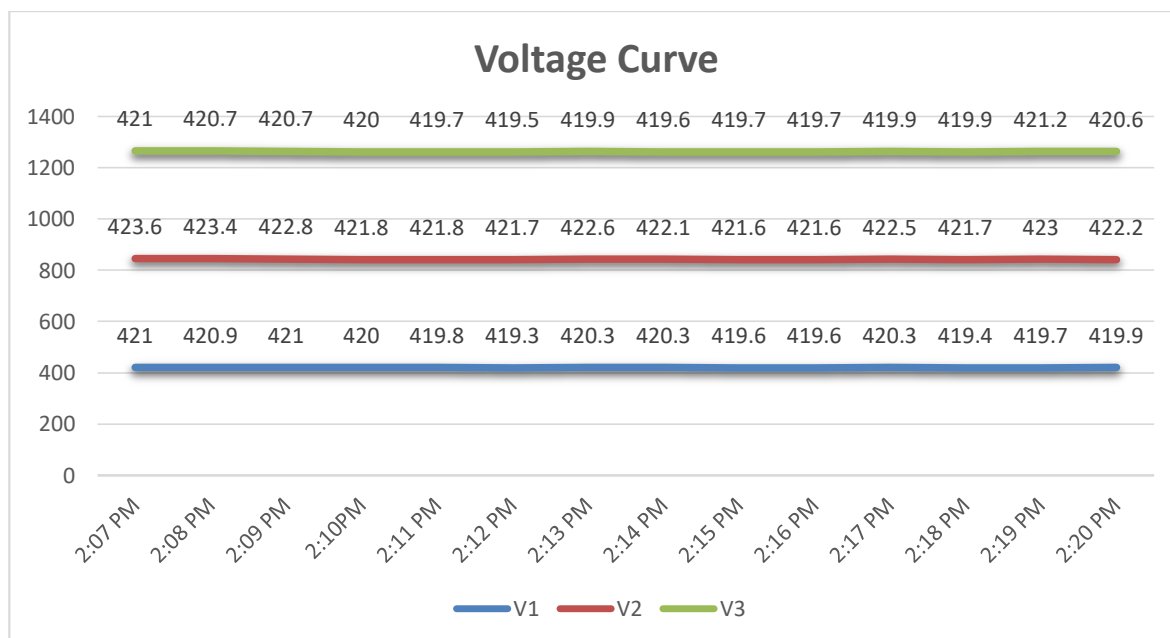
Table 4.8. 1 Load profile of Transformer

CSIT Transformer									
SN.	Name of the drive	Drive Status	Design	Measured Parameters			Calculated Parameters		
	Equipment Name		Transformer Rating KVA	Avg. Voltage (V)	Avg. Current (A)	PF	Calculated kVA	Calculated Values (KW)	Loading (%)
1	Transformer Near Boys Hostel	LT	250.00	419.84	95.33	0.91	69.32	63.08	27.73

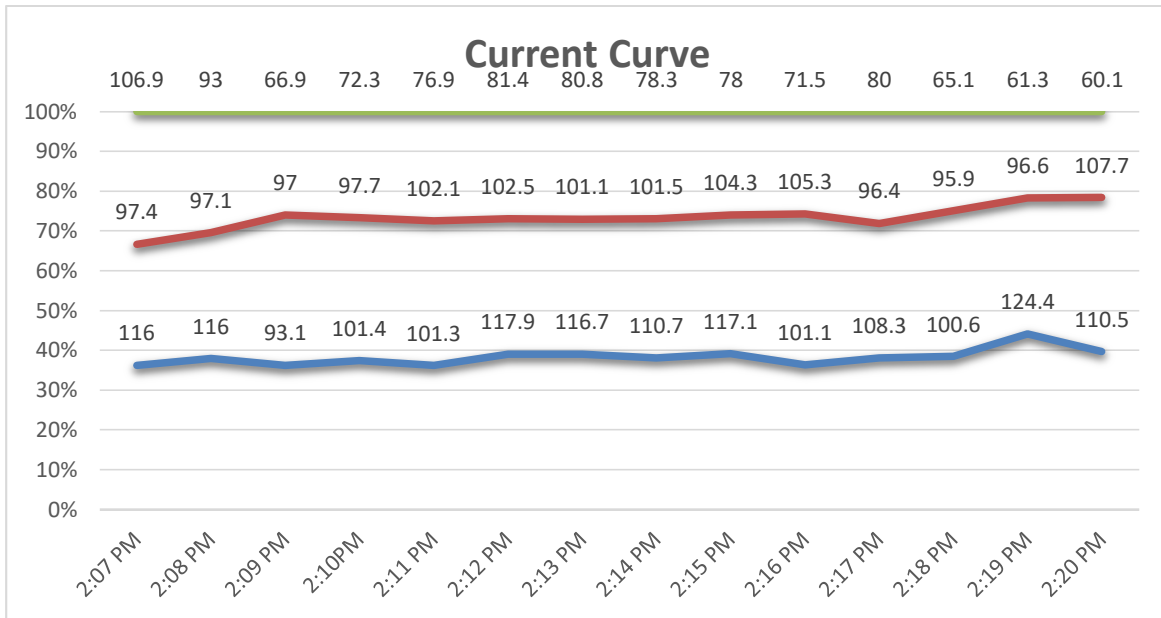
Observations:

1. Above table is based on recording of loading for Six hours.
2. Loading percentage of all transformers were very less because of non-availability of students because of semester break.

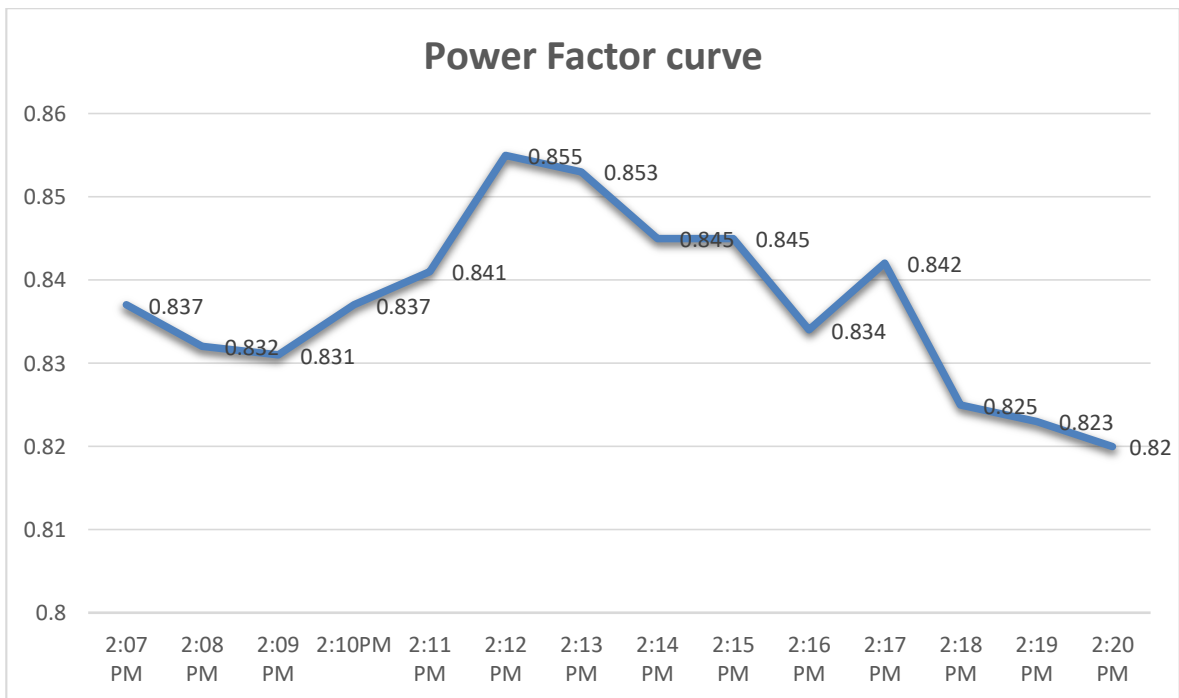
Graphical Representation of Voltage

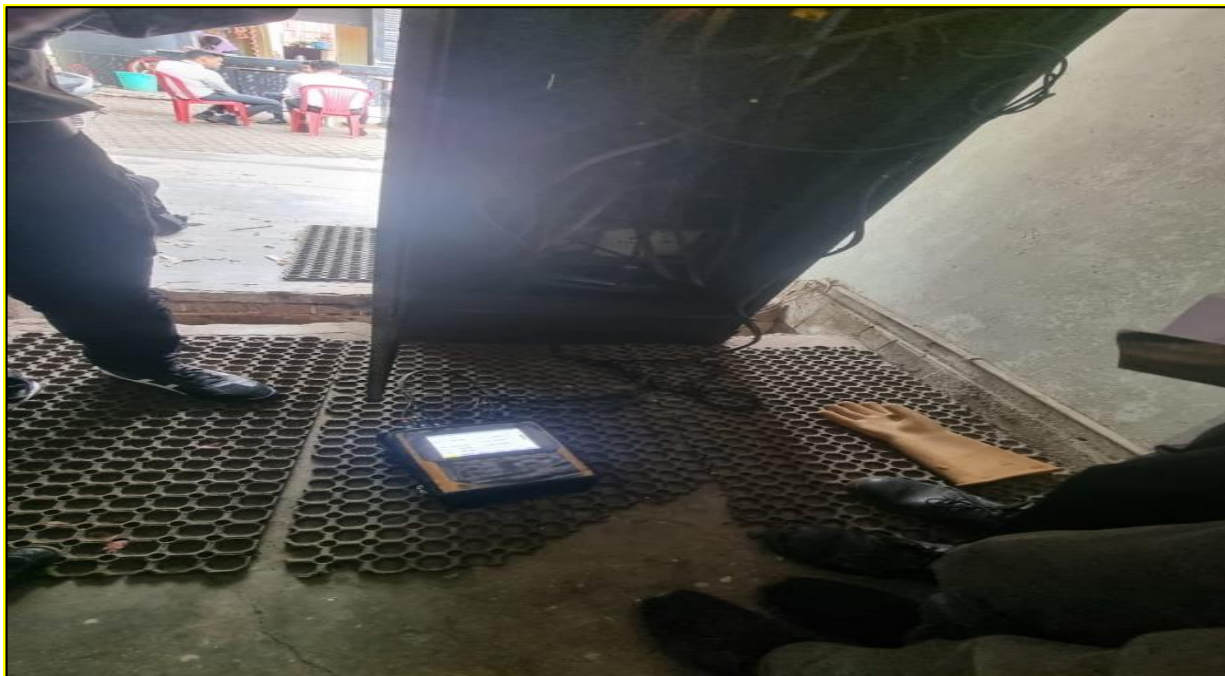


Graphical Representation of Current



Graphical Representation of Power factor





Recording of Electrical Parameters of Transformer



Cables Connection at the Transformer

4.9 POWER QUALITY

Power Quality & Harmonics

Equipment based on frequency conversion techniques generates harmonics. With the increased use of such equipment, harmonics- related problems have been enhanced. The harmonic currents generated by different types of loads travel back to the source. While traveling back to the source, they generate harmonic voltages, following simple Ohm's Law. Harmonic voltages, which appear on the system bus, are harmful to other equipment connected to the same bus. In general, sensitive electronic equipment connected to this bus will be affected. The Harmonics Level on the HT side of the Transformers was measured, details of which is as under: -

Particulars	TR
Voltage Harmonics (VTHD)	
"R"Phase	1.61
"Y"Phase	1.26
"B"Phase	1.17
Current Harmonics (ATHD)	
"R"Phase	13.38
"Y"Phase	12.62
"B"Phase	14

- Maximum Individual Frequency Voltage Harmonic: 3%
- Total Harmonic Distortion of the Voltage: 5%

harmonic current limitations

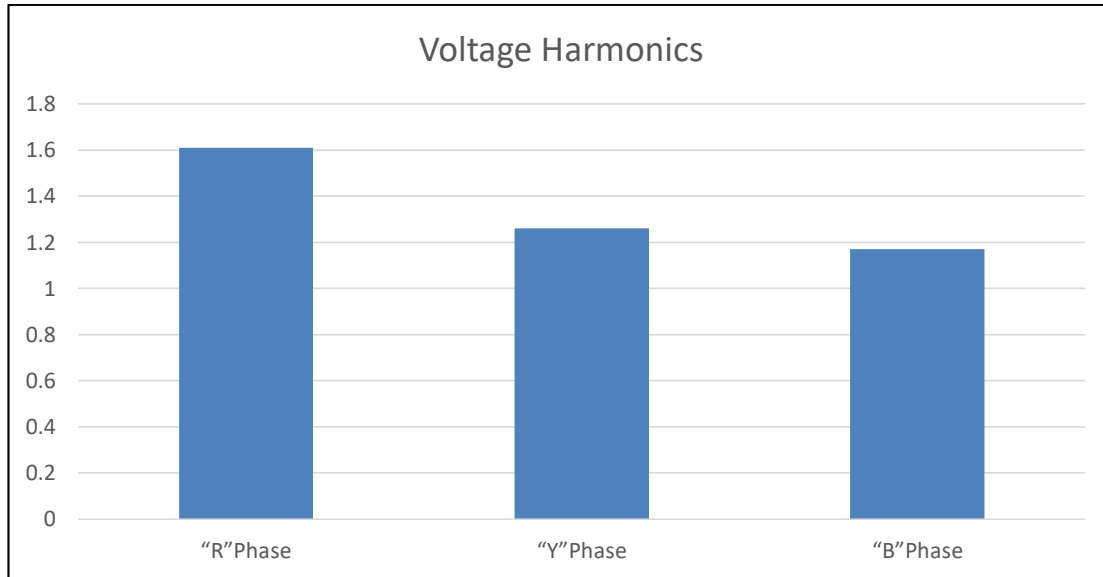
Maximum Harmonic Current Distortion in Percent of IL 120 Volt through 69 KV						
Individual Harmonic Order (Odd Harmonics)						
ISC/IL	h<11	11<h<17	17<h<23	23<h<35	35<h	TDD
<20*	4.0	2.0	1.5	0.6	0.3	5.0
20<50	7.0	3.5	2.5	1.0	0.5	8.0
50<100	10.0	4.5	4.0	1.5	0.7	12.0
100<1000	12.0	5.5	5.0	2.0	1.0	15.0
>1000	15.0	7.0	6.0	2.5	1.4	20.0

Even harmonics are limited to 25% of the odd harmonic limits
TDD refers to Total Demand Distortion based on the average demand current at the fundamental frequency and measured at the PCC (Point of Common Coupling).

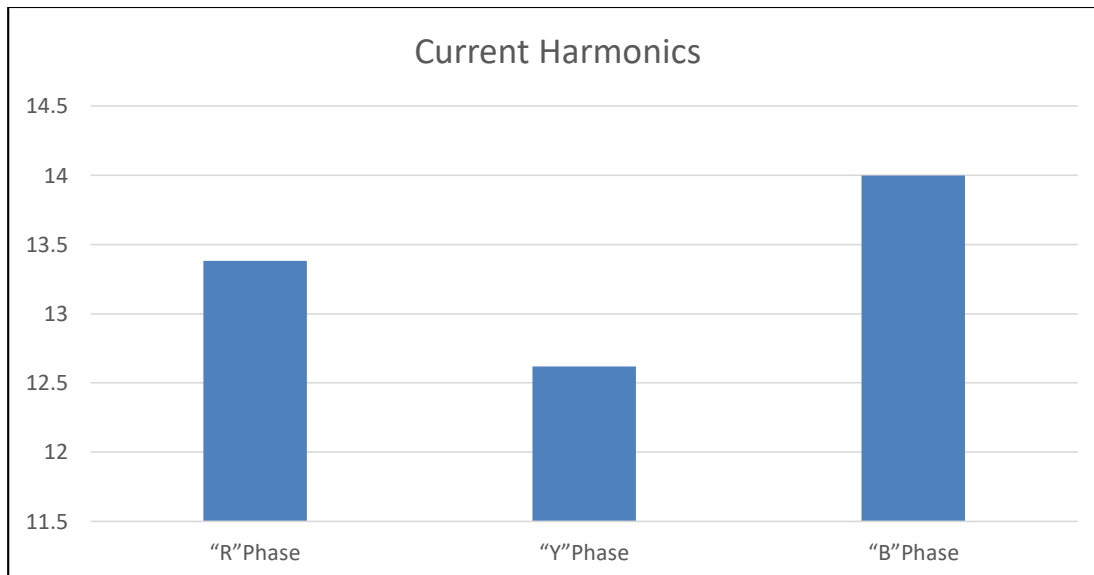
*All power generation equipment is limited to these values of current distortion regardless of ISC/ IL value.

ISC = Maximum short-circuit current at PCC.
IL = Maximum demand load current (fundamental) at the PCC.
h = Harmonic number.

Graphical Representation of Voltage Harmonics (V THD) of Main Incomer



Graphical Representation of Voltage Harmonics (V THD) of Main Incomer



Observations & Suggestions

As detailed above, the voltage harmonics levels were around 1.11-1.61% and the levels of the current harmonics were 12-14%. The Overall harmonics are within limits. If the Harmonics level is on the higher side, then appropriate harmonic filters may have to be installed in the system. Different technologies are available mitigating the harmonics from the system. These include Detuned or broadband harmonic filters: these filter banks are tuned to a frequency just

below predominant harmonic frequency is say, 5th, it is normal practice to tune the filters to 189 Hz, or 3.78th harmonic, in 50 Hz systems.

Active Harmonic Filters: these units are designed in such a manner that, they will inject harmonic frequencies in the system, which will be in anti-phase of the load harmonic frequencies. This will effectively free the source being loaded due to harmonics.

Major Causes of Harmonics

Devices that draw non-sinusoidal currents when a sinusoidal voltage is applied create harmonics. Frequently these are devices that convert AC to DC. Some of these devices are listed below:

- Electronic Switching Power Converters
- Computers, Uninterruptible power supplies (UPS), Solid-state rectifiers
- Electronic process control equipment, PLC's, etc.
- Electronic lighting ballasts, including light dimmer
- Reduced voltage motor controllers
- Arcing Devices
- Discharge lighting, e.g. Fluorescent, Sodium and Mercury vapor
- Transformers operating near saturation level
- Magnetic ballasts (Saturated Iron core)
- Induction heating equipment, Chokes, Motors, Appliances
- TV sets, air conditioners, washing machines, microwave ovens
- Fax machines, photocopiers, printers
- These devices use power electronics like SCRs, diodes, and thyristors, which are a growing percentage of the load in industrial power systems.

Many problems can arise from harmonic currents in a power system. Some problems are easy to detect; others exist and persist because harmonics are not suspected. Higher RMS current and voltage in the system are caused by harmonic currents, which can result in any of the problems listed below:

Blinking of Incandescent Lights	Transformer Saturation
Capacitor Failure	Harmonic Resonance
Circuit Breakers Tripping	Inductive Heating and Overload
Conductor Failure	Inductive Heating
Electronic Equipment Shutting down	Voltage Distortion
Flickering of Fluorescent Lights	Transformer Saturation
Fuses Blowing for No Apparent Reason	Inductive Heating and Overload
Motor Failures (overheating)	Voltage Drop
Electromagnetic Load Failures	Inductive Heating
Overheating of Metal Enclosures	Inductive Heating
Power Interference on Voice Communication	Harmonic Noise
Transformer Failures	Inductive Heating

4.10 SOLAR POWER IN COLLEGE

There were two Solar System Installed of Capacity 100 KW each in CSIT at –

- 1 Engineering Block
- 2 Administrative Building

It is worth noting that during the duration of audit, one of the solar stations was not in use as there were issues reported, i.e., of battery failure and some damaged panels.



Solar System Installed at Institute Building



Solar System and Installed Devices at the Institute

Observation: -

1. It is Observed that there are total 2 units having capacity 100 KWP of Solar power installed at the college out of which only 1 unit is in operating condition and other unit is in non-operation damage condition.

Recommendations: -

1. It is recommended to repair or replace the old unit and start the operation of 100 KWp solar panel for generation of Renewable Energy.

4.11 GENERAL TIPS FOR ENERGY CONSERVATION IN DIFFERENT UTILITIES SYSTEMS

4.11.1 ELECTRICITY

- Schedule your operations to maintain a high load factor
- Minimize maximum demand by tripping loads through a demand controller
- Use standby electric generation equipment for on-peak high load periods.
- Correct power factor to at least 0.99 under rated load conditions.
- Set transformer taps to optimum settings.
- Shut off unnecessary computers, printers, and copiers at night.

4.11.2 MOTORS

- Properly size to the load for optimum efficiency.
- (High efficiency motors offer of 4 - 5% higher efficiency than standard motors)
- Check alignment.
- Provide proper ventilation, (For every 10°C increase in motor operating temperature over recommended peak, the motor life is estimated to be halved)
- Check for under-voltage and over-voltage conditions.
- Balance the three-phase power supply.
- (An Imbalanced voltage can reduce 3 - 5% in motor input power)
- Demand efficiency restoration after motor rewinding.

4.11.3 FANS

- Use smooth, well-rounded air inlet cones for fan air intakes.
- Avoid poor flow distribution at the fan inlet.
- Minimize fan inlet and outlet obstructions.
- Clean screens, filters, and fan blades regularly
- Use aero foil-shaped fan blades.
- Minimize fan speed.

- Use low-slip or flat belts.
- Check belt tension regularly.
- Eliminate variable pitch pulleys.
- Use variable speed drives for large variable fan loads.
- Use energy-efficient motors for continuous or near-continuous operation
- Eliminate leaks in ductwork.
- Minimize bends in ductwork
- Turn fans off when not needed

4.11.4 PUMPS

- Operate pumping near best efficiency point.
- Modify pumping to minimize throttling.
- Adapt to wide load variation with variable speed drives or sequenced control of smaller units.
- Stop running both pumps -- add an auto-start for an on-line spare or add a booster pump in the problem area.
- Use booster pumps for small loads requiring higher pressures.
- Increase fluid temperature differentials to reduce pumping rates.
- Repair seals and packing to minimize water waste.
- Balance the system to minimize flows and reduce pump power requirements.
- Use siphon effect to advantage: don't waste pumping head with a free-fall (gravity) return.

4.11.5 LIGHTING

- Reduce excessive illumination levels to standard levels using switching, delamping, etc. (Know the electrical effects before doing delamping.)
- Aggressively control lighting with clock timers, delay timers, photocells, and/or occupancy sensors.
- Install efficient alternatives to incandescent lighting, mercury vapor lighting, etc.
- Efficiency (lumens/watt) of various technologies range from best to worst approximately

as follows: low pressure sodium, high pressure sodium, metal halide, fluorescent, mercury vapor, incandescent.

- Select ballasts and lamps carefully with high power factor and long-term efficiency in mind.
- Upgrade obsolete fluorescent systems to Compact fluorescents and electronic ballasts
- Consider lowering the fixtures to enable using less of them.
- Consider day lighting, skylights, etc.
- Consider painting the walls a lighter color and using less lighting fixtures or lower wattages.
- Use task lighting and reduce background illumination.
- Re-evaluate exterior lighting strategy, type, and control. Control it aggressively.
- Change exit signs from incandescent to LED.

4.11.6 DG SETS

- Optimize loading
- Use waste heat to generate steam/hot water/power an absorption chiller or preheat process or utility feeds.
- Use jacket and head cooling water for process needs
- Clean air filters regularly
- Insulate exhaust pipes to reduce DG set room temperatures.

4.11.7 WATER & WASTE WATER

- Recycle water, particularly for uses with less-critical quality requirements.
- Recycle water, especially if sewer costs are based on water consumption.
- Balance closed systems to minimize flow and reduce pump power requirements.
- Eliminate once-through cooling with water.
- Use the least expensive type of water that will satisfy the requirement.
- Test for underground water leaks. (It's easy to do over a holiday shutdown.)
- Check water overflow pipes for proper operating level.

- Automate blow down to minimize it.
- Provide proper tools for wash down- especially self-closing nozzles.
- Reduce flows at water sampling stations.
- Eliminate continuous overflow at water tanks.
- Promptly repair leaking toilets and faucets.
- Use water restrictors on faucets, showers, etc.
- Use self-closing type faucets in restrooms.

4.11.8 ENERGY MANAGEMENT STRATEGY

Energy Management should be seen as a continuous process. Strategies should be reviewed annually and revised as necessary. The key activities suggested have been outlined below:

Set Up an Energy Monitoring and Reporting System

Successful energy management requires the establishment of a system to collect/ analyze and report the energy costs and consumption pattern. This will enable an overview of energy use and its related costs, as well as facilitating the identification of savings that might otherwise not be detected. The system needs to record both historical and ongoing energy use, as well as cost information from billing data, and capable of producing summary reports on a regular basis. This information will provide the means by which trends can be analyzed and reviewed for corrective measures.

Implement A Staff Awareness and Training Program

A key ingredient to the success of an energy management program is maintaining a high level of awareness among staff. This can be achieved in a number of ways, including formal training, newsletters, posters and publications. It is important to communicate program plans and case studies that demonstrate savings, and to report results at least at 12-month intervals. Staff may need training from specialists on energy saving practices and equipment.

5. SOLID WASTE AUDIT

Solid waste is the unwanted or useless solid material generated from human activities in a residential, industrial, or commercial area. Solid waste management reduces or eliminates the adverse impact on the environment and human health. A number of processes are involved inefficiently managing waste for an organization. It is necessary to manage the solid waste properly to reduce the load on the waste management system. Solid waste generation and its management is a burning issue in current days. The rate of generation of solid waste is very high and yet we do not have adequate technology to manage the generated waste. Unscientific handling of solid waste can create threats to public health and environmental safety issues. Thus, it is necessary to manage solid waste properly to reduce the load on the waste management system. The purpose of this audit is to find out the quantity, volume, type, and current management practice of solid waste generation in the CSIT College campus. This report will help for further solid waste management and to go for green campus development.

5.1 GENERATION OF SOLID WASTE IN CSIT COLLEGE

The Institute has a clear policy for waste management. It encourages reuse and recycling of the waste, unless impossible. There are different types of waste are recorded such as paper waste, plastic waste, biodegradable waste etc. However biodegradable waste is recycled through the vermicomposting process previously, but now this process is not functioning. The wastes generated in the campus include

- (i) kitchen wastes,
- (ii) liquid waste (residential and eateries),
- (iii) sewage and sludge,
- (iv) plastic wastes,
- (v) cans and bottles,
- (vi) papers including packaging materials
- (vii) garden leaves and

The College is committed to ensuring that all forms of wastes generated are handled based on the RRRR (Reduce, Reuse, Recycle, Recover) principles following appropriate source segregation protocols including safe disposal of chemical and hazardous wastes. A preliminary survey reveals the domination of biodegradable components (volume basis) over the non-biodegradable counterparts on the campus. The students' hostels share the highest amount of solid waste mostly dominated by food/kitchen wastes (a substantial amount of papers, plastics, metals are also seen

with waste also generated in hostels) followed by residential areas, academic buildings, construction sites (occasionally), open areas including gardens and roads.

5.2 WASTE MANAGEMENT

Biodegradable Waste Management

Handling of the solid waste:

College has taken initiative for Biodegradable Waste Management to compost using processes like Dry & Wet Waste Management.

The mess wardens and the servers are strict towards the waste of food. The students are encouraged to take food as per their appetite. Though in a small quantity the vegetable peels and leftover food from the mess is collected in drums and is taken to nearby domestic cattle shelter for feeding. This practice is done twice i.e. after the lunch and dinner at the hostel.

Vermicomposting Unit

Vermicomposting technology relies upon the conjoint action of earthworms and microorganisms to rapidly transform varied types of solid wastes. Considering the simplicity and flexibility of the technology, a vermicomposting unit was established in the College. The prime objectives are to recycle biodegradable waste fractions in a sustainable manner and curtail the cost of purchasing organic manure from the market for landscaping ventures. The compost is usually taken by nearby farmers for their fields.

Liquid waste:

The major amount of liquid waste is sewerage water from academic block and hostels. The institute has installed an STP for the treatment of waste water. The treated waste water is used for maintain the greenery at the campus using a separate supply line. The water treated water has contributed in growing a variety of plants. The lush green area that occupies approximately 40 % of the total area remains green throughout the year due to the proper supply of water from the STP. The dewatered sludge and compost is given to farmers for farming.

Initiatives taken by the College for Waste Management

- The stationary waste generated at CSIT is sent for recycling and reuse.
- Hazardous waste generated in a solid and liquid state during experiments in the laboratory is disposed of properly.
- Biodegradable waste is a major solid waste generated on campus which is further treated by vermicompost technology.

Recommendations

- Provision of installation of garbage unit should be introduced where the multilevel segregation of various wastes such as paper, construction, glass, metal scrap, and food waste should be done. Further various waste recycling plans for different types of waste should be introduced.
 - Recycling facilities should be introduced and should be supported by City Municipality and private suppliers, including glass, cans, white, colored, and brown paper, plastic bottles, batteries, print cartridges, cardboard, and furniture.



Dust bins

6. ENVIRONMENT QUALITY AUDIT

This includes the plants, greenery, and sustainability of the campus to ensure that the buildings conform to green standards. This also helps in ensuring that the Environmental Policy is enacted, enforced, and reviewed using various environmental awareness programs.

6.1 ENVIRONMENT QUALITY AUDIT

The CSIT campus is well known for its lush green vicinity and beauty. The campus has a green cover of about 40%. It comprises of herbs, shrubs and various trees. The fauna includes herbs like Tulsi, Haldi, croton, aloe vera, lemon grass etc.; shrubs like- amla, bottle-brush, eliche, lemon and trees like guava, mango, neem, bel etc. Activities organized to create greenery and its conservation at the College campus is as follows-

- Plantation of diversified species, Uses of medicinal plants, Identification of plants species
- Waste management plan and disposal facility

Various departmental activities are being carried out every year such as: -

- Plantations and other Landscaping Activities
- Maintenance of Gardens and Landscape
- Maintenance of Plantations

There are three types of vegetation in the College campus, which are-

- i) Big Trees.
- ii) Herbs
- iii) Shrubs

6.1.1 PLANTATIONS

Towards the sustainable land use practice, approximately 5446 plant saplings of different species have been planted in various sites through routine and special plantation drives. This program helps in encouraging an eco-friendly environment that provides pure oxygen within the institute and awareness among villagers. The plantation program includes various types of indigenous species of ornamental and medicinal wild plant species.

Herbs

S. No	Common Name	Botanical Name	Number
1	Bay Leaves	Laurus nobilis	1
2	Brilliantly hot	Capsicum annuum	1
3	Dalchini	Cinnamomum zeylanicum	1
4	Cloves	Syzygium aromaticum	1
5	Cardamom	Elettaria cardamomum	4
6	Aloe vera	Aloe vera	10
7	Kali Mirch	Piper nigrum	20
8	Kalanchoe pinnata	Bryophyllum pinnatum	50
9	Chirayita	Swertia	4
10	Caraway	Trachyspermum ammi	3
11	Wandering Jew	Commelina benghalensis	40
12	Tulsi	Ocimum tenuiflorum	20
13	Lemon grass	Cymbopogon citratus	1

Shrubs

S. No	Common Name	Botanical Name	Number
1	Century plant	Agave americana	100
2	Taupata	Coprosma	50
3	Copper leaf	Acalypha wilkesiana	500
4	Ming aralia	Polyscias fruticosa	10
5	Sissoo spinach	Alternanthera sessilis	20
6	Garden privet	Ligustrum ovalifolium	30
7	Banana tree	Musa paradisiaca Linn	2
8	Spider plant	Chlorophytum comosum	400
9	Mountain Blechnum	Blechnum Orientale	5
10	Mosaic Plant	Peppy flora	20
11	Bodhi tree	Ficus religiosa	15
12	Dudhmogre	Delo nix regia	3
13	Jasmine	Jasminum sambac	100
14	Ixora	Ixora coccinea	40
15	Pedilanthus	Euphorbia	200
16	Corn Palm	Dracaena	250
17	Golden Durante	Durante repens	1000
18	Bougainvillea	Nyctaginaceae	80
19	Bird-of-paradise	Strelitziaceae	5
20	Adenium	Adenium obesum	10
21	Lily	Lilium	5
22	Spurge	Euphorbia	100
23	Creeping zinnia	Zinnia	400
24	Sunflowers	Helianthus	100

25	Blanket flowers	Gaillardia	80
26	Love Leena	Beaucarnea recurvata	17
27	Mini tagar	Tabernaemontana coronaria nana	80
28	Cord moss	Funaria	2
29	Fig trees	Ficus	50
30	Chinese spinach	Amaranthus tricolor	200
31	Fenishpama	Mimosa pudica	7
32	Vridhadaru	Argyreia speciosa	8
33	Garlic Vine	Mansoa alliacea	4
34	Lobster-claws	Heliconias	5
35	Button flower	Centaurea cyanus	50
36	Verbena	Verbena officinalis	50
37	Rhoeo	Tradescantia spathacea	150
38	Ferns	Polypodiopsida	50
39	King sago	Cycas	5
40	Song of India	Dracaena reflexa	100
41	Mardhravita	Mandragora officinarum	2
42	Bamboo	Bambusa vulgaris	100
43	Puding	Croton	20
44	Sensitive plant	Mimosa pudica	30
45	Angelica	Angelica archangelica	5
46	Norwegian angelica	Angelica archangelica	10
47	Black lily	Dracunculus vulgaris	50
48	Areca palm	Dypsis lutescens	50
49	Ribbon grass	Phalaris arundinacea	50
50	February daphne	Daphne mezereum	10
51	Spider lily	Hymenocallis littoralis	20
52	Rose	Rosa	40
53	Miss lily	Lilium hybrid	30
54	Harsingar	Nyctanthes arbor-tristis	2
55	Rhapis palm	Rhapis excelsa	20

Trees

S. No	Common Name	Botanical Name	Number
1	Stone apple	Aegle marmelos	2
2	Bur flower	Neolamarckia cadamba	1
3	Banyan	Ficus benghalensis	2
4	Ficus religiosa	Moraceae	2
5	Rudraksha	Elaeocarpus angustifolius	1
6	Nerium Oleander	Cascabela thevetia	4
7	Gooseberry	Phyllanthus emblica	2
8	Mango	Mangifera indica	4

9	Golden shower	Cassia fistula	10
10	Water palm	Araliaceae	40
11	Palmyra Palm	Borassus flabellifer	50
12	Spanish cherry	Mimusops elengi	12
13	Almond	Prunus dulcis	70
14	Foxtail millet	Setaria italic	2
15	Indian cork tree	Millingtonia hortensis	40
16	Crepe myrtle	Lagerstroemia indica	12
17	Mehandi	Lawsonia inermis	1
18	Royal Poinciana	Delo nix regia	12
19	Christmas Tree	Araucaria columnaris	3
20	Chinaberry	Melia azedarach	2
21	Coconut	Cocos nucifera	3
22	Blackboard tree	Alstonia scholaris	25
23	Champak	Magnolia Champaca	15
24	Ashoka	Saraca asoca	3
25	Yellow Flame	Peltophorum	100
26	Nimba tree	Azadirachta indica	2
27	Sagwan	Tectona grandis	20
28	Nilgiri	Myrtaceous	40
29	Arjun	Terminalia arjuna	2
30	Jequirity bean	Abrus precatorius	15
31	Gum Arabic tree	Acacia senegal	2
32	Black plum	Syzygium cumini	1
33	Mulberries	Morus	2
34	Sandalwood	Santalum album	1
35	Leamon	Rutaceae	2
36	Black Currant	Carissa carandas	2
37	Jackfruit	Artocarpus heterophyllus	25
38	Madras thorn	Pithecellobium dulce	5
39	Rose mallow	Hibiscus	10
40	Bird Lime Tree	Cordia dichotoma	1



Green Area View near Labs



Green Area View from Workshop



Greenery at the backyard and play ground



Greenery along the hostel and staff quarters way



Greenery at E- Block

Maintenance of Gardens and Landscape

In addition to new plantation drives and landscaping/beautification activities, all essential maintenance work (like lawn, hedge, existing plants /shrubs, growing of seasonal flowers) for previously developed flower and other gardens, as well as other locations of the College campus, is done regularly under the supervision of Horticulture Section.

Maintenance of Plantations

Apart from the maintenance of gardens, all previously planted trees (like roadside and other plantations) in different locations of the College campus are regularly nurtured by cleaning, fertilization, watering, etc.

House Plants

House plants do not just look good – they can make us feel good, too. Studies have shown that house plants-

- Boost our mood, productivity, concentration, and creativity
- Reduce our stress, fatigue, sore throats, and colds
- Help clean indoor air by absorbing toxins, increasing humidity & producing oxygen
- Add life to a sterile space, give privacy and reduce noise levels

Campus Involvement

For sustainable use of resources and for the mission of “GO-GREEN” it is necessary that the students, faculty, and administration welcome it. CSIT has an environment that invites opportunities to better its community through campus organizations. The green initiative started on the campus many years ago. The College students are actively participating and solely concerned with the environment. These students under the guidance of faculties strive to create an environmentally friendly campus. Their purpose is to create awareness and eventually act on that awareness. College is also actively conducting environmental awareness programs on campus regularly.

6.2 AMBIENT AIR QUALITY AT CSIT

Sr. No	Parameters	Unit	Result	Requirement permissible limits as per NAAQS/CPCB
1	Particulate Matter, PM10	µg/m ³	85.40	100
2	Particulate Matter, PM 2.5	µg/m ³	40.5	60
3	Sulphur Dioxide (as O ₂)	µg/m ³	7.9	80
4	Carbon Monoxide (as O)	mg/m ³	0.883	04
5	Oxide of Nitrogen (as O ₂)	µg/m ³	23.40	80

Central Pollution Control Board, New Delhi has set guidelines to monitor and analyze the air pollution quality parameters. The trees cover on the campus are the leading sources to absorb CO₂ and release enough fresh O₂ across the College Campus. The result shows that CSIT Campus's air quality status is good as compared to other locations. It is identified that CSIT campus is a green campus. CSIT campus observed minimum air pollution as compared to other Ambient Air Pollution Centers located in different parts of the city.

7. CARBON FOOTPRINTS

Carbon is the basis of life on mother Earth. It is incorporated into the plants through photosynthesis, consumed by animal species through the food, present in the form of carbon dioxide (CO₂) in the atmosphere, locked into the rocks like limestone, and compressed into the different fossil fuels such as coal and oil. As CO₂ levels in the atmosphere continue to increase, most climate designs or projects that the oceans of the world and trees will keep soaking up more than half of CO₂. The plants on land and in the sea, taken up carbon over many years increased the percentage discharged during decay, and this increased carbon became locked away as fossil fuels beneath the surface of the planet. At the start of the 21st century, we brought growing concern about global warming, climate change, food security, poverty, and population growth. In the 21st century, more carbon has been released into the atmosphere than that has been absorbed. CO₂ is a principal component causing global warming. Atmospheric carbon dioxide levels have increased to 40 % from preindustrial levels to more than 390 parts per million CO₂. On this background, it is a need of time to cover the research areas interrelated with climate change.

Carbon foot prints

In today's world, one of the biggest issues faced by all of us is global warming. Global warming refers to an increase in the average global temperature of mother Earth. The main cause of global warming is an increase in the concentration of greenhouse gases (GHGs) in the atmosphere due to anthropogenic activities and their level is determined with the help of global warming potential (GWP) and expressed as Carbon Footprint (CF). Carbon Footprint is another phenomenon used for GHGs or carbon dioxide emission in terms of CO₂ equivalents. There are various definitions of carbon footprint are in literature. But the most recognized definition given by Wiedmann is “the Carbon footprint is the measure of carbon dioxide emissions directly or indirectly caused by an activity or accumulated over the life stages of a product.” In other words, “A carbon footprint is the total greenhouse gas (GHG) emissions caused directly and indirectly by an individual, organization, event or product.”

As CSIT is considered an institutional organization, various energy resources like electricity, solar rooftop systems are used. It is necessary to calculate the carbon footprint of the College to upgrade the Clean Developmental Mechanism (CDM) in various processes. All the data from the various sources were collected from all the sectors where energy resources are used. The collected data is calculated by using standard emission factors.

Efforts for Carbon Neutrality

Air pollution is a matter of serious concern on the campus owing to its urban location. CSIT College as a responsible institution understands the importance of its carbon footprint and developed a plan to reduce greenhouse gas emissions in all its activities. Strictly ban on burning of dried leaves and waste paper in College.

Electricity carbon footprint

In the College, electricity is used for various purposes like residential, office use, and laboratories. The total electricity used in the College liberates mass kg of CO₂ per year. The laboratory equipment consumes the highest electricity which emits a large amount of carbon CO₂ per year. The solar panels are installed on the roof of various buildings produce electricity from solar panels which further saves ample mass of CO₂ per year.

Paper footprint

The papers are used in the institution for various purposes like exam answer sheets, circulars, notices, office work, etc. The papers are responsible for the emission of CO₂. The College used a. On the CSIT Campus, various departments follow paperless methods of communication to reduce the footprint by the use of papers.

Conclusion

India's CO₂ emission is increased by an estimated 4.6 % in 2017, despite a turbulent year for its economy. The carbon footprint of the nation is measured per person; India's emissions are still very low – at only 1.8 tons of CO₂ per capita which is much lower than the world average of 4.2 tons. But those emissions have been increasing steadily, with an average growth rate over the past decade of 6 %. The universities are the organizations which are having large areas which consume high quantities of electricity and LPG for many purposes. The CSIT College Campus emits approx. 24,000 tons of CO₂ per year approximately. The present Clean Development Mechanism (CDM) practices to reduce the 18.10 tons CO₂ per year approximately.

Recommendations

- The food waste generated from College hostel mess, guest house, canteens, and staff quarters should be converted into biogas which can be further utilized for hostel kitchens.
- The solar battery-operated vehicles should be used on the campus to overcome the vehicle

footprint.

- Green computing or E- work is helping the organization to reduce footprint very effectively.
- The solar energy-based street lamps on campus will reduce carbon footprint.
- The awareness should be made among the faculty, students, and other employees regarding Clean Development Mechanism (CDM) to reduce the consumption of electricity and natural resources.
- “Carbon Sequestration” survey should be conducted on the campus. Carbon sequestration is a process of converting atmospheric carbon i.e. CO₂ into other sinks of carbon such as vegetation, soil, ocean, etc. in various forms to mitigate global warming audit is one of the important clauses of the Kyoto Protocol.

Carbon Sequestration

While transforming ourselves from regional universities to global universities, the need of such universities to face the global future challenges and try to find out possible solutions for them. It is a social and environmental responsibility of Government Institutes, Universities, National and International Organizations to respond positively to various global issues at the local level and should collate the generated knowledge into the society. Global warming and climate change are current environmental issues that need to be addressed scientifically and efficiently. As Universities are provided with skillful human resources supported by analytical infrastructure, it is our duty to bring such ideas into practice.

8. GREEN INITIATIVES

CSIT is located in the area which is one of the important wilder areas of Durg city with its precious biodiversity. The major portion is covered with vegetation. The College aims to protect and conserve its biodiversity, fresh and clean ambiance through many initiatives. The College has taken the following green initiatives to protect and conserve nature.

Plantation and Nurturing Programme

Many plantation drives are taken by the College on its campus. Every year on 5th June i.e. World Environment Day, the CSIT takes Plantation activity. College has taken many plantation drives. The Horticulture Section looks after tree plantation activities. The trees are watered by students of various Departments. They nurture these trees throughout the year. Students of various departments and students make the plantation and nurturing program successful. A total of 23 plant saplings of different species (like ornamental, fruit and medicinal plant, etc.) were planted in various sites of the College campus during this year's environment day program.

Green computing practice

Being an academic institution, papers are used for various purposes like exam answer sheets, circulars, notices, office work, document printing, and Xeroxing. Since the trees are cut for paper manufacturing, the sequestration of carbon is reduced increasing carbon footprint. To cut down the carbon footprint, the College administration and various departments follow paperless methods of communication by using emails, online forms submission, etc. The paperless work was helpful in reducing tons of CO₂. The tons of biomass are saved by this green computing practice.

Solar Electricity Generation

The College has installed a 200KW capacity Solar Power Plant for electricity generation which produces electricity and sends it to the local grid which is helpful for an electricity bill reduction. Most of the buildings are constructed considering the need for Light and ventilation which reduces the use of electricity. The air conditioners are used only in essential conditions in the laboratories and offices to reduce electricity consumption.

Conferences and workshops on Environmental Sustainability

CSIT College organizes Conferences and Workshops based on the theme of environmental sustainability.

Energy Conservation Day: -

On 14th December of 2022, which is National Energy Conservation Day was celebrated at CSIT-Durg premises. The CSVTU, Bhilai, and CREDA, Raipur jointly organized and conducted the activities, which included speech, essay, and quiz contests. The prizes that were awarded to the winners of the various competitions each were worth of Rs. 8,000 and consisted of the articles that are listed below, according to the category in which they had placed.

9. CONCLUSIONS

Energy Audit, Environmental & Green Audit is one of the important tools to check the balance of natural resources and their judicious use. Energy Audit, Environmental & Green Auditing is the process of identifying and determining whether institutional practices are eco-friendly and sustainable. It is a process of regular identification, quantification, documenting, reporting and monitoring of environmentally important components in a specified area. The main objective to carry out a green audit is to check the green practices followed by the College and to conduct a well-defined audit report to understand whether the College is on the track of sustainable development. After completing the audit procedure of the College for green practices, there are the following conclusions, recommendations, and Environmental Management Plan (EMP) which can be followed by the College in the future for keeping campus environment friendly.

- College takes efforts to dispose of majority of waste by proper methods. Green computing i.e. Online payment systems, online circulars, and examination procedures are helpful for reducing the use of papers and ultimately reducing carbon footprint.
- Reducing the use of one-time use plastic bottles, cups, folders, pens, bouquets, decorative items will be useful to solve the problem of plastic pollution to some extent.
- Biodegradable waste is used efficiently for composting and vermicomposting.
- Use of LED lamps and Tube Lights is to be encouraged.
- Toilets and bathrooms are consuming more water in the departments. The replacement of old taps can be beneficial for solving this issue.
- The overall ambient air quality on the campus is good while some air quality issues that may arise due to developmental activities on the campus should be addressed. The sound levels on the campus are good.

Key Recommendations & Environment Management Plan (EMP)

Following are some of the key recommendations for improving the campus environment and to be considered as Environment Management Plan (EMP).

- An environmental policy document has to be prepared with all the recommendations and current practices carried by the College.
- A frequent visit should be conducted to ensure that the generated waste is measured, monitored, and recorded regularly and information should be made available to the administration.
- The College should develop internal procedures to ensure its compliance with environmental legislation and responsibility should be fixed to carry out it in practice.
- The solid waste should be reused or recycled at maximum possible places.
- Installation of sensor-based electrification items like fans, lights, etc. can save electricity
- Regular checkups and maintenance of pipes, overhead tanks, and plumbing systems should be done by the engineering section to reduce overflow, leakages, and corruptions.
- Science laboratories large amount of water goes waste during the process of making distilled water; the system should develop to reuse this water for other purposes.
- No such processes or activities were observed at CSIT which can deteriorate the environmental quality.
- The said college is in continuous efforts to spread the environmental awareness programs among staff and students.
- It was also observed that the said college is keeping the environmental quality at priority in every developmental stage.

***DEVELOPMENT IS AN IMPORTANT ASPECT OF ANY
ORGANIZATION, COLLEGE, OR Institute. THIS
DEVELOPMENT BY CSIT IS ALWAYS ACHIEVED AT THE
EXPENSE OF ENVIRONMENTAL REHABILITATION.***

***WE ARE GLAD TO DECLARE THAT CSIT IS AN
ENVIRONMENT-FRIENDLY COLLEGE ALONG WITH MANY
GREEN DEVELOPMENT PROCESSES THAT ARE FAIRLY
PRACTICED BY THE COLLEGE.***

